

UTL MP-0072 SPECTRUM

Element Management System (EMS)

Operations, Administration & Maintenance (OAM) Guide

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1.	About This Guide	11
1.1	Overview	11
1.2	Customer Support and Services.....	11
1.3	UTL Support Contact Information	11
1.4	Customer Support Web Portals.....	11
2.	Introduction to UTL MP-0072 Spectrum EMS	12
2.1	Overview	12
2.2	UTL MP-0072 Spectrum EMS.....	12
2.3	UTL MP-0072 Spectrum EMS Features	12
2.4	FCAPS	13
2.5	UTL MP-0072 Spectrum EMS Network Diagram	13
3.	Web Client UI on UTL MP-0072 Spectrum EMS	14
3.1	Overview	14
3.2	Accessing the Web Client UI.....	14
3.3	Home Page Layout	15
3.4	Zone View.....	15
3.5	Circle View.....	16
3.6	Node View	16
4.	User Management	17
4.1	Add User	18
4.2	Remove User	19
4.3	Edit Users	21
4.4	Privileges	22
4.4.1	Roles Permissions	23
4.4.2	User Permissions.....	25
4.4.3	Add Role	26
4.5	Change Password	27
4.6	Edit Profile.....	28
5.	Online Management.....	30
5.1	Current Alarms.....	30
5.2	Amplifiers Diagnostics	31
5.3	All Client & Line Performance	32
5.4	Port Inventory	32
5.5	Card Inventory	33
5.6	SFP-XFP Inventory.....	34
5.7	Network View.....	35
6.	Report Management	35
6.1	Network Summary.....	36
6.2	Node Status	37
6.3	Network Status History	38
6.4	User Login & Logout.....	39

6.5	Click-Event Details	40
6.6	Card Presence	41
6.7	Card Jack In & Jack Out.....	42
6.8	Performance History	43
6.9	Domain Alarm History.....	44
7.	Card Management	45
7.1	MD-ROADM.....	46
7.1.1	Alarms.....	46
7.1.2	Direction Settings.....	46
7.1.2.1	Direction View	46
7.1.2.2	Add/Delete Ports	47
7.1.2.3	Block/Un-Block Channels	47
7.1.2.4	Configuration Settings	47
I.	Summary.....	48
II.	Attenuation.....	48
III.	Card Details	49
7.1.3	Alarm Severity	49
7.2	OMTP12	49
7.2.1	Configuration Settings	50
7.2.1.1	Client Traffic	50
7.2.1.2	Card Inventory	50
7.2.1.3	Loopback.....	51
7.2.1.4	ALS & FEC	51
7.2.1.5	Switching.....	52
7.2.1.6	Threshold Settings.....	52
7.2.1.7	TTI Settings	53
I.	Line-1 OTU TTI Message	53
II.	Line-1 ODU TTI Message.....	53
III.	Line-2 OTU TTI Message	54
IV.	Line-2 ODU TTI Message	54
7.2.1.8	Wavelength Settings.....	55
7.2.2	Alarms.....	55
7.2.2.1	STM64 Alarms.....	56
7.2.2.2	10G WAN Alarms.....	56
7.2.2.3	10G LAN Alarms	57
7.2.2.4	OTU2 Alarms.....	58
I.	Client OTU2 Alarms.....	58
II.	Line OTU2 Alarms	58
7.2.3	Performance	59
7.2.3.1	Optical Performance	59

7.2.3.2	OTU-2 Performance	59
I.	OTU Performance	59
II.	ODU Performance	60
III.	FEC Performance.....	60
7.2.3.3	STM64 Performance	61
7.2.3.4	10G WAN Performance	61
7.2.3.5	10G LAN Performance.....	62
7.2.4	Alarm Severity	62
7.2.4.1	OTU2 Alarm Severity	62
7.2.4.2	ODU-DeMultiplex Alarm Severity	63
7.2.4.3	Optical Alarm Severity	63
7.2.4.4	SDH Alarm Severity	64
7.2.4.5	10G WAN Alarm Severity	64
7.2.4.6	10G LAN Alarm Severity	64
7.3	OMMP82	65
7.3.1	Configuration Settings	65
7.3.1.1	Client Traffic	65
7.3.1.2	Bandwidth Chart	65
7.3.1.3	Card Inventory	66
7.3.1.4	Loopback	66
7.3.1.5	ALS & FEC	67
7.3.1.6	Threshold Settings.....	67
7.3.1.7	TTI Settings	68
I.	Line-1 OTU TTI Message	68
II.	Line-1 ODU TTI Message.....	68
III.	Line-2 OTU TTI Message	69
IV.	Line-2 ODU TTI Message	69
7.3.1.8	Wavelength Settings	70
7.3.1.9	Switching	70
7.3.1.10	Auto Negotiation	71
7.3.2	Alarms.....	71
7.3.2.1	STM16 Alarms.....	71
7.3.2.2	STM4 Alarms.....	72
7.3.2.3	STM1 Alarms.....	72
7.3.2.4	GigE Alarms	73
7.3.2.5	OTU1 Alarms.....	73
7.3.2.6	OTU2 Alarms.....	74
I.	Line-1 Alarms	74
II.	Line-2 Alarms	74
7.3.3	Performance	75
7.3.3.1	STM16 Performance	75

7.3.3.2	STM4 Performance	75
7.3.3.3	STM1 Performance	76
7.3.3.4	GigE Performance.....	76
7.3.3.5	OTU1 Performance	77
7.3.3.6	Line Performance	77
I.	OTU Performance	77
II.	ODU Performance	78
III.	FEC Performance.....	78
7.3.3.7	OPTICAL Performance	79
7.3.4	Alarm Severity	79
7.3.4.1	SDH Alarm Severity	79
7.3.4.2	GigE Alarm Severity	80
7.3.4.3	Optical Alarm Severity	80
7.3.4.4	OTU2 Alarm Severity	81
7.4	OBA.....	81
7.4.1	Alarms.....	81
7.4.2	Settings.....	82
7.4.3	Performance	82
7.4.4	Alarm Severity	83
7.5	OPA.....	83
7.5.1	Alarms.....	83
7.5.2	Settings.....	84
7.5.3	Performance	84
7.5.4	Alarm Severity	85
7.6	SCC.....	85
7.6.1	Alarms.....	86
7.6.2	Settings.....	86
7.6.2.1	IP Address	86
7.6.2.2	Card Details	87
7.6.2.3	ALS	87
7.6.2.4	Ethernet Port Access.....	87
7.6.2.5	Software Upload.....	88
7.6.2.6	NTP Settings	88
7.6.3	Performance	89
7.6.4	Alarm Severity	89
7.7	SCU.....	90
7.7.1	Settings.....	90
7.8	BMU	90
7.8.1	Settings.....	91
7.9	PS1 & PS2	91

7.9.1	Alarms.....	91
7.10	Buzzer	91
7.10.1	Settings.....	92
7.11	Fan.....	92
7.11.1	Alarms.....	92
7.12	OMTP11	93
7.12.1	Configuration Settings	93
7.12.1.1	Client Traffic.....	93
7.12.1.2	Loopback	93
7.12.1.3	ALS & FEC.....	94
7.12.1.4	Threshold Settings.....	94
7.12.1.5	TTI Settings	94
I.	Line OTU TTI Message	94
II.	Line ODU TTI Message.....	95
7.12.1.6	Wavelength Settings	95
7.12.1.7	Card Inventory	96
7.12.2	Alarms.....	96
7.12.2.1	STM64 Alarms	96
7.12.2.2	10G WAN Alarms	97
7.12.2.3	10G LAN Alarms	97
7.12.2.4	OTU2 Alarms	98
I.	Client Alarms	98
II.	Line Alarms	99
7.12.3	Performance	99
7.12.3.1	Optical Performance.....	100
7.12.3.2	Client OTU Performance	100
I.	OTU Performance	100
II.	ODU Performance	101
III.	FEC Performance.....	101
7.12.3.3	Line Performance.....	101
I.	OTU Performance	101
II.	ODU Performance	102
III.	FEC Performance.....	102
7.12.3.4	STM64 Performance.....	102
7.12.3.5	10G WAN Performance.....	103
7.12.3.6	10G LAN Performance	103
7.12.4	Alarm Severity	104
7.12.4.1	Optical Alarm Severity.....	104
7.12.4.2	OTU2 Alarm Severity.....	104
7.12.4.3	STM64 Alarm Severity	105

7.12.4.4	10G WAN Alarm Severity	105
7.12.4.5	10G LAN Alarm Severity	105
7.13	OMMP81	105
7.13.1	Configuration Settings	106
7.13.1.1	Client Traffic	106
7.13.1.2	Bandwidth Chart	106
7.13.1.3	ALS & FEC	107
7.13.1.4	Loopback	107
7.13.1.5	Auto Negotiation	108
7.13.1.6	Threshold Settings	108
7.13.1.7	TTI Settings	109
I.	Line-1 OTU TTI Message	109
II.	Line-1 ODU TTI Message	109
V.	Line-2 OTU TTI Message	109
VI.	Line-2 ODU TTI Message	110
7.13.1.8	Wavelength Settings	110
7.13.1.9	Card Inventory	111
7.13.2	Alarms	111
7.13.2.1	STM16 Alarms	111
7.13.2.2	STM4 Alarms	112
7.13.2.3	STM1 Alarms	112
7.13.2.4	GigE Alarms	113
7.13.2.5	OTU1 Alarms	114
7.13.2.6	OTU2 Alarms	114
7.13.3	Performance	115
7.13.3.1	STM16 Performance	115
7.13.3.2	STM4 Performance	115
7.13.3.3	STM1 Performance	116
7.13.3.4	GigE Performance	116
7.13.3.5	OTU1 Performance	117
7.13.3.6	Line Performance	117
I.	Line-1 OTU Performance	117
II.	Line-1 ODU Performance	118
III.	Line-1 FEC Performance	118
7.13.3.7	Optical Performance	118
7.13.4	Alarm Severity	119
7.13.4.1	SDH Alarm Severity	119
7.13.4.2	GigE Alarm Severity	119
7.13.4.3	OTU1 Alarm Severity	120
7.13.4.4	ODU-Demultiplex Alarm Severity	120
7.13.4.5	Optical Alarm Severity	120

7.14	OMTP11T.....	121
7.14.1	Configuration Settings	121
7.14.1.1	ALS & FEC	121
7.14.1.2	Card Inventory	122
7.14.1.3	Loopback	122
7.14.1.4	Threshold Settings.....	123
7.14.1.5	TTI Message	123
7.14.1.6	Wavelength.....	124
7.14.2	Alarms.....	124
7.14.2.1	Line1 Alarms	125
7.14.2.2	Line2 Alarms	125
7.14.3	Performance	126
7.14.3.1	Optical Performance	126
7.14.3.2	OTU2 Performance	126
7.14.3.3	Line-1 OTU Performance.....	127
7.14.3.4	Line-1 ODU Performance	127
7.14.3.5	Line-1 FEC Performance.....	128
7.14.3.6	Line-2 OTU Performance.....	128
7.14.3.7	Line-2 ODU Performance	128
7.14.3.8	Line2 FEC Performance.....	129
7.14.4	Alarm Severity	129
7.14.4.1	OTU Alarm Severity	129
7.14.4.2	Optical Alarm Severity	130
7.15	OLA.....	130
7.15.1	Alarms.....	130
7.15.2	Settings	131
7.15.3	Performance	131
7.15.4	Alarm Severity.....	132
7.16	OPS.....	132
7.16.1	Alarms.....	133
7.16.2	Settings.....	133
7.16.3	Performance	134
8.	Tree Management.....	134
9.1	Zone.....	134
8.1.1	Add Zone.....	135
8.1.2	Remove Zone	135
9.2	Circle.....	136
8.2.1	Add Circle.....	137
8.2.2	Remove Circle.....	138
9.3	Node	139
8.3.1	Add Node	139

8.3.2	Edit Node	139
8.3.2.1	De-Configure	140
8.3.2.2	Modify	142
8.3.3	Remove Node	144
8.3.4	Add OPS (Optical Protection Switch).....	144
8.3.5	Edit OPS.....	145
8.3.6	Remove OPS.....	146
9.	Network Planning Tool	147
9.1	Inserting a node.	148
9.2	Inserting a Link.	149
9.3	Link View	149
9.4	Routes View	150
9.5	Traffic Matrix Design.....	150
9.6	Traffic Normalization.....	151
10.	Software Upgrade Management	151

1. About This Guide

1.1 Overview

This guide describes the UTL MP-0072 Spectrum EMS for 40/80-Channel Dense Wavelength Division Multiplexing (DWDM) with channel bit-rates from 10 to 100Gbps. This guide is intended for technical personnel responsible for installing and maintaining the UTL MP-0072 Spectrum EMS, Version 2.0 Software.

The UTL MP-0072 Spectrum EMS Guide is intended for the following audience:

- System Engineers who are responsible for provisioning & configuring the UTL MP-0072 System.
- Maintenance Engineers whose responsibilities include replacing modules, upgrading software and keeping the UTL MP-0072 Spectrum EMS in working state.

1.2 Customer Support and Services

If you encounter difficulty in using this UTL product, you may contact our support personnel by:

- EMAIL (Preferred Method) – Email us at the addresses listed below. Outline your problem in detail. Please include your return email address and a telephone number.
- TELEPHONE – Contact us via telephone at the number listed below, and request Technical Support. Our offices are open Monday to Saturday, 9:00 a.m. to 5:30 p.m. (IST).
- WEB PORTAL – Saving vital time for your support team, UTL provide direct access to the web portal for fault escalating. Please visit our website about customer support web portals.

1.3 UTL Support Contact Information

Product	UTL MP-0072 Spectrum EMS
Email	nfsdwdmsupport@utlindia.com
Phone	080 28524050 Ext 226 (Monday to Saturday, 9 a.m. to 5.30 p.m.)

1.4 Customer Support Web Portals

Our customer support web portals help customers to log-in for assistance from our support team. To find out more about our Customer Support Web Portals, visit <http://www.utlindia.com>.

2. Introduction to UTL MP-0072 Spectrum EMS

2.1 Overview

This chapter provides a product overview of the UTL MP-0072 Spectrum EMS platform for 40/80-Channel Dense Wavelength Division Multiplexing (DWDM) with channel bit-rates from 10 to 100Gbps.

2.2 UTL MP-0072 Spectrum EMS

UTL MP-0072 Spectrum EMS enables users to rapidly install, monitor and manage UTL DWDM MP-0072 network elements in linear/ring/mesh topology from a single point of EMS contact. The main advantage of the topology is simplicity. Another advantage is scalability.

UTL MP-0072 Spectrum EMS supports a full set of management features and is specifically designed to monitor and manage UTL 40/80-Channel DWDM product. The UTL MP-0072 Spectrum EMS is a standalone, turn-key device that includes all the hardware and software needed to provide a highly-available, centralized, and scalable management system for multiple platforms based on UTL DWDM hardware. UTL MP-0072 Spectrum EMS is easily configurable to manage UTL DWDM family products that support Simple Network Management Protocol (SNMPv2c).

UTL MP-0072 Spectrum EMS supports management functions through an intuitive, easy-to-use and Web-client based graphical user interface (GUI).

The system can record all log-ins for a period of at least SIX months and thereafter back up is possible.

The system can store more than 10000 performance records and 500 fault alarm records for restoration in the EMS, on loss of connectivity.

2.3 UTL MP-0072 Spectrum EMS Features

- Intuitive web client based GUI.
- FCAPS model as defined by the ITU-T standards.
- Access to DWDM Network Elements is provided through SNMPv2c.
- Real-time accessibility to Fault and Performance status.
- All collected data is stored in highly-available MySQL RDBMS.
- Allows operator management of alarm states, maintaining an audit log of all activities.
- Allows collection & display of various parameters from managed devices.
- Report generation of historical data.
- Authentication, authorization access to appropriate network resources.
- Involves configuration of the network devices.
- Gather and store configurations from network devices (this can be done locally or remotely).

2.4 FCAPS

FCAPS is an ISO model and stands for the Fault, Configuration, Accounting, Performance and Security functions that are required to manage a network.

Abbreviation	Meaning
F	Fault detection and correction
C	Configuration and operation
A	Accounting and billing
P	Performance assessment and optimization
S	Security assurance and protection

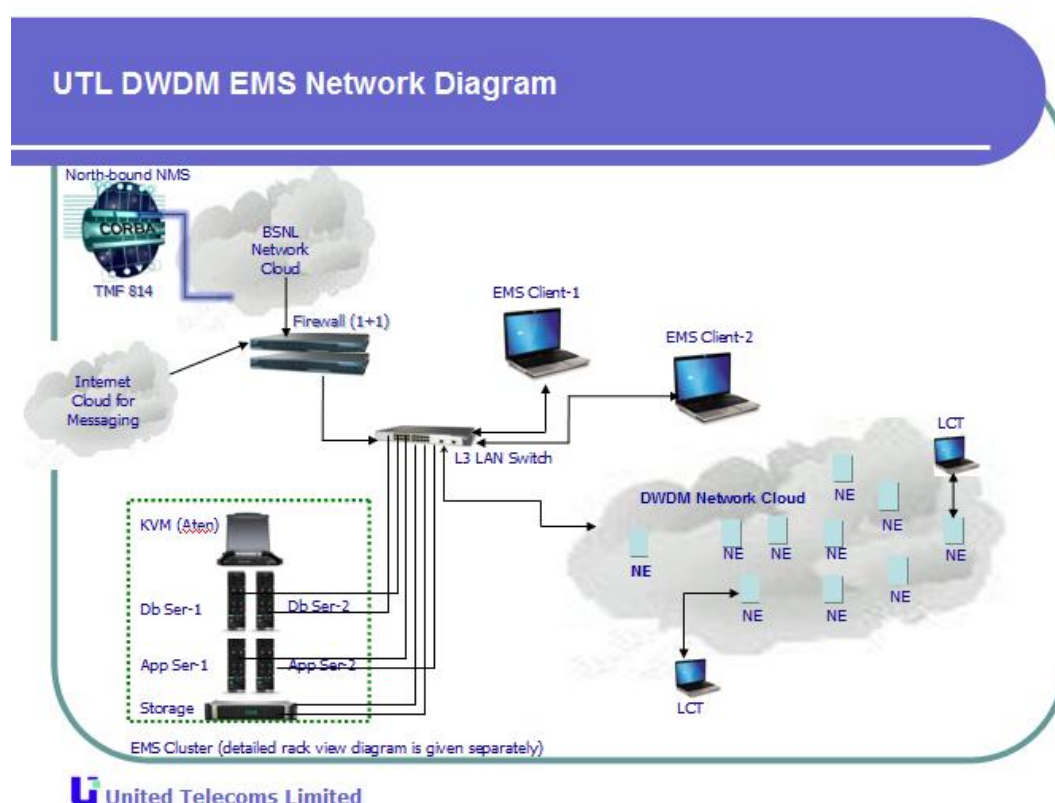
Figure The FCAPS model of network management.

In conclusion, FCAPS to be very useful in understanding various network management functions. However, we need to keep in mind that FCAPS is just a framework that helps us understand better. It does not drive product functionality or user need. These are based on the type and size of the network and the type of business. Many types of user needs may be solved using a single product functionality which may fall under different categories. For example, one of the functionalities is to test the functionality of a certain service. This can be used for troubleshooting. It can also be used for validating provisioning or to take performance measurements. This is exactly the reason why sometimes the same network management product offers functionalities that fall under different FCAPS categories.

2.5 UTL MP-0072 Spectrum EMS Network Diagram

The UTL DWDM network elements in the topology communicate with their respective EMS. This is depicted in the sample network diagram below.

The Element management system (EMS) accesses the UTL MP-0072 topology through the RJ45 10/100 Mbps Ethernet connector on the front panel of the SCC card from any node from respected zone.



3. Web Client UI on UTL MP-0072 Spectrum EMS

3.1 Overview

This chapter provides an introduction to the UTL MP-0072 Spectrum EMS Web Client User Interface (UI), which is used to configure UTL MP-0072 Spectrum EMS parameters, define the network elements, and monitor the status of the managed network elements. It describes how to access the Web Client UI, the layout of the Web Client UI screens, how to access online help, and the typical workflow you would follow to use UTL MP-0072 Spectrum EMS.

3.2 Accessing the Web Client UI

The initial page requires a login and password. Multiple users are configured by default, as indicated in the Table below:

User ID	Permission Level
User	Lowest
Manager	Medium
Administrator	Highest

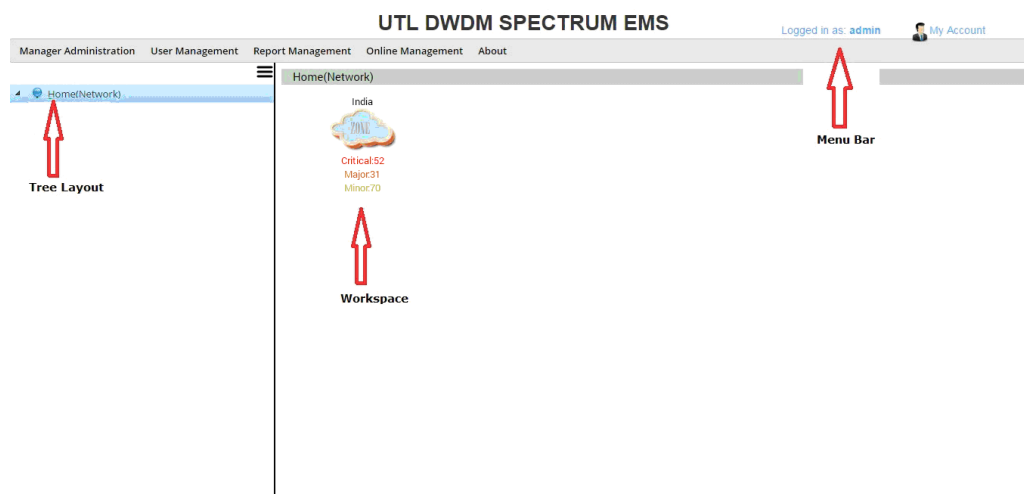
The only user with complete permission to change any parameter is the user **admin** (Administrator). You should login in as **admin** when configuring UTL MP-0072 Spectrum EMS for the first time and use this session to change passwords, add or remove users, and configure the desired access permissions.

Start the UTL MP-0072 Spectrum EMS by double-clicking the icon provided on the Desktop.

Log-in to the EMS as admin@utl.in using the password '**admin1234**' (default login).

3.3 Home Page Layout

The Home View displays all network zones in a graphical form. Double click on specific icon to display zone view.

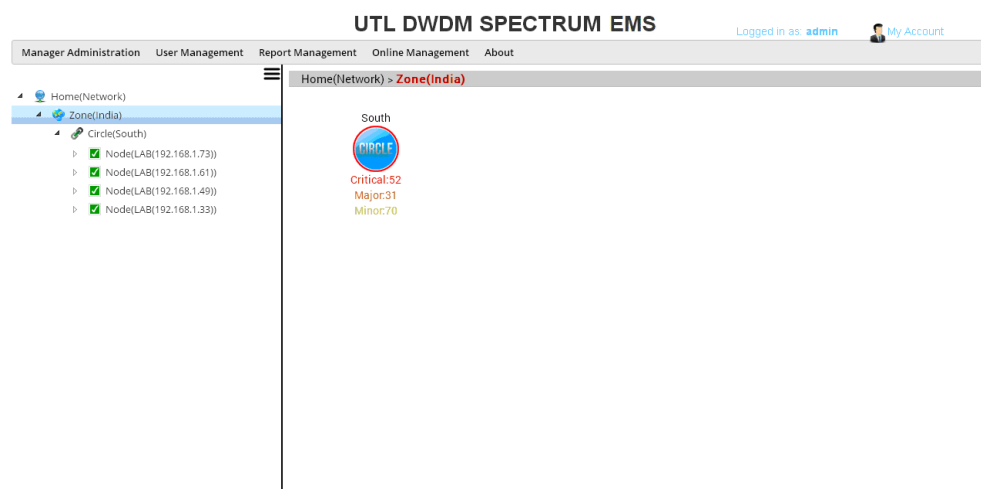


DESCRIPTION

Parameter	Description
Tree Layout	It contains a list which is arranged in one after another.
Menu Bar	It is allocated at the top for quick reference.
Title Bar	It is at the top which displays the name of the page.
Workspace	It is the place to work (create or edit).

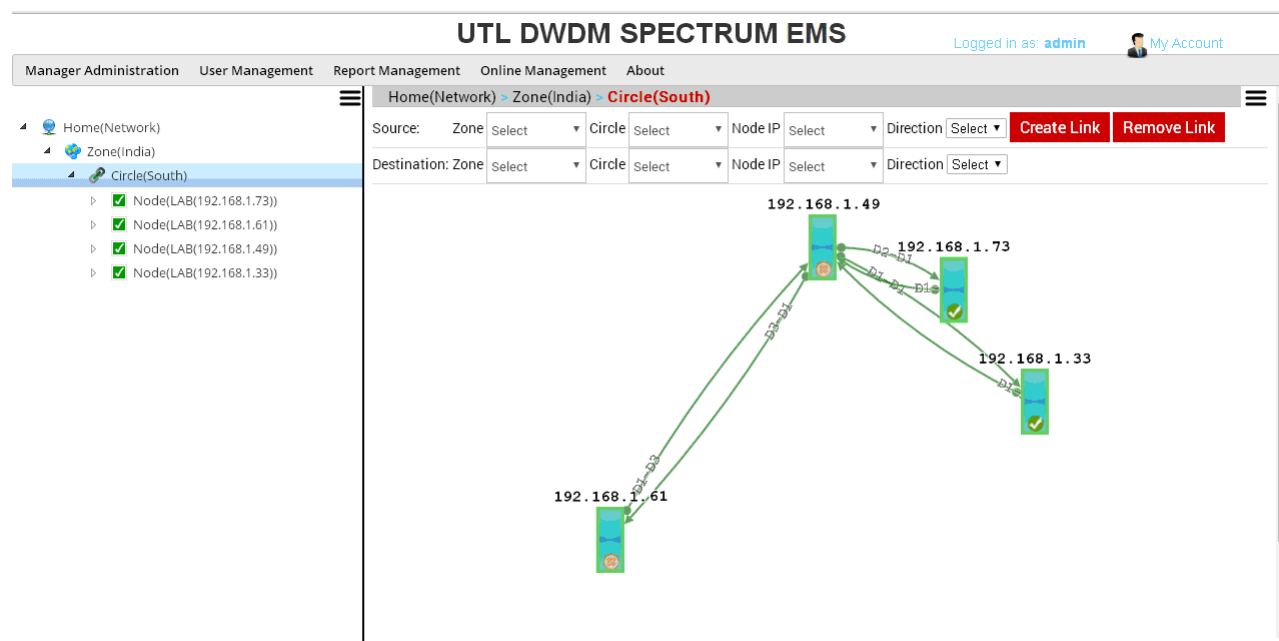
3.4 Zone View

The Zone View displays all circles in a graphical form. Double click on specific icon to display circle view.



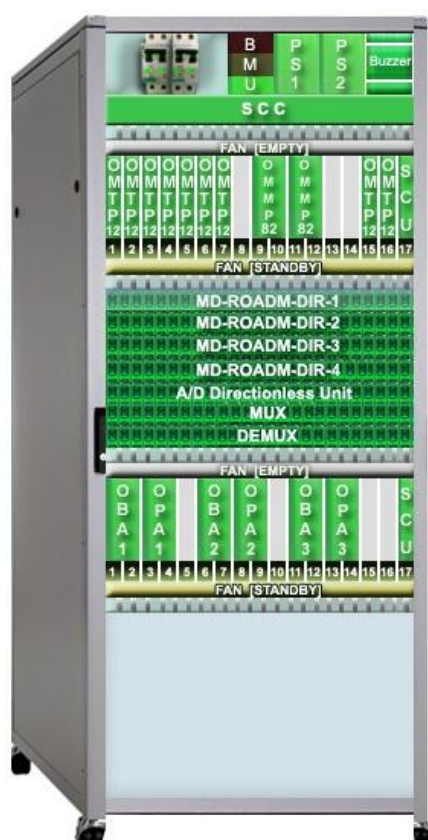
3.5 Circle View

Selecting a circle branch object in the network tree View launches the corresponding circle topology view in the workspace area. Double click on any network node to access.



3.6 Node View

Displays the graphical rack view of node. Various operations on the network elements can be performed using the right-side buttons provided on each of the objects in the node rack view.



Selected Card Details



Card	MD-ROADMPLUS-DIR-1
------	--------------------

Status	Accessible
--------	------------

Alarms Not Present

To view card alarms

Alarm

To do card configurations

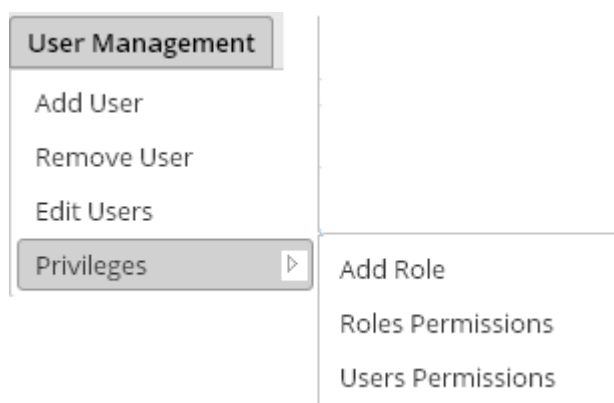
Settings

To do card alarm severity settings

Alarm Severity

4. User Management

User Management is an authentication feature that provides administrators with the ability to identify and control the state of **users** logged into the network. The User Management pull down menu is shown below.



Click on **Menu->User Management** for the following options under User Management:

- 1.Add User
- 2.Remove User
- 3.Edit User
- 4.Privileges
 - 4.1 Roles & Permissions
 - 4.2 Add Role

4.3 Users Permissions

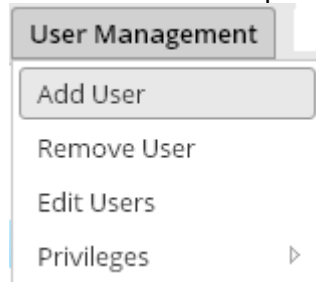
5. Change Password

6. Edit Profile

Select the appropriate menu type from the drop-down menu.

4.1 Add User

Click on **Menu-> User Management -> Add User** option.



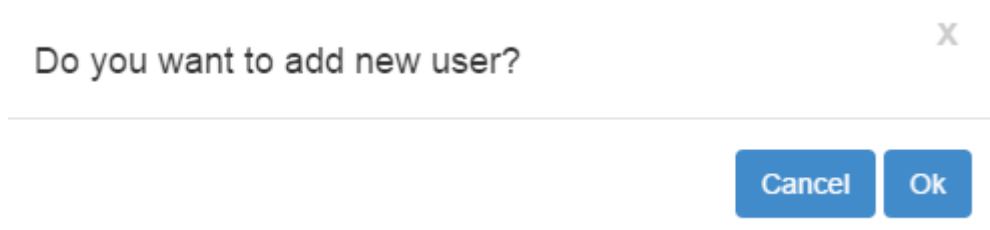
Click on the **Add User** menu option, then Add User dialog as shown in below.

A screenshot of the 'Add User' dialog box. The dialog has a blue header bar with the text 'Add User'. Below the header, there are several input fields with labels to their left: 'User id:' with a text box containing 'Email address'; 'User Name:' with a text box containing 'User Name'; 'Role:' with a dropdown menu showing 'Select'; 'Password:' with a text box containing 'Password'; 'Re-Type Password:' with a text box containing 'Re-Type Password'; 'Client ip:' with a text box containing three dots; 'Contact:' with a text box containing 'Contact Number'; and 'Location:' with a text box containing 'Location'. At the bottom right of the dialog, there are two blue buttons: 'Add User' and 'Close'.

- Enter the User Id, must be a mail id (ex: info@utlindia.com).
- Enter the User Name you want, at least one character.
- Select the role of the new user.
- Enter the password, and it should contain 8 to 15 characters.
- Enter the Re-Type password, and it should contain 8 to 15 characters and must match with the password.

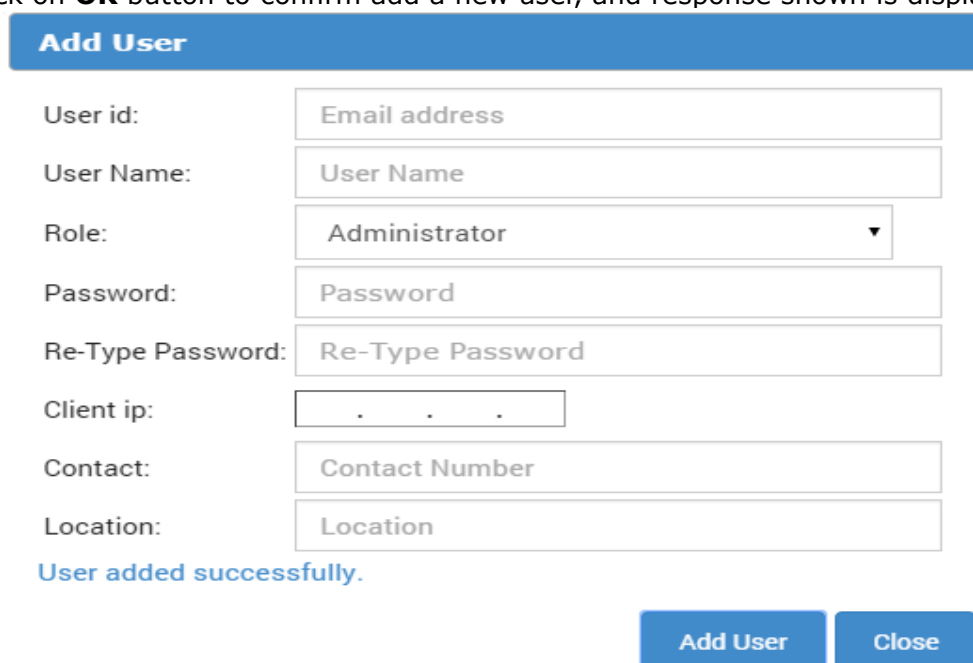
- Enter the Client IP.
- Enter Contact Number, must be 10 digits (ex: 9876543210).
- Enter Location Name, at least one character.

Click on **Add User** button to add a new user. The confirmation dialog shown in below figure is presented.



A confirmation dialog box with the title "Do you want to add new user?". It has a close button (X) in the top right corner. At the bottom, there are two buttons: "Cancel" and "Ok".

Click on **OK** button to confirm add a new user, and response shown is displayed below.



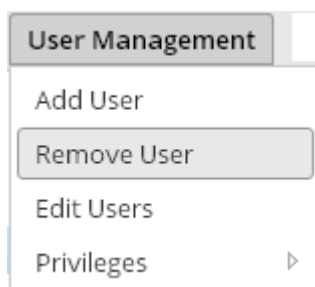
The "Add User" form is displayed. It has a blue header bar with the text "Add User". Below the header, there are several input fields and a dropdown menu:

- User id: Email address
- User Name: User Name
- Role: Administrator (dropdown menu)
- Password: Password
- Re-Type Password: Re-Type Password
- Client ip: . . .
- Contact: Contact Number
- Location: Location

Below the input fields, the text "User added successfully." is displayed in blue. At the bottom right, there are two buttons: "Add User" and "Close".

4.2 Remove User

Click on **Menu-> User Management -> Remove User** option.



A menu titled "User Management" is shown. It contains the following options: "Add User", "Remove User" (highlighted), "Edit Users", and "Privileges" (with a right-pointing arrow).

Click on the **Remove User** menu option, then Remove User dialog will be displayed as shown below.

Remove User

User id:

manager@utl.in ▼

User Name:

manager

Role:

Manager

ClientIP:

192.168.21.1

Permission:

partial

Remove User

Close

- Select the user from drop down list.
- Finally click on **Remove User** button.

The confirmation dialog shown in below figure is presented.

Do you want to remove user?

X

Cancel

Ok

Click on **Ok** button to update user information.

Following response window will be displayed:

Remove User

User id:

admin@utl.in ▼

User Name:

admin

Role:

Administrator

ClientIP:

192.168.21.1

Permission:

partial

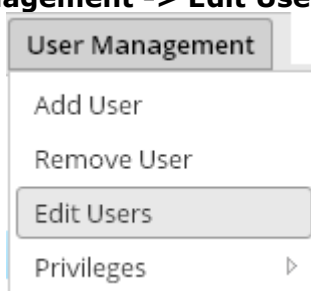
User removed successfully.

Remove User

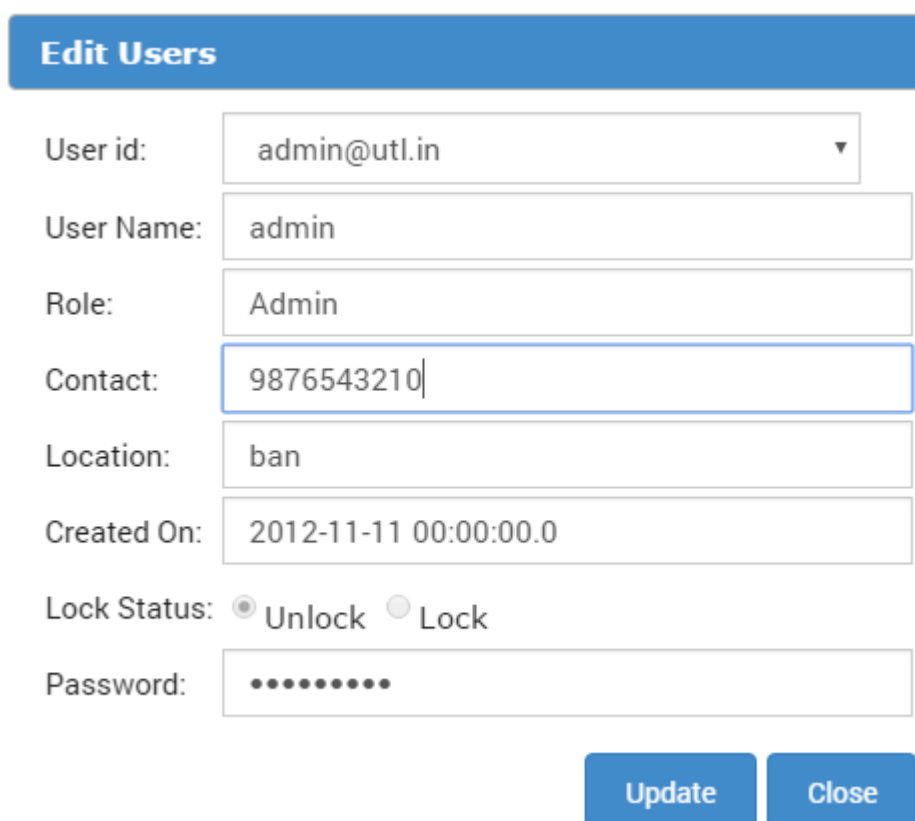
Close

4.3 Edit Users

Click on **Menu-> User Management -> Edit User** option.



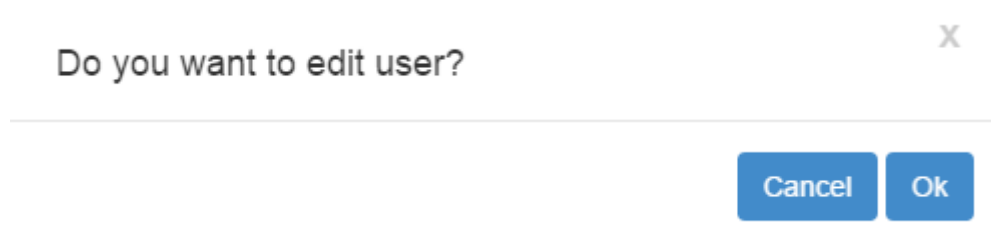
Click on the **Edit User** menu option, then Edit User dialog will be displayed as shown below.

A screenshot of the 'Edit Users' dialog box. It has a blue header bar with the text 'Edit Users'. Below the header, there are several input fields: 'User id:' with a dropdown menu showing 'admin@utl.in'; 'User Name:' with a text box containing 'admin'; 'Role:' with a text box containing 'Admin'; 'Contact:' with a text box containing '9876543210'; 'Location:' with a text box containing 'ban'; 'Created On:' with a text box containing '2012-11-11 00:00:00.0'; 'Lock Status:' with two radio buttons, 'Unlock' (selected) and 'Lock'; and 'Password:' with a text box containing ten dots. At the bottom right, there are two blue buttons: 'Update' and 'Close'.

By using this option user can edit only three things (User Name, Contact and Location).

- Enter the User Name you want, at least one character.
- Enter Contact Number, must be 10 digits (ex: 9876543210).
- Enter Location Name, at least one character.

Click on **Update User** button to update the present login user.
The confirmation dialog shown in below figure is presented.

A screenshot of a confirmation dialog box. It has a title bar with a close button (X). The main text of the dialog is 'Do you want to edit user?'. At the bottom right, there are two blue buttons: 'Cancel' and 'Ok'.

Click on **Ok** button to confirm user update.

Following success message window will be displayed:

Edit Users

User id:	admin@utl.in
User Name:	admin
Role:	Admin
Contact:	9876543210
Location:	ban
Created On:	2012-11-11 00:00:00.0
Lock Status:	☒ Unlock ☐ Lock
Password:	••••••••

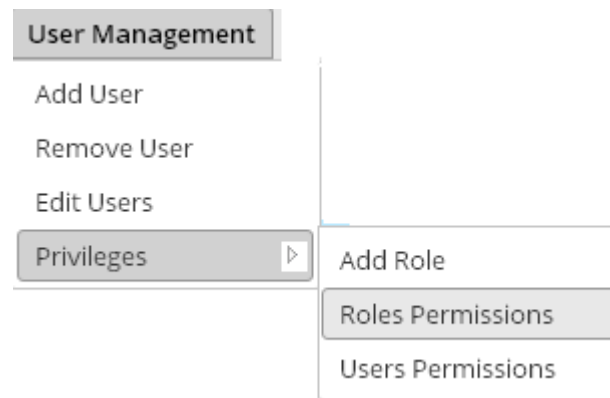
admin user updated successfully.

Update

Close

4.4 Privileges

Privileges are a special right or advantage that only one person or group has. Click on **Menu-> User Management -> Roles Operations** option.



4.4.1 Roles Permissions

A user's role determines what they can and cannot do in a system. Each role has a default set of permissions. The following dialog describes these permissions. In general, Administrators have full rights to all tasks. Click on the **Roles & Permissions** menu option, then Roles & Permissions dialog will be displayed as shown below.

The screenshot shows a web interface titled "Roles And Permissions". It contains several sections for configuring permissions for a selected role (currently "Operator").

- Role Details:** A dropdown menu shows "Operator". A red note states: "Note: Admin role provides full control access and permissions by default."
- Permissions:**
 - Zone:** "Add Zone" (unchecked), "Remove Zone" (unchecked).
 - Circle:** "Add Circle" (unchecked), "Remove Circle" (unchecked), "Edit Circle" (checked).
 - Node:** "Add Node" (unchecked), "Remove Node" (unchecked), "Edit Node" (checked).
 - Card Functionalities:** "Card Configurations" (unchecked), "Card Severity Settings" (checked).
 - Report Management:**
 - Row 1: "Network Summary" (unchecked), "Node Status" (unchecked), "Network Status History" (unchecked), "User Login & Logout" (unchecked).
 - Row 2: "Click Event Details" (unchecked), "Card Presence" (unchecked), "Card JackIn & JackOut" (unchecked), "Performance History" (checked).
 - Row 3: "Domain Alarm History" (unchecked).
 - Online Management:** "Current Alarms" (unchecked), "Card Inventory" (unchecked).

At the bottom, there are links for "Check all" and "Uncheck all", and a blue "Save" button.

Select required permissions and click on **save** button to update the permissions.

The confirmation dialog shown in below figure is presented.

A confirmation dialog box with the title "Do you want to save permissions ?" and a close button (X) in the top right corner. At the bottom right, there are two buttons: "Cancel" and "Ok".

Click on **Ok** button to update permissions.

Following success message window will be displayed:

Roles And Permissions

Role Details
Role: Operator

Note: Admin role provides full control access and permissions by default.

Permissions

Zone

☐ Add Zone
☐ Remove Zone

Circle

☐ Add Circle
☐ Remove Circle
☒ Edit Circle

Node

☐ Add Node
☐ Remove Node
☒ Edit Node

Card Functionalities

☐ Card Configurations
☒ Card Severity Settings

Report Management

☐ Network Summary

☐ Node Status

☐ Network Status History

☐ User Login & Logout

☐ Click Event Details

☐ Card Presence

☐ Card JackIn & JackOut

☒ Performance History

☐ Domain Alarm History

Online Management

☐ Current Alarms
☐ Card Inventory

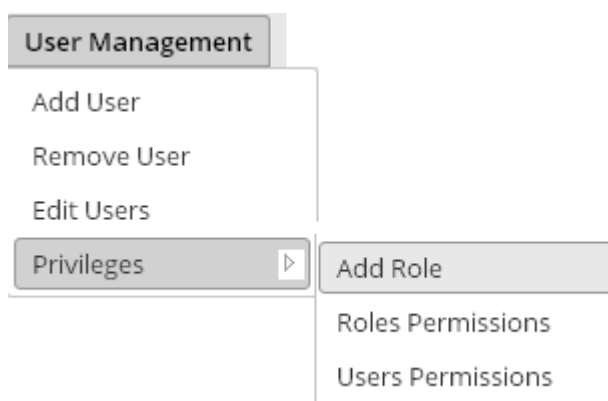
Check all | Uncheck all

Save

Permissions saved successfully.

4.4.2 User Permissions

Click on **Menu-> User Management -> Privileges ->Users Permissions** option.



Select User: Select ▼

Permission Nodes

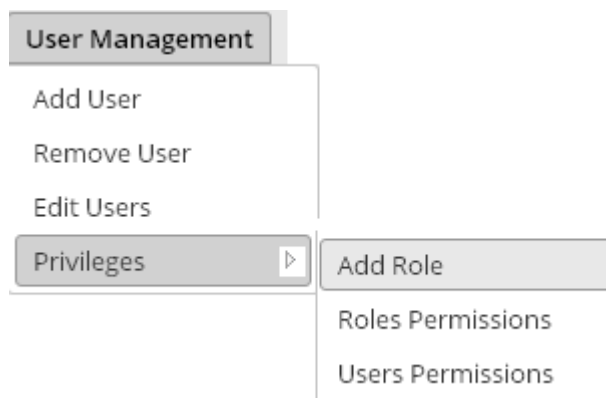
S.No	Zone Name	Circle Name	Node IP
No data available!			

Zone: India ▼ Circle: UTL ▼

NodeIP: ▼ Add

4.4.3 Add Role

Click on **Menu-> User Management -> Roles Operations** option.

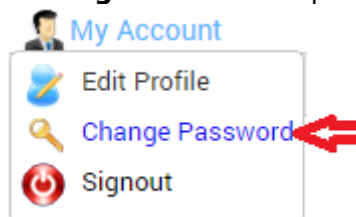


Click on the **Roles Operations -> Add Role** menu option, then Roles Operations dialog will be displayed as shown below.

Roles Operations			
Roles Details		+ Add new record	
ID	Role Name		
1	Admin		
2	Operator		
3	User		

4.5 Change Password

Click on **Menu-> My Account -> Change Password** option.



Click on the **Change Password** menu option, then Change Password dialog will be displayed as shown below.

Change Password

Current User Name:

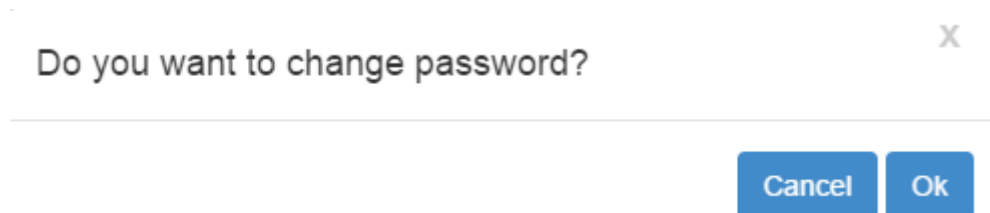
Current Password:

New Password:

Confirm Password:

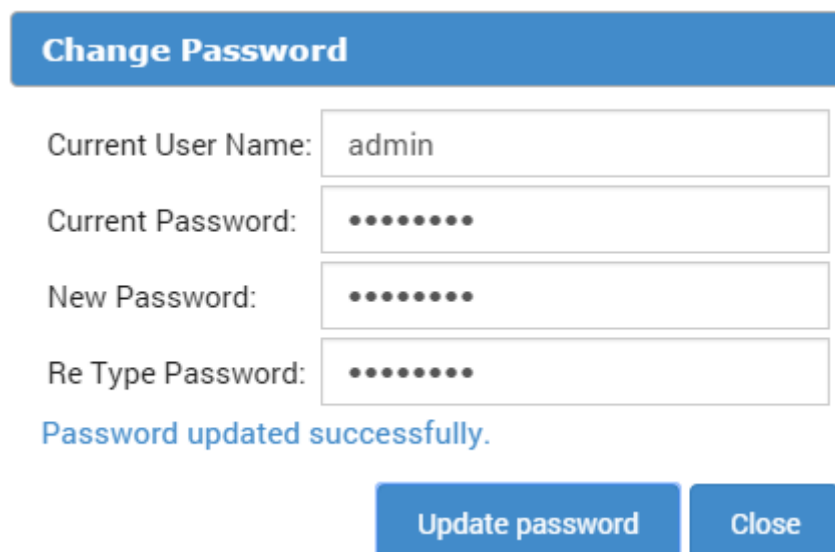
- Enter the current login password.
- Enter the password, and it should contain 8 to 15 characters.
- Enter the Re-Type password, and it should contain 8 to 15 characters and must match with the password.

Finally click on **Update password** button. The confirmation dialog shown in below figure is presented.



A confirmation dialog box with a title bar containing a close button (X). The main text asks "Do you want to change password?". At the bottom right, there are two buttons: "Cancel" and "Ok".

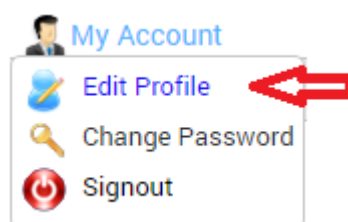
Click on **OK** button to confirm update password, and success response window will be displayed as shown below.



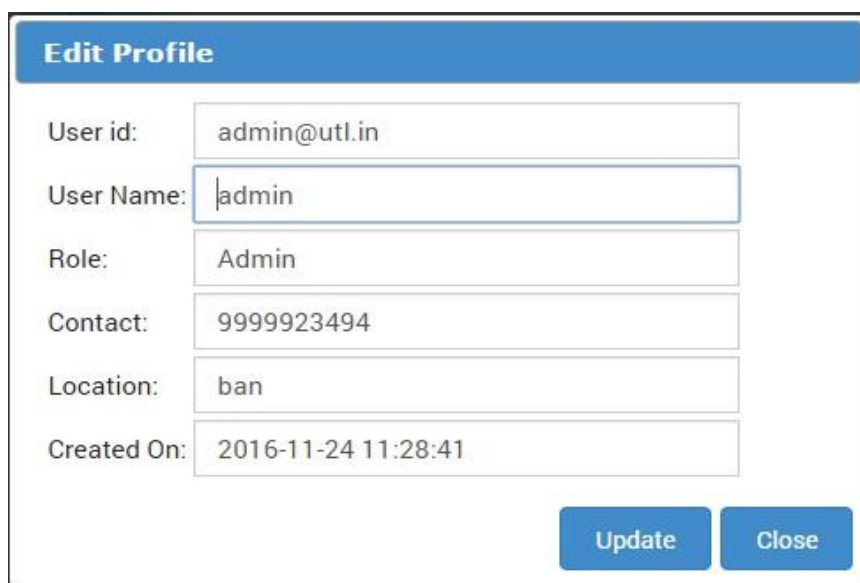
A "Change Password" dialog box with a blue header. It contains four input fields: "Current User Name:" with the value "admin", "Current Password:" with masked dots, "New Password:" with masked dots, and "Re Type Password:" with masked dots. Below the fields, a blue message states "Password updated successfully.". At the bottom right, there are two buttons: "Update password" and "Close".

4.6 Edit Profile

Click on **Menu-> My Account -> Edit Profile** option.



Click on the **Edit Profile** menu option, then Update User dialog will be displayed as shown below.



The 'Edit Profile' dialog box contains the following fields and values:

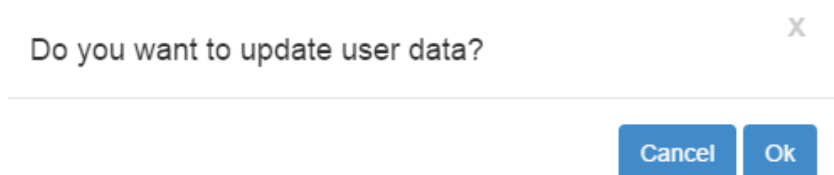
Field	Value
User id:	admin@utl.in
User Name:	admin
Role:	Admin
Contact:	9999923494
Location:	ban
Created On:	2016-11-24 11:28:41

Buttons: Update, Close

By using this option user can edit only three things (User Name, Contact and Location).

- Enter the User Name you want, at least one character.
- Enter Contact Number, must be 10 digits (ex: 9876543210).
- Enter Location Name, at least one character.

Click on **Update User** button to update the present login user.
The confirmation dialog shown in below figure is presented.

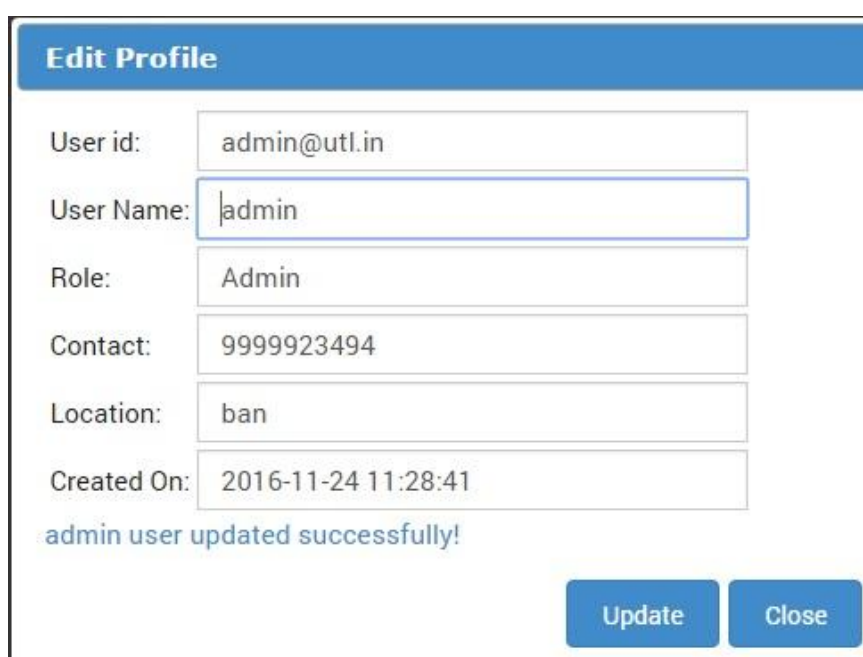


The confirmation dialog box contains the following text and buttons:

Do you want to update user data? X

Buttons: Cancel, Ok

Click on **Ok** button to confirm user update.
Following success message window will be displayed for the user:



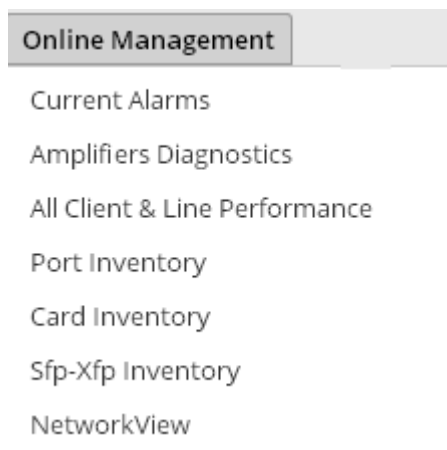
The 'Edit Profile' dialog box contains the same fields and values as the first figure, with the addition of a success message:

admin user updated successfully!

Buttons: Update, Close

5. Online Management

UTL EMS provides real-time availability of any network node. It allows nodes to be monitored continuously to get the real-time availability status of each of them. It can also monitor critical performance, amplifier diagnostics, port inventory, XFP/SFP inventory and current alarms. The online management pull down menu is shown below.



5.1 Current Alarms

Click on **Menu-> Online Management -> Current Alarms** option to display only the active alarms retrieved from the node.



To View current Alarms:

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.

Then click the **Search** button. Select filters based on the requirement.

Current(Live) Alarms Report

Select Zone: India Select Circle: NFS Select Node/IP: 192.168.1.33 **Search** ☐ Auto Refresh

Active Filters:

Filter by : Card Name: Select Card Name Rack Number: Select Rack Number Subrack Number: Select Subrack Slot Number: Select Slot Port Number: Select Port Number

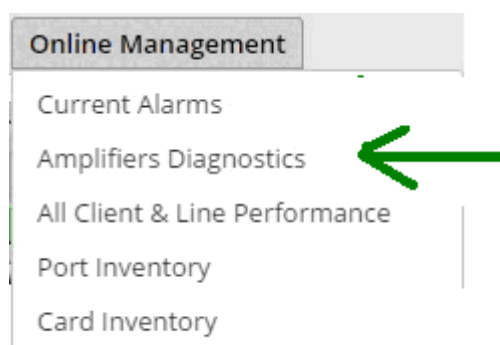
Show 10 entries **Copy** **CSV** **Excel** **PDF** **Print** Search:

S.No	Card Name	Rack Number	Subrack Number	Slot Number	Port Number	Alarm Name	Severity	Alarm Generated Date
1	10GOMTP12	1	3	1	1	line1 XFP Input Power Failure Alarm	Critical	2016-11-17 20:17:49
2	10GOMTP12	1	3	1	1	line1 OTU Loss of Signal Payload Alarm	Critical	2016-11-17 20:17:49
3	10GOMTP12	1	3	3	1	line1 XFP Input Power Failure Alarm	Critical	2016-11-17 20:32:02
4	10GOMTP12	1	3	3	1	line1 OTU Loss of Signal Payload Alarm	Critical	2016-11-17 20:32:02
5	10GOMTP12	1	3	1	1	client1 XFP Input Power Failure Alarm	Critical	2016-11-17 20:17:49
6	10GOMTP12	1	3	1	1	client1 SDH Loss of Signal Payload Alarm	Critical	2016-11-17 20:17:49
7	10GOMTP12	1	3	3	1	client1 XFP Presence Absence Notification	Critical	2016-11-17 20:32:02
8	10GOMTP12	1	3	3	1	client1 SDH Loss of Signal Payload Alarm	Critical	2016-11-17 20:32:01
9	10GOMTP12	1	3	1	2	line2 XFP Input Power Failure Alarm	Critical	2016-11-17 20:17:49
10	10GOMTP12	1	3	1	2	line2 OTU Loss of Signal Payload Alarm	Critical	2016-11-17 20:17:49

Showing 1 to 10 of 16 entries Page to go: **GO** **First** **Previous** **1** **2** **Next** **Last**

5.2 Amplifiers Diagnostics

Click on **Menu-> Online Management -> Amplifiers Diagnostics** option to display the amplifiers diagnostics retrieved from the node.



Amplifiers Diagnostics

Select Zone: Select Circle: Select Node:

Active Filters:

Filter by : Node IP Address: Rack Number: Card Name:

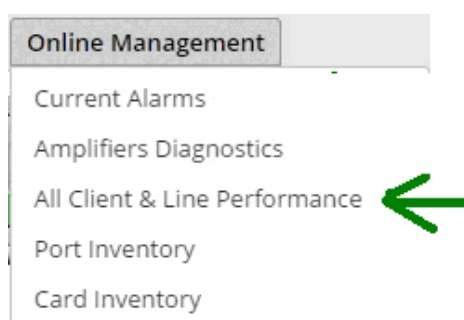
Show entries Search:

S.No	IP Address	Terminal	Card	In	Mid-Out	Gain	Mode	Mid-In	Out	Gain-2	Mode-2
No data available in table											

Showing 0 to 0 of 0 entries Page to go:

5.3 All Client & Line Performance

Click on **Menu-> Online Management -> All Client & Line Performance** option to display the client & line performance statistics retrieved from the node.



All client and line live optical performance

Select Zone: Select Circle: Select Node/IP: Select Card:

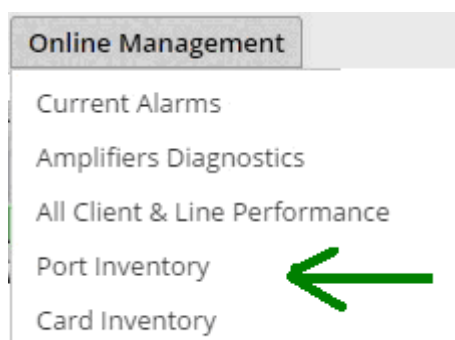
Show entries Search:

S.No	Rack	Subrack	Slot	CL1 In	CL1 Out	Line1 In	Line1 Out	Line2 In	Line2 Out
1	1	3	1	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
2	1	3	2	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
3	1	3	3	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
4	1	3	4	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
5	1	3	5	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
6	1	3	6	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
7	1	3	7	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port
8	1	3	8	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port	xfp_absent_for_configured_port

Showing 1 to 8 of 8 entries Page to go:

5.4 Port Inventory

Click on **Menu-> Online Management -> Port Inventory** option to display the ports inventory retrieved from the node.



Port Correlation Inventory Report

Select Zone:INDIA

Select Circle:KTK

Select NodeID:192.168.21.46

Select Card:10GOMTP12

Search

Active Filters:

Filter by : Rack Number

Select Rack Number

Subrack Number

Select Subrack Number

Slot Number

Select Slot

Show25

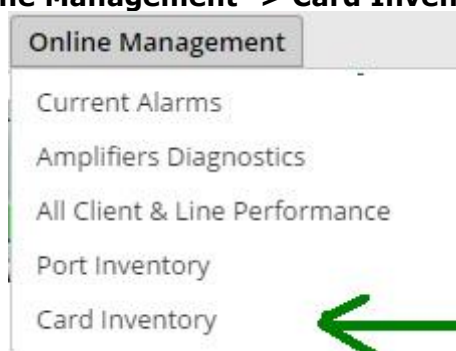
entries

Search:

S.No	Rack	Subrack	Slot Number	Port1	DateAndTime
1	1	3	1	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
2	1	3	2	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
3	1	3	3	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
4	1	3	4	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
5	1	3	5	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
6	1	3	6	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
7	1	3	7	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27
8	1	3	8	STM64[xfp_absent_for_configured_port] [SERVICE-DOWN]	2017-02-09 14:41:27

5.5 Card Inventory

Click on **Menu-> Online Management -> Card Inventory** option.



Click on the **Card Inventory** menu option, then Card Inventory report will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot.

To View card inventory:

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.
- Select card from drop down list.

Then click the **Search** button. Select filters based on the requirement.

Card Inventory Report

Select Zone: Select Circle: Select NodeID: Select Card: **Search**

Active Filters:

Filter by: Card Name: Shelf Number: Slot Number:

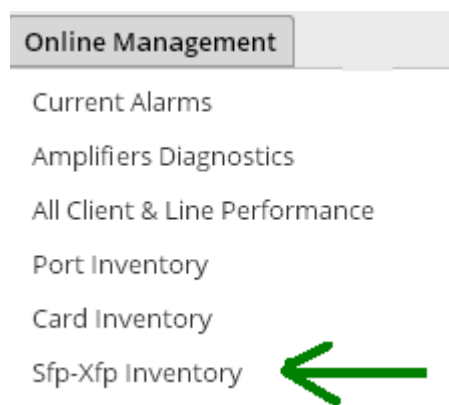
Show entries **Copy** **CSV** **Excel** **pdf** Search:

S.No	Card Name	Shelf Number	Slot Number	S/W Version	H/W Version	Serial Number
No data available in table						

Showing 0 to 0 of 0 entries Page to go: **GO** **First** **Previous** **Next** **Last**

5.6 SFP-XFP Inventory

The small form-factor pluggable (SFP) is a compact, hot-pluggable optical module transceiver used for both telecommunication and data communications applications. The XFP (10 Gigabit Small Form Factor Pluggable) is a standard for transceivers for high-speed computer network and telecommunication links that use optical fiber. Click on **Menu-> Online Management -> Sfp-Xfp Inventory** option to display the Sfp-Xfp inventory retrieved from the node.



Sfp-Xfp Inventory Report

Select Zone: Select Circle: Select NodeID: Select Card: **Search**

Active Filters:

Filter by: Rack Number: Subrack Number: Slot Number:

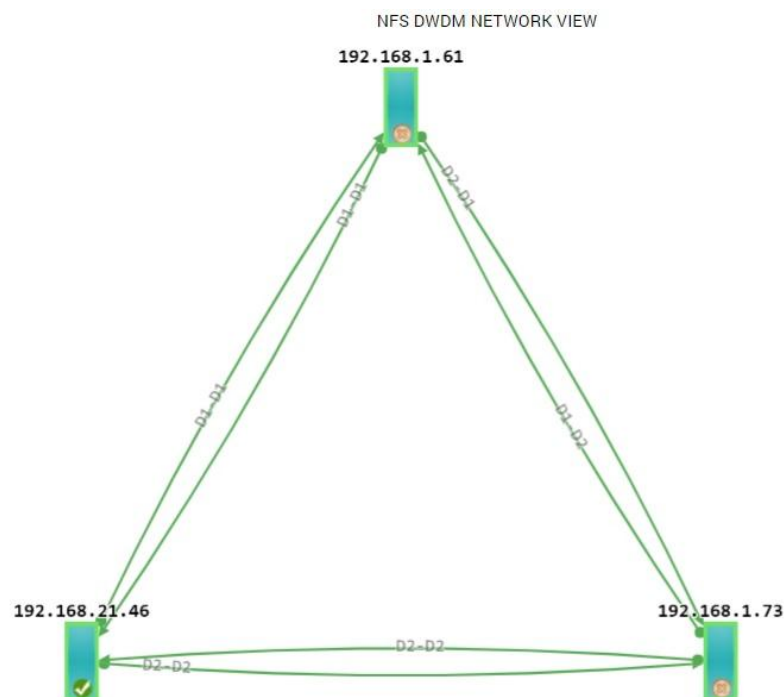
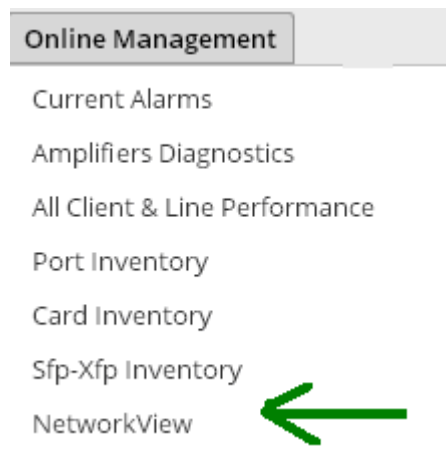
Show entries Search:

S.No	Rack	Subrack	Slot Number	Client1	Client2	Client3	Client4	Client5	Client6	Client7	Client8	Line1	Line2	DateAndTime
1	1	3	10	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	2017-02-09 14:43:25
2	1	3	12	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	2017-02-09 14:43:25

Showing 1 to 2 of 2 entries Page to go: **GO** **First** **Previous** **1** **Next** **Last**

5.7 Network View

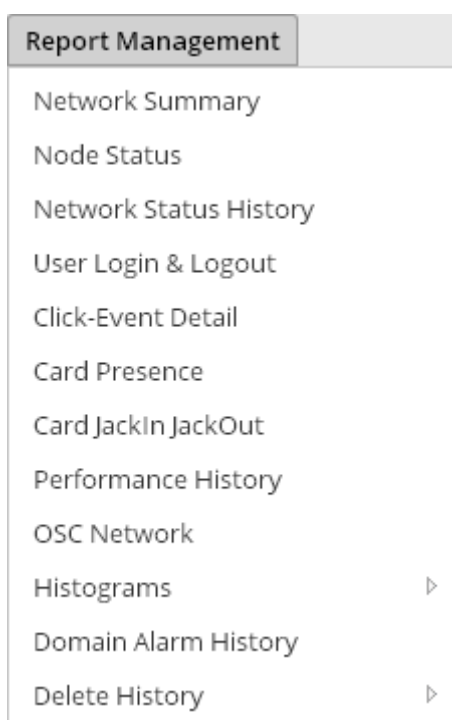
Click on **Menu-> Online Management -> Network View** option to display the network topology.



6. Report Management

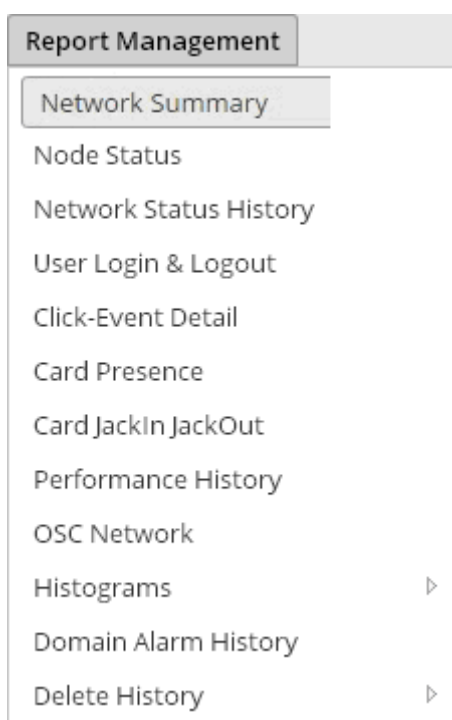
Use the Report Management module to view security and usage reports for devices. These reports can provide useful information about your network.

The report management pull down menu is shown below.



6.1 Network Summary

Click on **Menu-> Report Management -> Network Summary** option.



Click on the **Network Summary** menu option, then Network Summary report will be displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.

Then click the **Search** button. Select filters based on the requirement.

Network Summary Report

Select Zone:
 Select Circle:
 Select NodeIP:

Active Filters: Karnataka x UTL R&D -80CH DWDM x 192.168.21.2 x 192.168.21.4 x 192.168.21.6 x 192.168.21.8 x 192.168.21.16 x 192.168.21.18 x DWDM x 80 x

Filter by : Zone Name: Circle Name: Node IP: Node Type: Terminal Type:

Show entries

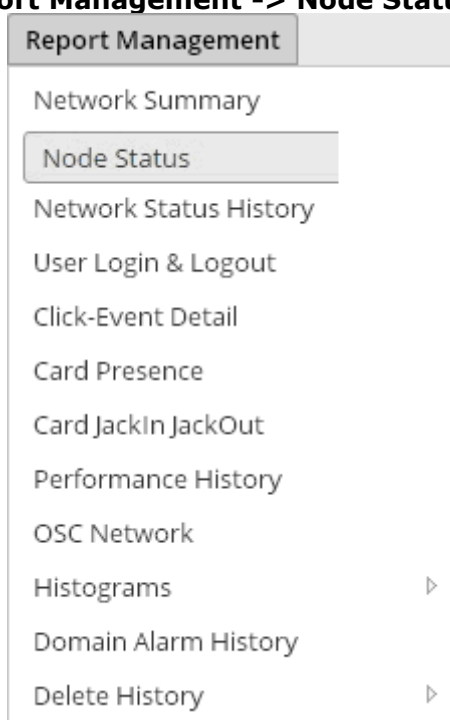
 Search:

S.No	Zone Name	Circle Name	Node IP	Node Type	Channel Type	Location	Contact Person	Phone Number

Showing 1 to 2 of 2 entries (filtered from 12 total entries) Page to go:

6.2 Node Status

Click on **Menu-> Report Management -> Node Status** option.



Click on the **Node Status** menu option, then Node Status report will be displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.

Then click the **Search** button. Select filters based on the requirement.

Node Status Report

Select Zone:
 Select Circle:
 Select NodeIP:

Active Filters: Karnataka x UTL R&D -80CH DWDM x 192.168.21.2 x 192.168.21.4 x 192.168.21.6 x

Filter by: Zone Name: Circle Name: Node IP:

Show entries

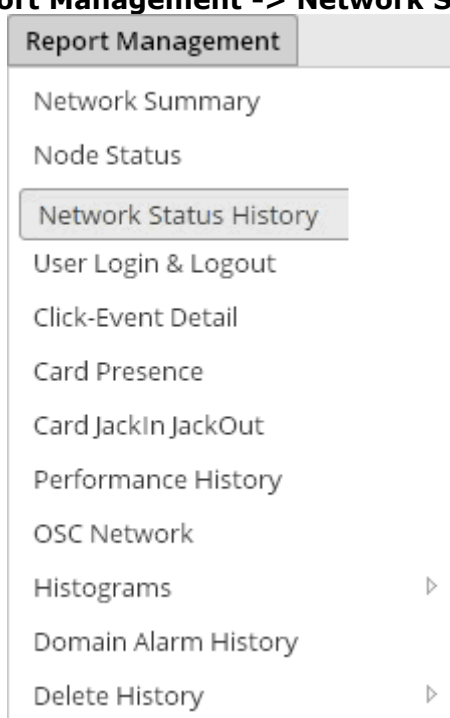
 Search:

S.No	Zone Name	Circle Name	Node IP	Node Type	Location	Node Status

Showing 1 to 3 of 3 entries (filtered from 12 total entries) Page to go:

6.3 Network Status History

Click on **Menu-> Report Management -> Network Status History** option.



Click on the **Network Status History** menu option, then Network Status History report will be displayed as shown below.

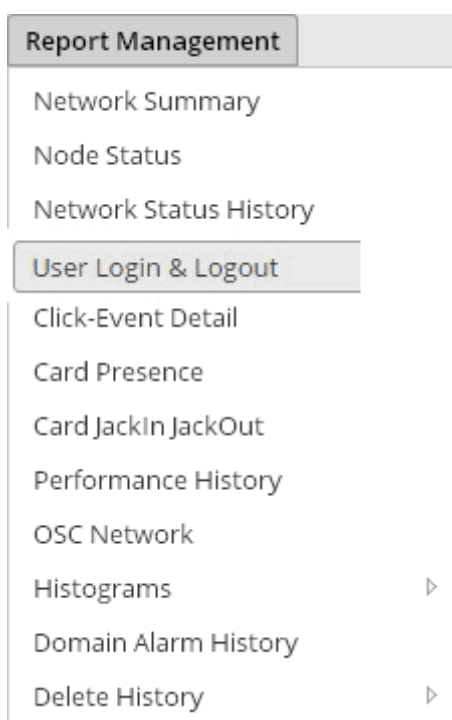
- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement.

[illegible]

6.4 User Login & Logout

Click on **Menu-> Report Management -> User Login & Logout** option.



Click on the **User Login & Logout** menu option, then User Login & Logout report will be displayed as shown below.

- Select user ID from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement.

Login And Logout Report

Select User ID: All ▼ Select From Date: 2015-03-10 11:5:36 Select To Date: 2016-3-2 11:5:36 Search

Active Filters: ALL x ALL x

Filter by : User Name Select User Name User Login IP: Select IP

Show 10 ▼ entries Copy CSV Excel pdf Search:

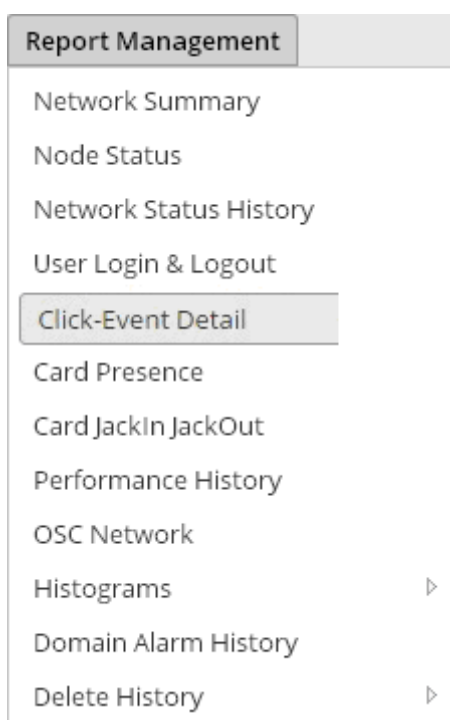
S.No	User Name	User Login IP Address	Login Time	Logout Time

Showing 1 to 10 of 1,377 entries Page to go: GO

Previous 1 2 3 4 5 ... 138 Next

6.5 Click-Event Details

Click on **Menu-> Online Management -> Click-Event Details** option.



Click on the **Click-Event Details** menu option, then Click-Event Details report will be displayed as shown below.

- Select user id from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement.

Click Event Details Report

Select User ID: All

Select From Date: 2014-03-26 11:6::21

Select To Date: 2016-3-2 11:6:21

Search

Active Filters: ALL x ALL x

Filter by : User Name: Select User Name Role: Select Role

Show 10 entries

Copy CSV Excel pdf

Search:

S.No	User Name	Role	Event Description	Time Stamp

Showing 1 to 10 of 18,716 entries

Page to go: GO

First

Previous

1

2

3

4

5

...

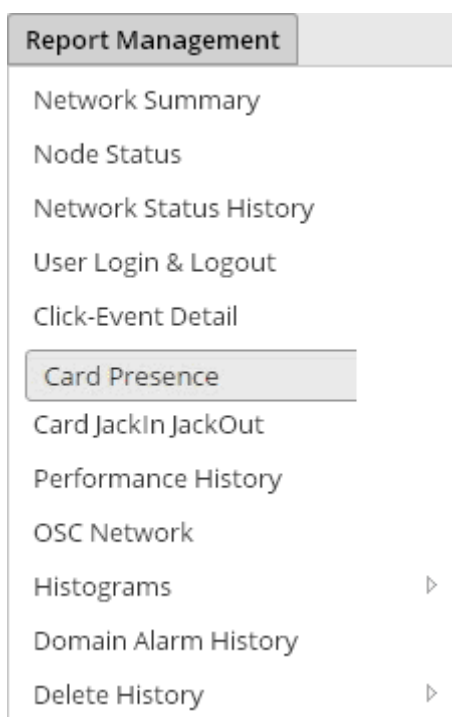
1872

Next

Last

6.6 Card Presence

Click on **Menu-> Report Management -> Card Presence** option.



Click on the **Card Presence** menu option, then Card Presence report will be displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.

Then click the **Search** button. Select filters based on the requirement.

Card Presence Report

Select Zone:
 Select Circle:
 Select NodeIP:
Search

Active Filters: 192.168.21.2 x 192.168.21.4 x Shelf-1 x Shelf-3 x SCU x 4x2.5G x slot-9 x slot-12 x slot-13 x

Filter by : Node IP: Shelf Number: Slot Number: Card Name:

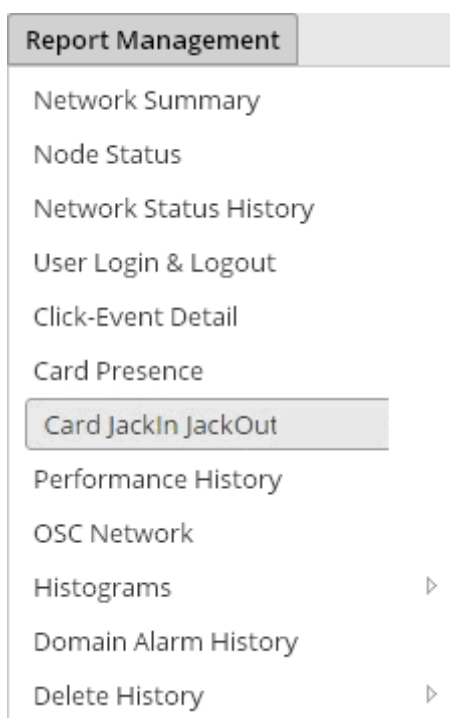
Show entries
 Copy **CSV** **Excel** **pdf**
 Search:

S.No	Zone Name	Circle Name	Node IP	Terminal Number	Shelf Number	Slot Number	Card Name

Showing 1 to 8 of 8 entries (filtered from 1,125 total entries) Page to go: **GO** **Previous** **1** **Next**

6.7 Card Jack In & Jack Out

Click on **Menu-> Online Management -> Card Jack in & Jack Out** option.



Click on the **Card Jack in & Jack out Report** menu option, then Card Jack in & Jack Out report is displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement.

Card Jack-In Jack-Out Report

Select Zone: Karnataka

Select Circle: UTL R&D-80CH DWDM

Select Node/IP: 192.168.21.2

Select From Date: 2014-03-25 11:9:31

Select To Date: 2016-3-2 11:9:31

Active Filters: ALL x Terminal-1 x Shelf-1 x Shelf-3 x Shelf-4 x Other x slot-1 x slot-6 x slot-9 x slot-11 x slot-12 x slot-16 x slot-2 x slot-5 x slot-7 x slot-14 x slot-15 x 4x2.5G x 8x1G x

Search

Filter by : Node IP Address: Select Node IP

Terminal Number: Select Terminal

Shelf Number: Select Shelf

Slot Number: Select Slot

Card Name: Select Card Name

Show 10 entries

Copy
CSV
Excel
pdf

Search:

S.No	Node IP Address	Terminal Number	Shelf Number	Slot Number	Card Name	Card JackIn Time	Card JackOut Time

Showing 1 to 4 of 4 entries (filtered from 82 total entries)

Page to go: 1 GO

First

Previous

1

Next

Last

6.8 Performance History

Click on **Menu-> Report Management -> Performance History** option.

- Report Management
 - Network Summary
 - Node Status
 - Network Status History
 - User Login & Logout
 - Click-Event Detail
 - Card Presence
 - Card JackIn JackOut
 - Performance History
 - OSC Network
 - Histograms
 - Domain Alarm History
 - Delete History

Click on the **Performance History** menu option, then 24-Hour performance report is displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.
- Select card from drop down list.
- Select rack from drop down list.
- Select shelf from drop down list.
- Select slot from drop down list.
- Select type from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement.

Performance History Report

Select Zone: India ▼ Select Circle: DELHI ▼ Select Node IP: 192.168.31.21 ▼ Select Card: Select ▼ Select Rack: Select ▼ Select Subrack: Select ▼ Select Slot: Select ▼ Select Type: Select ▼

Select From Date: 2016-10-28 13:20:11 Select To Date: 2016-11-29 13:20:11 **Search**

Active Filters:

Filter by : Zone Name: Select Zone Name Circle Name: Select Circle Name Node IP: Select IP Address DWDM Type: Select DWDM Type Node Type: Select Node Type

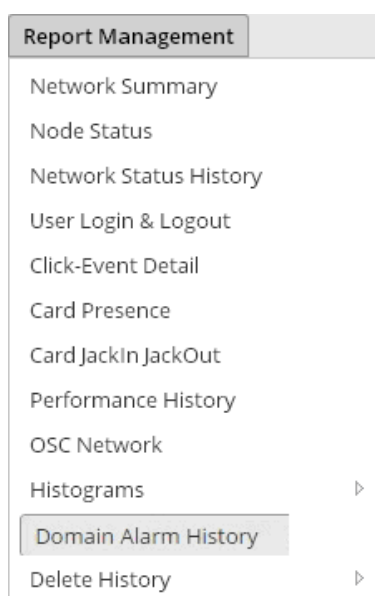
Show 10 ▼ entries **Copy** **CSV** **Excel** **pdf** Search:

S.No	Zone Name	Circle Name	Node IP	DWDM Type	Node Type	Location	Contact Person	Phone Number
No data available in table								

Showing 1 to 1 of 1 entries Page to go: **GO** **First** **Previous** **1** **Next** **Last**

6.9 Domain Alarm History

Click on **Menu-> Report Management -> Domain Alarm History** option.



Click on the **Domain Alarm History** menu option, then Domain Alarm History report is displayed as shown below.

- Select zone from drop down list.
- Select circle from drop down list.
- Select NE from drop down list.
- Select from date.
- Select to date.

Then click the **Search** button. Select filters based on the requirement

Alarms History Report

Select Zone: Karnataka ▼ Select Circle: UTL R&D-80CH DWDM ▼ Select Node/IP: 192.168.21.2 ▼ Select From Date: 2014-03-12 11:11:51 Select To Date: 2016-3-2 11:11:51

Active Filters: 8x1G x MSTP(10G LAN) x 4x2.5G x OPA x Terminal-1 x Shelf-1 x slot-14 x slot-1 x slot-2 x slot-5 x Client-1 Input Optical Power Failure x

Client-1 Output Optical Power Failure x

Filter by : Card Name: Select Card Name Terminal Number: select Terminal Shelf Number: select Shelf Slot Number: select Slot Alarm Type: select Alarm Type

Show 10 entries [Copy](#) [CSV](#) [Excel](#) [pdf](#) Search:

S.No	Card Name	Terminal Number	Shelf Number	Slot Number	Alarm Type	Severity	Alarm Generated Time	Alarm Cleared Time

Showing 1 to 2 of 2 entries (filtered from 1,143 total entries) Page to go: [GO](#) [Previous](#) [1](#) [Next](#)

7. Card Management

The UTL DWDM transport network is built using 10G (or better) systems with multi degree WSS based ROADMs, pre-amplifiers, mux/demux, amplifiers, transponders and muxponders with multi rate client interfaces. It supports in-service remote configuration and management of wavelengths. The client side interfaces are STM1, STM4, STM16, STM-64 and GigE (As per IEEE 802.3 and 802.3 ah standards) and 10G Ethernet LAN and WAN PHY as per 802.3ae. The system is capable to interface OTU - 1, OTU-2 interfaces as per requirement. Please refer UTL MP0072 DWDM technical manual for more information about cards.

UTL DWDM systems contain below types of cards.

1. MD-ROADM 1x2 (Multi-Degree Reconfigurable Optical Add Drop Multiplexer)
2. MD-ROADM 1x4 (Multi-Degree Reconfigurable Optical Add Drop Multiplexer)
3. A/D Directionless Unit (2D/4D)
4. OMTP11 (Optical Multi Transponder 1-Client 1-Line)
5. OMTP11T (Optical Multi Transponder 2-Lines)
6. OMTP12 (Optical Multi Transponder 1-Client 2-Lines)
7. OMTP22 (Optical Multi Transponder 2-Clients 2-Lines)
8. OMMP81 (Optical Multi Service Muxponder 8-Clients 1-Line)
9. OMMP82 (Optical Multi Service Muxponder 8-Clients 2-Lines)
10. OBA (Optical Booster Amplifier)
11. OPA MSA (Optical Power Amplifier with Mix Stage Access)
12. OLA (Optical Line Amplifier)
13. OPS (Optical Protection Switch)
14. SCC (System Control Card)
15. SCU (Subrack Control Unit)
16. BMU (Bay Monitoring Unit)

17. PSU (Power Supply Unit)
18. BUZZER
19. Fan
20. MUX/DEMUX

7.1 MD-ROADM

The DWDM transport network is built using 10G systems with multi degree WSS based ROADMs. The following options are available under MD-ROADM:

1. Alarms.
2. Settings.
3. Alarm Severity.

Select MD-ROADM card on node view, then FCAPS buttons will be enabled.

7.1.1 Alarms

Click on **Alarms** button, then alarms dialog will be displayed as shown below.

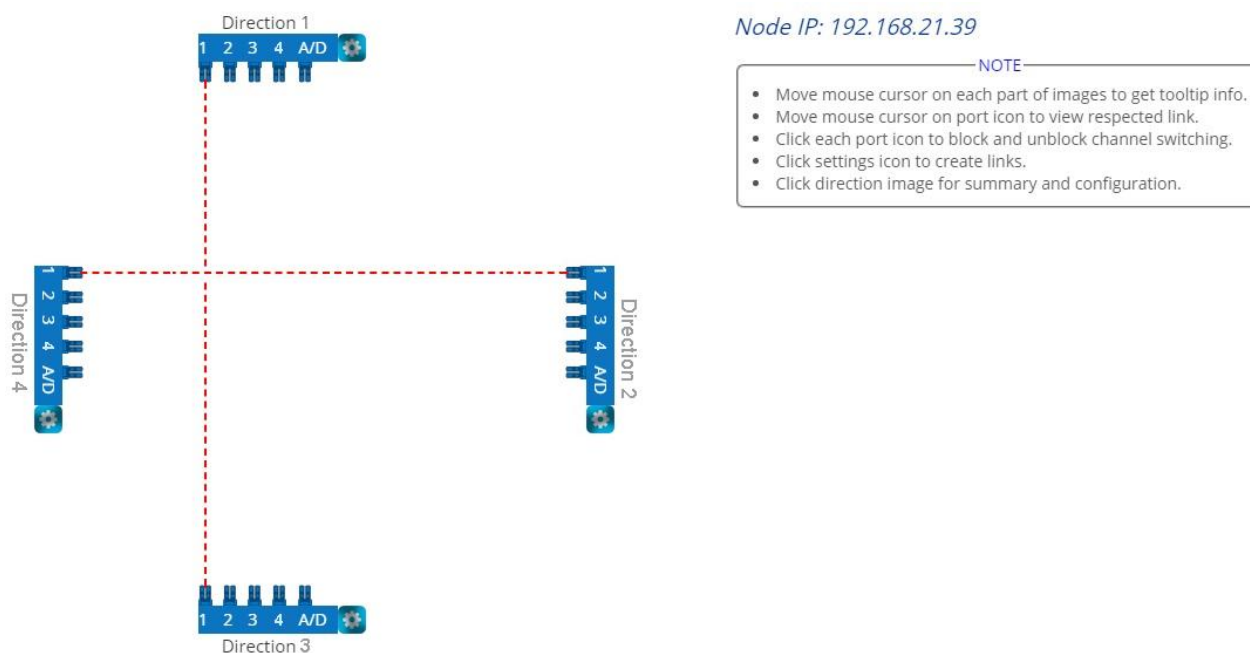


This screen will automatically notify an alarm.

7.1.2 Direction Settings

Click on **Settings** button, then settings dialog will be displayed as shown below.

7.1.2.1 Direction View



7.1.2.2 Add/Delete Ports

Direction 1	Directions	Ports		
1	Dir-2 ▼	Port-1 ▼	Add	Delete
2	Free ▼	Free ▼	Add	Delete
3	Free ▼	Free ▼	Add	Delete
4	Free ▼	Free ▼	Add	Delete

Refresh All

7.1.2.3 Block/Un-Block Channels

Channel	Block/Un-Block
1	☒
2	☒
3	☐
4	☐
5	☐
6	☐
7	☐
8	☐
9	☐
10	☐
11	☐
12	☐
13	☐
14	☐
15	☐

refresh

Note: Tick to Un-Block

7.1.2.4 Configuration Settings

I. Summary

- II. Attenuation
- III. WSS Power
- IV. Card Details
- V. Software Upgrade

I. Summary

Click on a button to see summary details.

Summary

Direction-1 -- Linked Channels

1

2

3

4

A/D

Dir-2
Port-1

Refresh

II. Attenuation

Click on a apply button to set attenuation.

Direction- 1

M
D
R
O
A
D
M

Port No	Channel No	Attenuation	Apply
4	1	0.0 ▼	Apply
4	2	0.0 ▼	Apply
4	3	0.0 ▼	Apply
4	4	0.0 ▼	Apply
4	5	0.0 ▼	Apply
4	6	0.0 ▼	Apply

Refresh

III. Card Details

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

Card Details

Serial No.

UTLMPHFCLA0518ROCA-000098

Refresh

HW Version

1.0

Refresh

SW Version

3.0

Refresh

Card Reset

Soft Reset

Warning: The card software will reboot

7.1.3 Alarm Severity

Click on **Alarm Severity** button, then alarm severity dialog will be displayed as shown below. You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Alarm Severity

WSS Module Connection Alarm

Critical ▾

Refresh

Data Retrieved Successfully..

7.2 OMTP12

The following data traffic options are available under OMTP12 card:

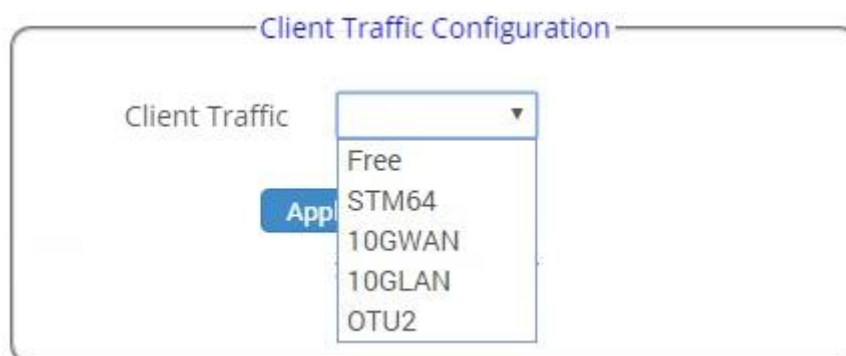
1. STM 64.
2. 10G WAN.
3. 10G LAN.
4. OTU 2.

Select OMTP12 card on node view, then FCAPS buttons will be enabled.

7.2.1 Configuration Settings

7.2.1.1 Client Traffic

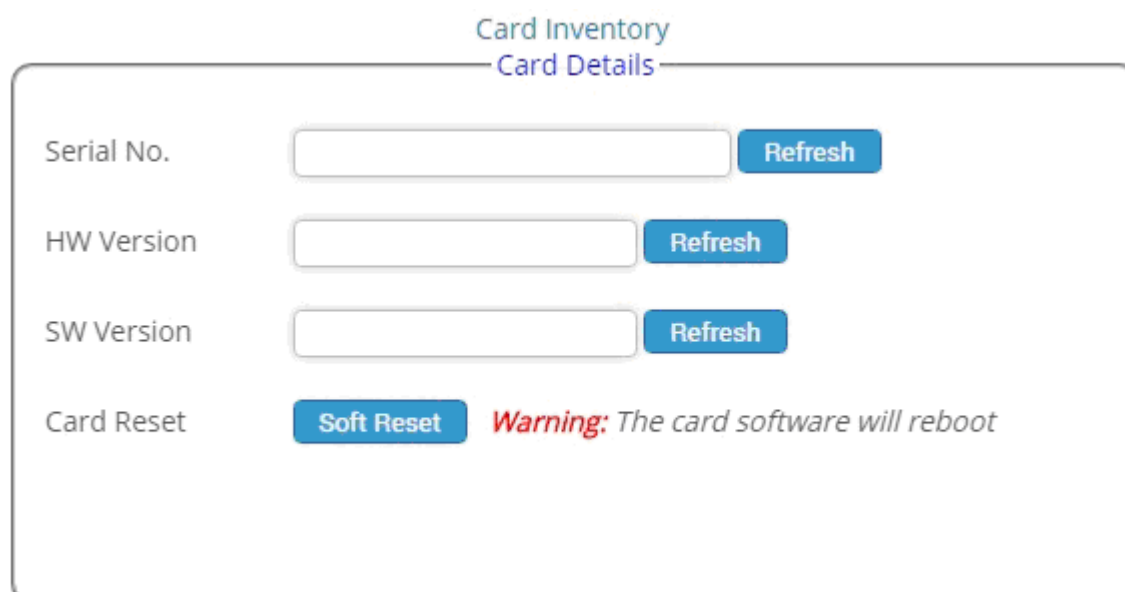
Click on **Settings** button, then settings dialog will be displayed as shown below. The user can set any traffic like STM64, 10G WAN, 10G LAN and OTU2 for this card.



The image shows a dialog box titled "Client Traffic Configuration". Inside, there is a label "Client Traffic" next to a dropdown menu. The dropdown menu is open, showing a list of options: "Free", "STM64", "10GWAN", "10GLAN", and "OTU2". To the left of the dropdown menu, there is a blue button labeled "Apply".

7.2.1.2 Card Inventory

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.



The image shows a dialog box titled "Card Inventory" with a subtitle "Card Details". It contains four rows of information, each with a text input field and a "Refresh" button:

- Serial No. [input field] Refresh
- HW Version [input field] Refresh
- SW Version [input field] Refresh

At the bottom, there is a "Card Reset" section with a "Soft Reset" button and a warning message: *Warning: The card software will reboot*.

7.2.1.3 Loopback

Fiber optic loopback is known as the routing of electronic signals, digital data streams, or flows of items back to their source without intentional processing or modification. In terms of telecommunication, loopback is a hardware or software method to feed a received signal or data back to the sender. It is very useful for solving physical connection problems.

If you want to set the loopback, Select loopback type and click on **Apply** button.

The screenshot displays the 'Loopback' configuration section with three panels: 'Client Loopback', 'Line-1 Loopback', and 'Line-2 Loopback'. Each panel contains three radio button options: 'No Loopback', 'Near-End Loopback', and 'Far-End Loopback'. Below the options in each panel are 'Apply' and 'Refresh' buttons. The 'Loopback' title is centered above the panels.

7.2.1.4 ALS & FEC

User can set the Laser ON/OFF, Automatic Laser Shutdown (ALS) and forward error correction (FEC) to correct bit errors in the received data using below mentioned dialog.

- Select configuration type and click on **Apply** button.

Laser Configuration

Client Laser

☐ LASER OFF

☐ LASER ON

Apply Refresh

Line-1 Laser

☐ LASER OFF

☐ LASER ON

Apply Refresh

Line-2 Laser

☐ LASER OFF

☐ LASER ON

Apply Refresh

ALS & FEC

Client ALS

☒ ALS Disable

☐ ALS Enable

Apply Refresh

Line-1 ALS

☒ ALS Disable

☐ ALS Enable

Apply Refresh

Line-2 ALS

☒ ALS Disable

☐ ALS Enable

Apply Refresh

OTU2 FEC Configuration

Line-1 FEC

☐ No FEC

☐ Normal FEC

Apply Refresh

Line-2 FEC

☐ No FEC

☐ Normal FEC

Apply Refresh

7.2.1.5 Switching

The user can select switching mode either automatic or manual for line path. Also user can select operational line path either Line1 or Line2.

- Select mode type and click on **Apply** button.

Switching mode

Mode of switching : ▼

Apply Refresh

Operational line port

Receive Path : ▼ Refresh

Apply Refresh

7.2.1.6 Threshold Settings

You can also configure the BER thresholds that raise or clear a signal degrade alarm and the time interval for the thresholds. If the BER thresholds and interval are not configured, the default values are used.

- Select threshold settings and click on **Apply** button.

Threshold

Client SDH Threshold Settings

B2 SF:

B2 SD:

Line-1 Performance Threshold Settings

OTU SD:

ODU SD:

Client WAN Threshold Settings

B2 SF:

B2 SD:

Line-2 Performance Threshold Settings

OTU SD:

ODU SD:

Client OTU2 Threshold Settings

OTU SD:

ODU SD:

7.2.1.7 TTI Settings

The Trail trace identifier (TTI) is used to identify the signal from the source to the destination within the network.

- Please refer below Line-1 and Line-2 screens for OTU/ODU TTI message settings. Fill the required configuration and click on **Apply** button.

I. Line-1 OTU TTI Message

TTI Message

☒ Line-1
 ☐ Line-2

Line1 OTU TTI Message Settings

☒ OTU TTI
 ☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit			
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear
OTU Expected			
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear

II. Line-1 ODU TTI Message

TTI Message

☒ Line-1
☐ Line-2

Line1 ODU TTI Message Settings

☐ OTU TTI
☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

III. Line-2 OTU TTI Message

TTI Message

☐ Line-1
☒ Line-2

Line2 OTU TTI Message Settings

☒ OTU TTI
☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

IV. Line-2 ODU TTI Message

TTI Message

☐ Line-1
 ☒ Line-2

Line2 ODU TTI Message Settings

☐ OTU TTI
 ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

7.2.1.8 Wavelength Settings

UTL DWDM transport systems include pluggable and software-tunable transceiver modules capable of operating on 40 or 80 channels. UTL DWDM tunable transceivers are able to function on various wavelengths and to adjust wavelengths according to user needs, making them prevalent among DWDM systems.

- Select the required channel and click on **Apply** button.
- If tunable card then only the **Apply** button will be enabled.

Wavelength

☒ 40 Channel
 ☐ 80 Channel

40 Channel Wavelength Selection

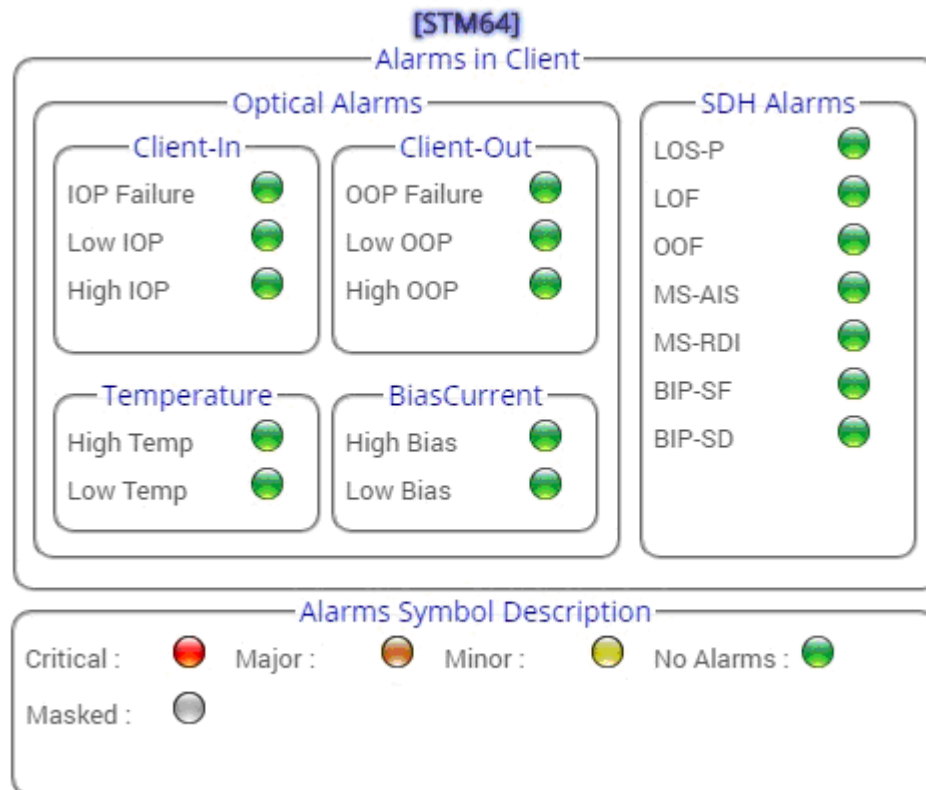
Line-1 Channel	1	Wavelength		Apply Refresh
Line-2 Channel	1	Wavelength		Apply Refresh

7.2.2 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements. Therefore, alarms need to be obtained to monitor and manage alarms at the card, node, or network level. In the card, node, or network view, click the Alarms tab to display the alarms for that card, node, or network.

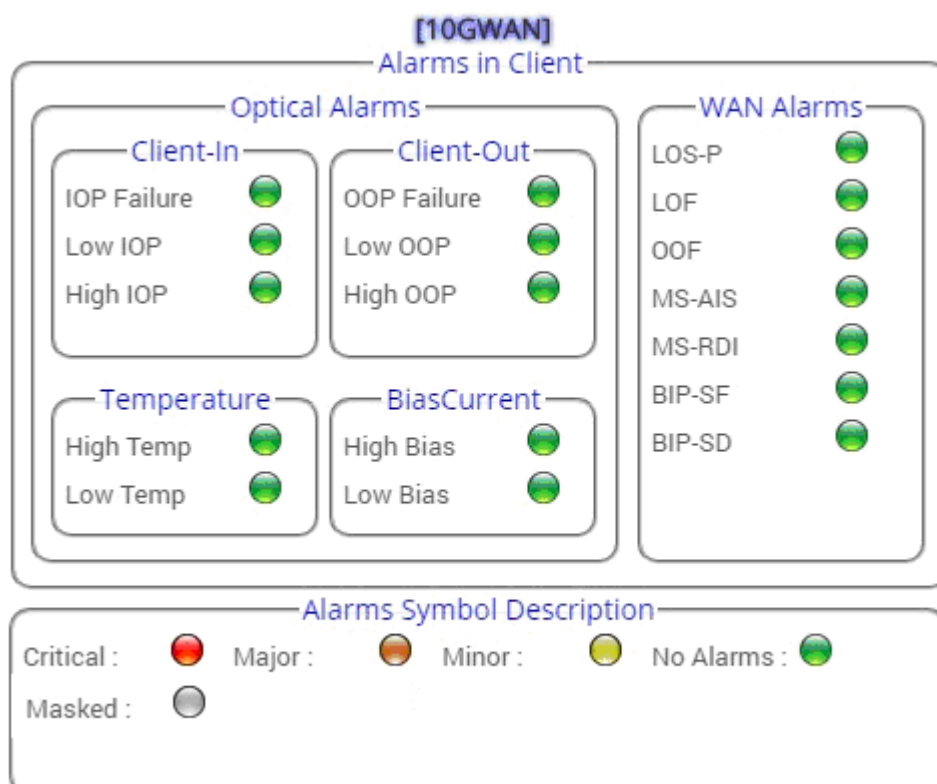
7.2.2.1 STM64 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the STM64 alarms.



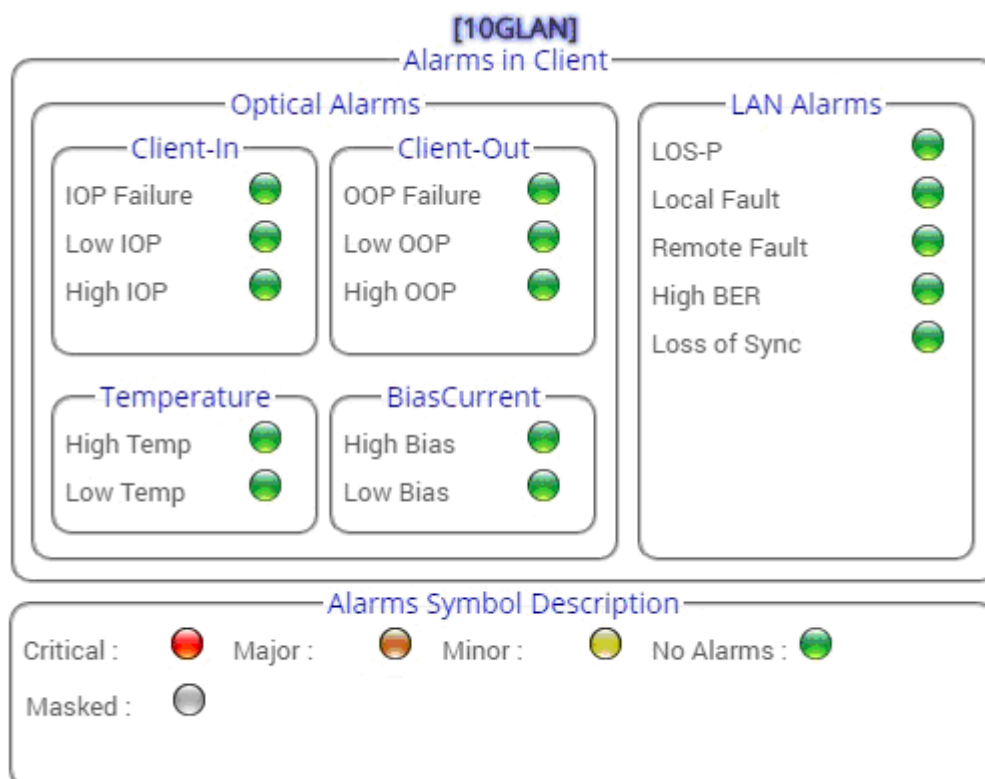
7.2.2.2 10G WAN Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the WAN alarms.



7.2.2.3 10G LAN Alarms

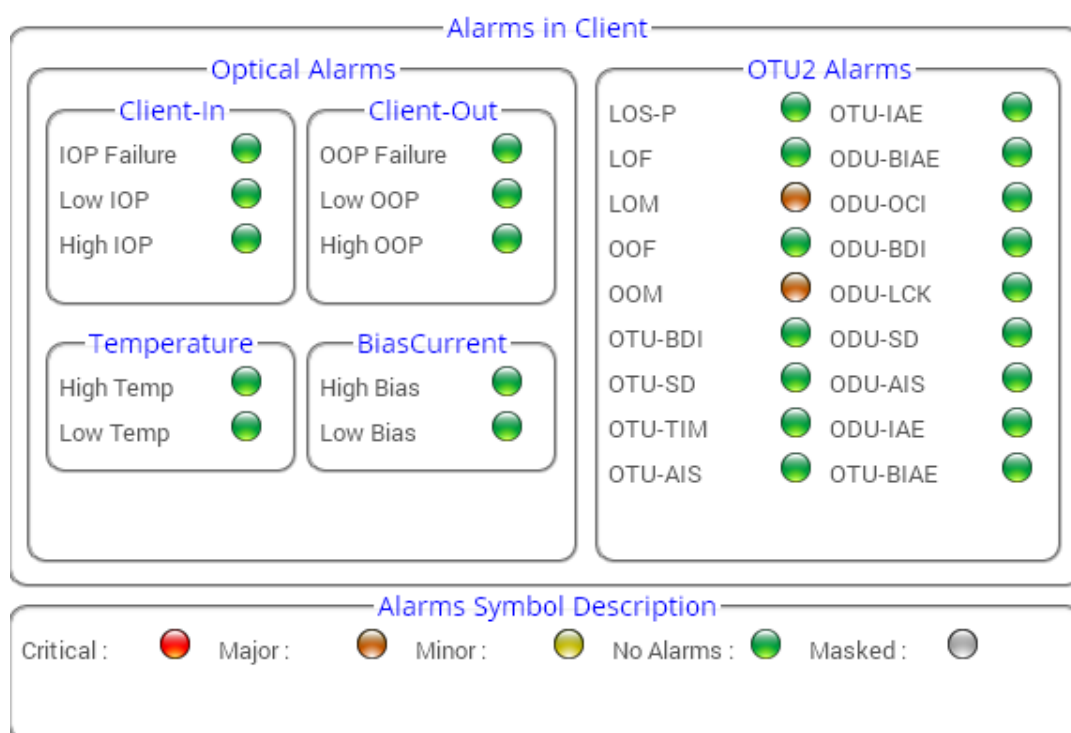
The screen uses evident colour changes on the GUI to help you quickly locate the LAN alarms.



7.2.2.4 OTU2 Alarms

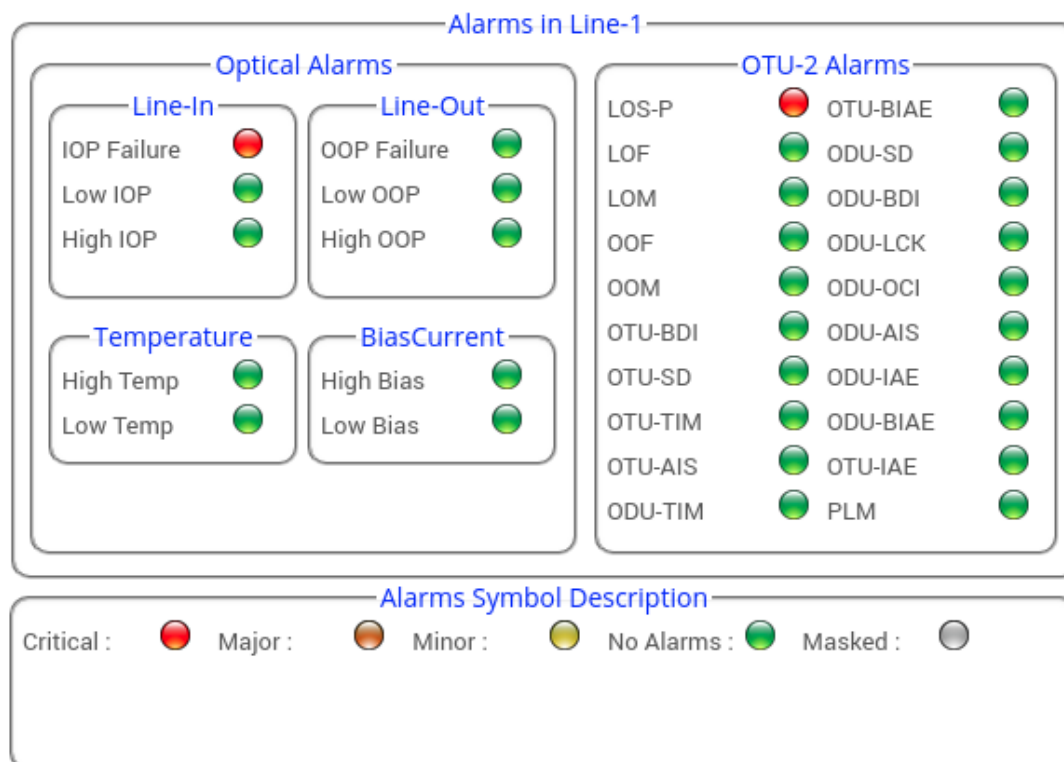
I. Client OTU2 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the client OTU2 alarms.



II. Line OTU2 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the Line-1 or Line-2 OTU2 alarms.



7.2.3 Performance

Click on **Performance** button, then performance dialog is displayed as shown below.

UTL DWDM products contain fiber optic components, including fiber optic transmitters and receivers. These components are extremely susceptible to contamination by particles of dirt or dust, which can obstruct the optic path and cause performance degradation. To ensure optimum product performance, it is important that all optics and connector ferrules be kept clean.

7.2.3.1 Optical Performance

This screen collects optical performance data of the card.

	Client	Line-1	Line-2
Rx Power(dBm)	no_input_power	no_input_power	no_input_power
Tx Power(dBm)	no_output_powe	0.90	-3.55
TxBias (mA)	0.00	64.76	60.50
TxTemp(Degree)	33.79	36.38	44.96
TxOSNR(dB)		43.95	NaN

7.2.3.2 OTU-2 Performance

I. OTU Performance

This screen collects OTU performance data of the card.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Error Free Seconds	0	Error Free Seconds	24
Background Block Errors	0	Background Block Errors	0
Unavailable Seconds	24	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0

[Re-initializes all the counters]

II. ODU Performance

This screen collects ODU performance data of the card.

ODU-Near End-Performance		ODU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Error Free Seconds	67	Error Free Seconds	67
Background Block Errors	0	Background Block Errors	0
Unavailable Seconds	0	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Error	0		
TCM-2 BIP Error	0		

[Re-initializes all the Line port counters]

[Refresh](#) [Re-initialize](#)

III. FEC Performance

This screen collects FEC performance data of the card.

FEC-Performance	
Zeros Corrected Bits	0
Ones Corrected Bits	0
Uncorrected Words	0
Corrected Bits	0

[Re-initializes all the Line port counters]

[Refresh](#) [Re-initialize](#)

7.2.3.3 STM64 Performance

This screen collects STM 64 performance data of the card.

[STM64] SDH Performance-Client

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	210
B1 Error Count	11760000
B2 Error Count	1296960000
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.2.3.4 10G WAN Performance

This screen collects WAN performance data of the card.

[10GWAN] WAN Performance-Client

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	70
B1 Error Count	3920000
B2 Error Count	432319228
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.2.3.5 10G LAN Performance

This screen collects LAN performance data of the card.

[10GLAN]
10G LAN Performance(Client)

UnderSize Frames Received	0	Pause Frames Received	0
Fragment Frames Received	0	Broadcast Frames Received	0
Over Size Frames Received	0	Multicast Frames Received	0
Frames Received	0	Unicast Frames Received	0
Frames Transmitted	0	Unicast Frames Transmitted	0
FrameCheck Seq. Errors Received	0	Broadcast Frames Transmitted	0
		Multicast Frames Transmitted	0

7.2.4 Alarm Severity

You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Some alarms with default severity are displayed on EMS as follows

Loss of I/P power - Critical, Loss of signal – Critical, Loss of frame - Critical , Input High Power – Major, OTU/ODU Signal Degrade – Minor etc.

7.2.4.1 OTU2 Alarm Severity

You can assign or modify the severity of OTU2 alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OTN-Alarm Severity

LOS-P	Critical ▼	OTU-BIAE	Critical ▼
LOF	Critical ▼	ODU-OCI	Critical ▼
LOM	Critical ▼	ODU-SD	Critical ▼
OOF	Critical ▼	ODU-BDI	Critical ▼
OOM	Critical ▼	ODU-LCK	Critical ▼
OTU-BDI	Critical ▼	ODU-TIM	Critical ▼
OTU-SD	Critical ▼	ODU-AIS	Critical ▼
OTU-TIM	Critical ▼	ODU-IAE	Critical ▼
OTU-AIS	Critical ▼	ODU-BIAE	Critical ▼
OTU-IAE	Critical ▼	PLM	Critical ▼

7.2.4.2 ODU-DeMultiplex Alarm Severity

You can assign or modify the severity of ODU-Demultiplex alarms. The severity levels that you can assign are: Critical, Major, and Minor.

The screenshot shows a configuration window titled "ODU-DeMultiplex". Inside, there is a label "ODU LOFLOM" followed by a red dropdown menu currently set to "Critical". Below this is a blue "Refresh" button.

7.2.4.3 Optical Alarm Severity

You can assign or modify the severity of optical alarms. The severity levels that you can assign are: Critical, Major, and Minor.

The screenshot shows a configuration window titled "Optical Alarms". It contains four sections, each with a title and a list of alarms with their severity levels:

- Input**: IOP Failure (Critical), Low IOP (Critical), High IOP (Critical).
- Output**: OOP Failure (Critical), Low OOP (Critical), High OOP (Critical).
- Temperature**: High Temp (Critical), Low Temp (Critical).
- Bias**: High Bias (Critical), Low Bias (Critical).

A blue "Refresh" button is located at the bottom right of the window.

7.2.4.4 SDH Alarm Severity

You can assign or modify the severity of SDH alarms. The severity levels that you can assign are: Critical, Major, and Minor.

SDH-Alarms Severity

LOS	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SD	Critical ▼
OOF	Critical ▼	BIP-SF	Critical ▼
MS-AIS	Critical ▼		

[Refresh](#)

- Select the severity type and click on **Apply** button.

7.2.4.5 10G WAN Alarm Severity

You can assign or modify the severity of WAN alarms. The severity levels that you can assign are: Critical, Major, and Minor.

WAN Alarms Severity

LOS-P	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SF	Critical ▼
OOF	Critical ▼	BIP-SD	Critical ▼
MS-AIS	Critical ▼		

[Refresh](#)

7.2.4.6 10G LAN Alarm Severity

You can assign or modify the severity of LAN alarms. The severity levels that you can assign are: Critical, Major, and Minor.

LAN Alarms Severity

LOS-P	Critical ▼
LOSSync	Critical ▼
High Bit Errors	Critical ▼
Local Fault	Critical ▼
Remote Fault	Critical ▼

[Refresh](#)

7.3 OMMP82

The following data traffic options are available under OMMP82 card:

1. STM16
2. STM4
3. STM1
4. GigE
5. OTU1

Select OMMP82 card on node view then FCAPS buttons will be enabled.

7.3.1 Configuration Settings

7.3.1.1 Client Traffic

Click on **Settings** button, then Traffic configuration dialog is displayed as shown below. The user can set any traffic like STM16, STM4, STM1, 1GigE and OTU2 for this card.

Client Traffic Configuration

Port-1	STM16 ▼	Apply
Port-2	STM16 ▼	Apply
Port-3	STM16 ▼	Apply
Port-4	Free ▼	Apply
Port-5	Free ▼	Apply
Port-6	Free ▼	Apply
Port-7	Free ▼	Apply
Port-8	Free ▼	Apply

Refresh All

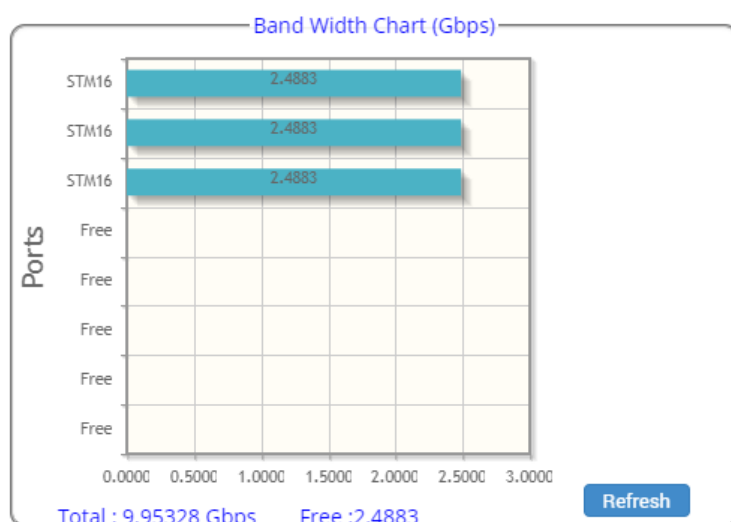
SFP Status:

Port1 - SFP(2.50 Gbps) Present - Port Configured
 Port2 - SFP(2.50 Gbps) Present - Port Configured
 Port3 - SFP(2.50 Gbps) Present - Port Configured
 Port4 - SFP Absent - Port UnConfigured

Port5 - SFP Absent - Port UnConfigured
 Port6 - SFP Absent - Port UnConfigured
 Port7 - SFP Absent - Port UnConfigured
 Port8 - SFP Absent - Port UnConfigured

7.3.1.2 Bandwidth Chart

The bandwidth of all client ports is displayed as shown below.



7.3.1.3 Card Inventory

Click on **Settings->Card Inventory**, then card details dialog is displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

Card Inventory
Card Details

Serial No.		Refresh
HW Version		Refresh
SW Version		Refresh
Card Reset	Soft Reset	Warning: The card software will reboot

If you want to reset the card click the **Soft Reset** button and click yes.

7.3.1.4 Loopback

Fiber optic loopback is known as the routing of electronic signals, digital data streams, or flows of items back to their source without intentional processing or modification. In terms of telecommunication, loopback is a hardware or software method to feed a received signal or data back to the sender. It is very useful for solving physical connection problems.

If you want to set the loopback, Select loopback type and click on **Apply** button.

Loopback

Client Diagnostic Loopback

Port-1	Clear Loopback	Apply	Refresh
Port-2	Clear Loopback	Apply	Refresh
Port-3	Clear Loopback	Apply	Refresh
Port-4	Clear Loopback	Apply	Refresh
Port-5	Clear Loopback	Apply	Refresh
Port-6	Clear Loopback	Apply	Refresh
Port-7	Clear Loopback	Apply	Refresh
Port-8	Clear Loopback	Apply	Refresh

Refresh All

Line-1 Loopback

Line: Clear Loopback

Apply Refresh

Line-2 Loopback

Line: Clear Loopback

Apply Refresh

7.3.1.5 ALS & FEC

User can set the Laser ON/OFF, Automatic Laser Shutdown (ALS) and forward error correction (FEC) to correct bit errors in the received data using below mentioned dialog.

- Select configuration type and click on **Apply** button.

ALS & FEC

Client-SFP Laser Configuration

Port-1	LASER ON	Apply	Refresh
Port-2	LASER ON	Apply	Refresh
Port-3	LASER ON	Apply	Refresh
Port-4	LASER ON	Apply	Refresh
Port-5		Apply	Refresh
Port-6		Apply	Refresh
Port-7		Apply	Refresh
Port-8		Apply	Refresh

Refresh All

ALS Configuration

Port-1	ALS Disable	Apply	Refresh
Port-2	ALS Disable	Apply	Refresh
Port-3	ALS Disable	Apply	Refresh
Port-4	ALS Disable	Apply	Refresh
Port-5	ALS Disable	Apply	Refresh
Port-6	ALS Disable	Apply	Refresh
Port-7	ALS Disable	Apply	Refresh
Port-8	ALS Disable	Apply	Refresh

Refresh All

Client-OTU1 FEC Configuration

Port-1: Normal FEC

Apply Refresh

Refresh All

Line-XFP Laser Configuration

Line-1	LASER ON	Apply	Refresh
Line-2	LASER ON	Apply	Refresh

Line ALS Configuration

Line-1	ALS Disable	Apply	Refresh
Line-2	ALS Disable	Apply	Refresh

Line-OTU2 FEC Configuration

Line-1	Super FEC	Apply	Refresh
Line-2	Super FEC	Apply	Refresh

7.3.1.6 Threshold Settings

You can also configure the BER thresholds that raise or clear a signal degrade alarm and the time interval for the thresholds. If the BER thresholds and interval are not configured, the default values are used.

- Select threshold settings and click on **Apply** button.

Threshold

Client SDH B2 SF Threshold

Select Port Client-2 ▼

B2 SF 1e-5 ▼ Apply Refresh

Client OTU1 Threshold Settings

Select Port Client-1 ▼

OTU SD 1e-7 ▼ Apply Refresh

ODU SD 1e-7 ▼ Apply Refresh

Line OTU2 Threshold Settings

Select Port Line-1 ▼

OTU SD 1e-7 ▼ Apply Refresh

ODU SD 1e-7 ▼ Apply Refresh

Client SDH B2 SD Threshold

Select Port Client-2 ▼

B2 SD 1e-7 ▼ Apply Refresh

7.3.1.7 TTI Settings

The Trail trace identifier (TTI) is used to identify the signal from the source to the destination within the network. Please refer below Line-1 and Line-2 screens for OTU/ODU TTI message settings.

I. Line-1 OTU TTI Message

- Fill the required configuration and click on **Apply** button.

TTI Message

☒ Line-1
☐ Line-2

Line-1 OTU TTI Message Settings

☒ OTU TTI
☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear
OTU Expected	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear

II. Line-1 ODU TTI Message

Fill the required configuration and click on **Apply** button.

TTI Message

☒ Line-1 ☐ Line-2

Line-1 ODU TTI Message Settings

☐ OTU TTI ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

III. Line-2 OTU TTI Message

Fill the required configuration and click on **Apply** button.

TTI Message

☐ Line-1 ☒ Line-2

Line-2 TTI Message Settings

☒ OTU TTI ☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

IV. Line-2 ODU TTI Message

Fill the required configuration and click on **Apply** button.

TTI Message

☐ Line-1
 ☒ Line-2

Line-2 ODU TTI Message Settings

☐ OTU TTI
 ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

7.3.1.8 Wavelength Settings

UTL DWDM transport systems include pluggable and software-tunable transceiver modules capable of operating on 40 or 80 channels. UTL DWDM tunable transceivers are able to function on various wavelengths and to adjust wavelengths according to user needs, making them prevalent among DWDM systems.

- Select the required channel and click on **Apply** button.
- If tunable card then only the **Apply** button enabled.

Wavelength

☒ 40 Channel
 ☐ 80 Channel

40 Channel Wavelength Selection

Line-1 Channel	1	Wavelength	Apply Refresh
Line-2 Channel	1	Wavelength	Apply Refresh

Xfp Status: Line1 - **Present** Line2 - **Present**

7.3.1.9 Switching

The user can select switching mode either automatic or manual for line path. Also user can select operational line path either Line1 or Line2.

- Select mode type and click on **Apply** button.

Switching

Switching mode

Mode of switching : Automatic ▼

Apply Refresh

Current Operational Receive Path

Receive Path : Line-1 ▼ Refresh

7.3.1.10 Auto Negotiation

The user can apply auto negotiation either disable or enable for client ports.

Auto Negotiation

Client Port Auto-Negotiation Settings

Port-5 Enable ▼ Apply Refresh

Refresh All

7.3.2 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements. Therefore, alarms need to be obtained to monitor and manage alarms at the card, node, or network level. In the card, node, or network view, click the Alarms tab to display the alarms for that card, node, or network.

7.3.2.1 STM16 Alarms

The screen gives a complete picture of the system STM16 alarms.

[STM16]

Alarms in Client-2

Optical Alarms

Client In

IOP Failure ●

Low IOP ●

High IOP ●

Client Out

OOP Failure ●

Low OOP ●

High OOP ●

Temperature

High Temp ●

Low Temp ●

BiasCurrent

High Bias ●

Low Bias ●

STM16 Alarms

LOS-P ●

LOF ●

OOF ●

MS-AIS ●

MS-RDI ●

BIP-SD ●

BIP-SF ●

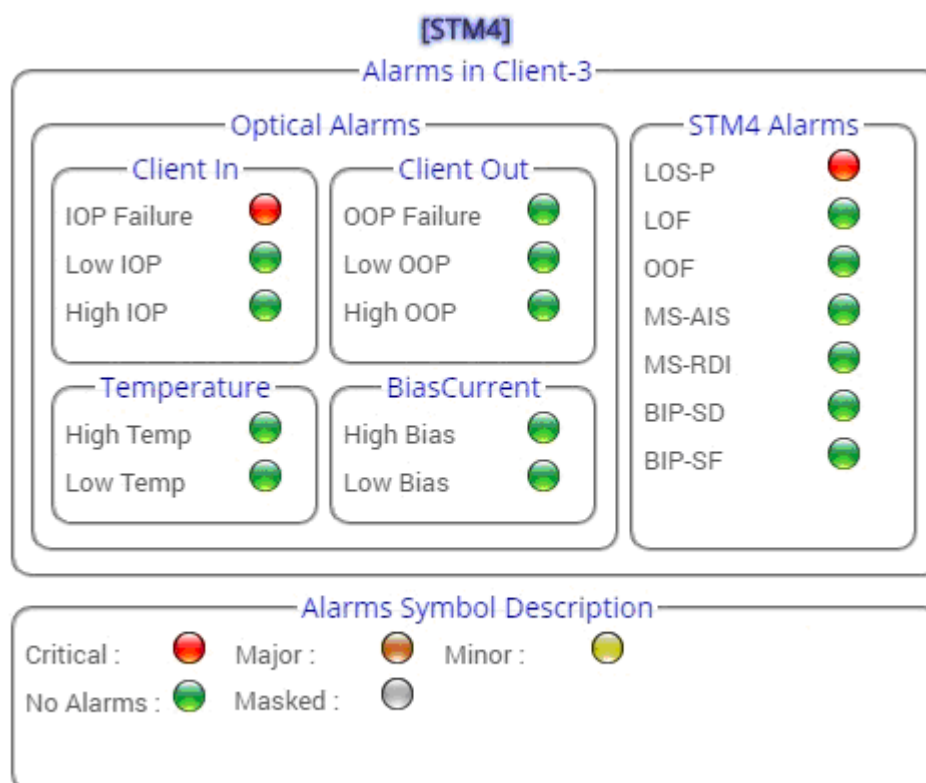
Alarms Symbol Description

Critical : ● Major : ● Minor : ●

No Alarms : ● Masked : ●

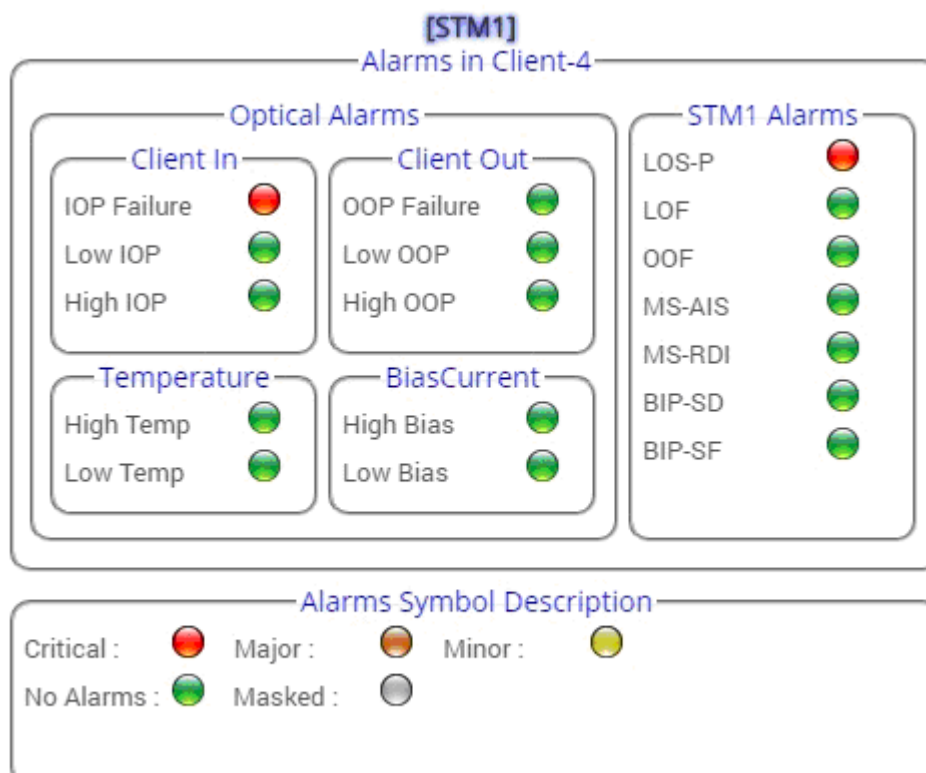
7.3.2.2 STM4 Alarms

The screen gives a complete picture of the system STM4 alarms.



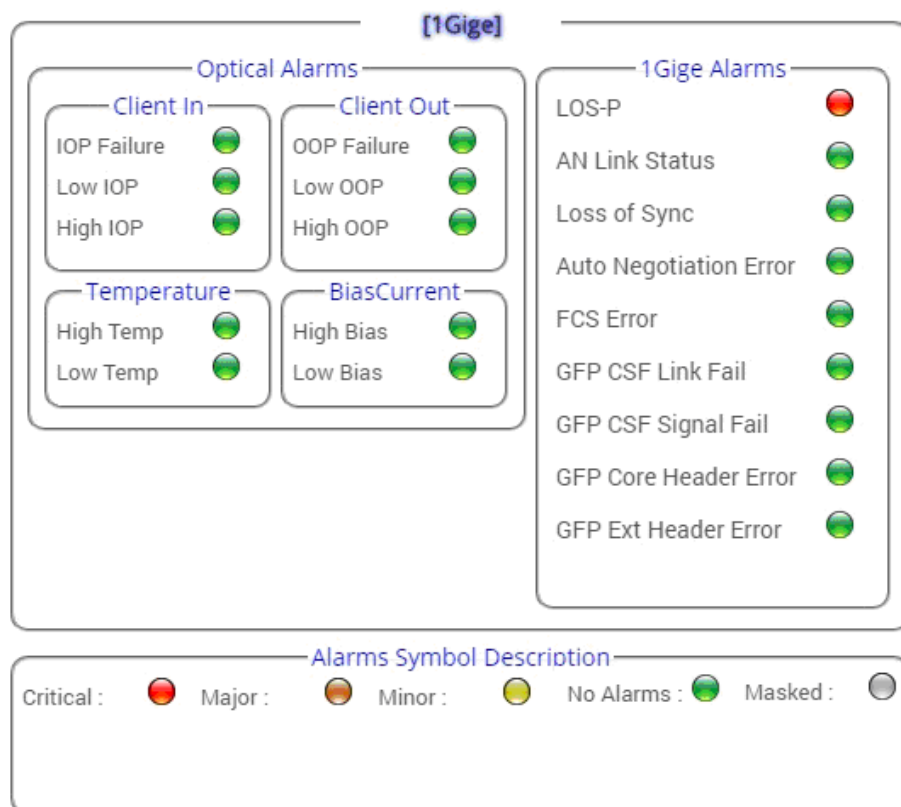
7.3.2.3 STM1 Alarms

The screen gives a complete picture of the system STM1 alarms.



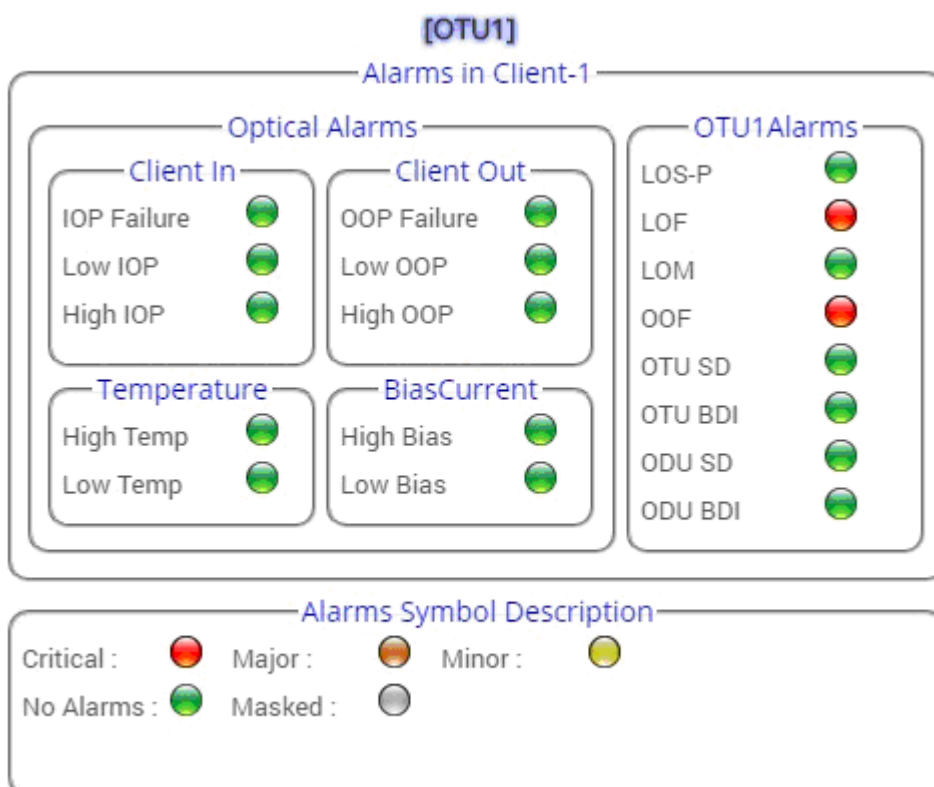
7.3.2.4 GigE Alarms

The screen gives a complete picture of the system GigE alarms.



7.3.2.5 OTU1 Alarms

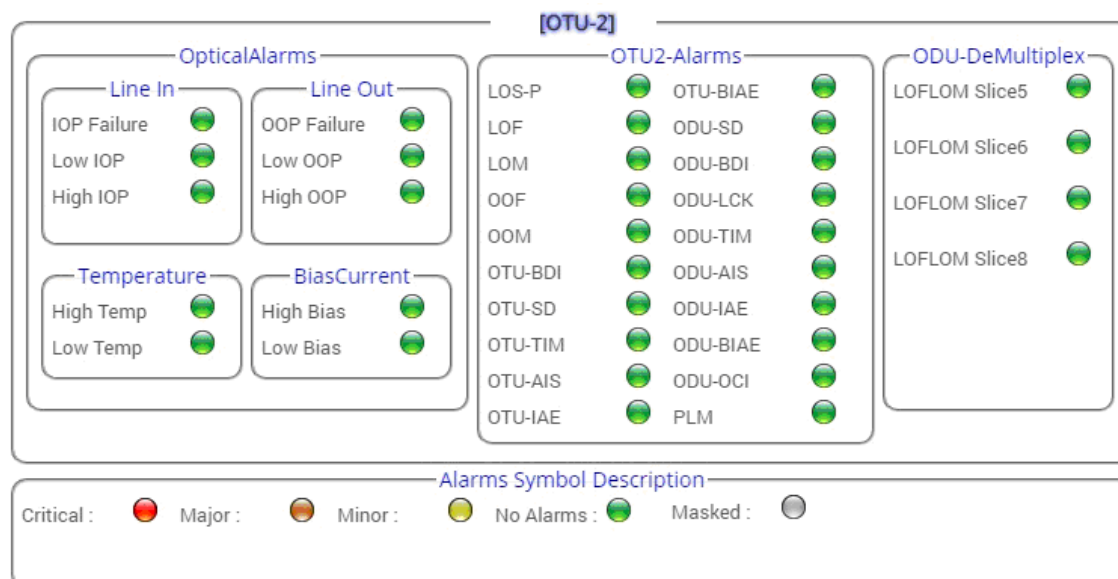
The screen gives a complete picture of the system OTU1 alarms.



7.3.2.6 OTU2 Alarms

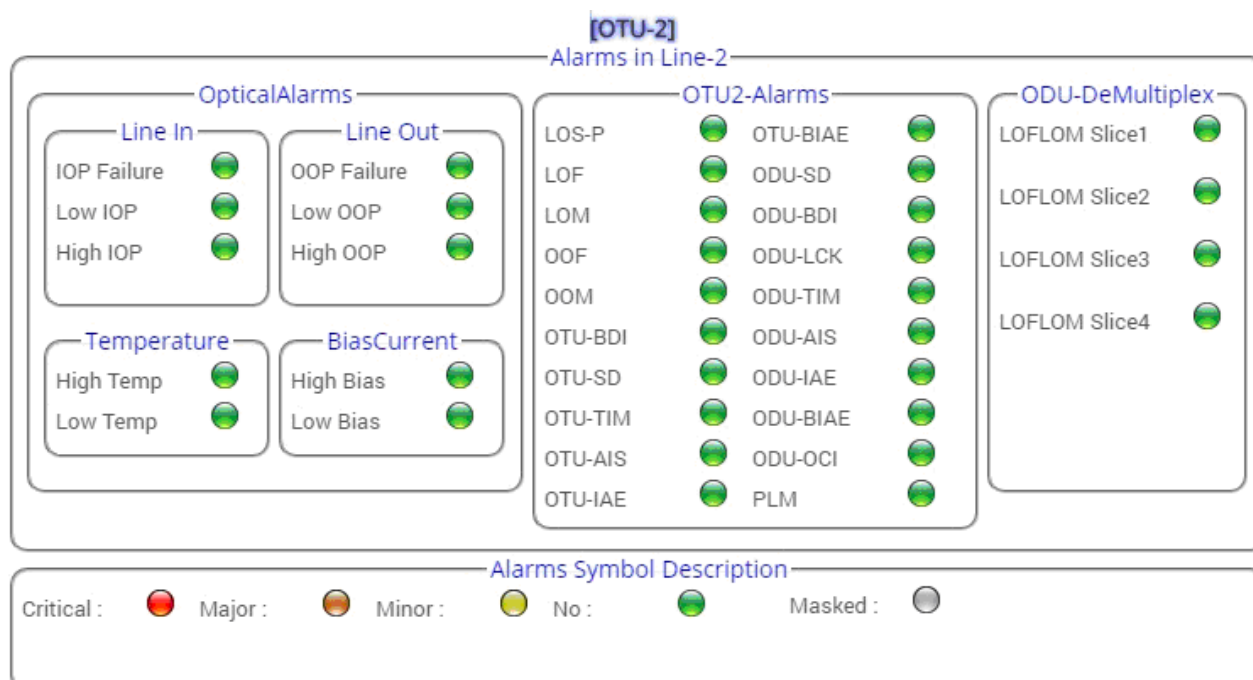
I. Line-1 Alarms

The screen gives a complete picture of the system Line-1 OTU2 alarms.



II. Line-2 Alarms

The screen gives a complete picture of the system Line-2 OTU2 alarms.



7.3.3 Performance

Click on **Performance** button, then performance dialog is displayed as shown below.

7.3.3.1 STM16 Performance

This screen collects optical performance data of the card.

STM16-Performance in Client-2

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	81912
B1 Error Count	1965949026
B2 Error Count	127789159079
MS-REI Error Count	74705948409
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.3.3.2 STM4 Performance

This screen collects OTN performance data of the card.

STM-4 Performance in Client-1

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	81741
B1 Error Count	1961854128
B2 Error Count	30082298092
MS-REI Error Count	37275891114
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.3.3.3 STM1 Performance

This screen collects STM4 performance data of the card.

STM-1 Performance in Client-3

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	81946
B1 Error Count	2622343610
B2 Error Count	8522712245
MS-REI Error Count	333362
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.3.3.4 GigE Performance

This screen collects GigE performance data of the card.

1Gige-Performance in Client-2

Frames Transmitted	0	OverSize Frames Received	0
Frames Received	0	Vlan Tagged Frames Transmitted	0
Vlan Tagged Frames Received	0	Fragment Frames Received	0
Pause Frames Received	0	OverSize Frames Transmitted	0
Unicast Frames Received	0	FrameCheck Seq Errors Received	0
Multicast Frames Received	0	Frames Received With NoDefect	0
Broadcast Frames Received	0	GfpCore Header ErrorCount	0
UnderSize Frames Received	0	GfpExtention Header ErrorCount	0
GfpFrameCount With SingleBitError	0		

[Re-initializes all the counters]

7.3.3.5 OTU1 Performance

This screen collects OTU1 performance data of the card.

OTU1-Performance in Client-4

Error Seconds	0	Ones Corrected Errors	7917563
Severe Error Seconds	0	Uncorrected Words	10718894577
Background Block Errors	0	Section Monitoring BIP	1668321147
Error Free Seconds	0	Path Monitoring BIP	1668320178
Unavailable Seconds	82022	SM Backward Error Counter	1483153289
TCM-1 Counter	1668314649	PM Backward Error Counter	590316892
TCM-2 Counter	1668317892	Corrected Bit Counter	71863192
Zeros Corrected Errors	63945629		

[Retrieves counter values] **Refresh** [Re-initializes all the counters] **Re-initialize**

7.3.3.6 Line Performance

I. OTU Performance

This screen collects OTU performance data of the card for Line-1/Line-2.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Background Block Errors	0	Background Block Errors	0
Error Free Seconds	0	Error Free Seconds	0
Unavailable Seconds	402	Unavailable Seconds	0
Bit Interleaved Parity	26279535	Backward Error Indication	16092584
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0

Refresh

[Re-initializes all the counters] **Re-initialize**

II. ODU Performance

This screen collects ODU performance data of the card for Line-1/Line-2.

ODU-Near End-Performance		ODU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Background Block Errors	0	Background Block Errors	0
Error Free Seconds	1	Error Free Seconds	1
Unavailable Seconds	497	Unavailable Seconds	497
Bit Interleaved Parity	32570895	Backward Error Indication	19690695
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Errors	32570895		
TCM-2 BIP Errors	32570895		

[Refresh](#)

[Re-initializes all the counters] [Re-initialize](#)

III. FEC Performance

This screen collects FEC performance data of the card for Line-1/Line-2.

FEC-Performance	
Zeros Corrected Bits	293407
Ones Corrected Bits	758191
Uncorrected Words	0
Corrected Bits	1051598

[Refresh](#)

[Re-initializes all the counters] [Re-initialize](#)

7.3.3.7 OPTICAL Performance

This screen collects optical performance data of the card.

Optical Performance

Client

	Client-1	Client-2	Client-3	Client-4	Client-5	Client-6	Client-7	Client-8
Rx Power(dBm)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
Tx Power(dBm)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
TxBias (mA)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
TxTemp(Degree)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh

Line1

Rx Power(dBm)	no_input_pow	TxBias (mA)	43.65	TxOSNR	43.53
Tx Power(dBm)	0.34	TxTemp(Degree)	35.70	Refresh	

Line2

Rx Power(dBm)	xfp_is_absent	TxBias (mA)	xfp_is_absent	TxOSNR	NaN
Tx Power(dBm)	xfp_is_absent	TxTemp(Degree)	xfp_is_absent	Refresh	

7.3.4 Alarm Severity

Click on **Alarm Severity** button, then alarm severity dialog will be displayed as shown below. You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Some alarms with default severity are displayed on EMS as follows

Loss of I/P power - Critical, Loss of signal – Critical, Loss of frame - Critical ,
Input High Power – Major, OTU/ODU Signal Degrade – Minor etc.

7.3.4.1 SDH Alarm Severity

You can assign or modify the severity of SDH alarms. The severity levels that you can assign are: Critical, Major, and Minor.

SDH-Alarms Severity

LOS	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SD	Critical ▼
OOF	Critical ▼	BIP-SF	Critical ▼
MS-AIS	Critical ▼		

Refresh

- Select the severity type and click on **Apply** button

7.3.4.2 GigE Alarm Severity

You can assign or modify the severity of GigE alarms. The severity levels that you can assign are: Critical, Major, and Minor.

1GIGE-Alarms Severity

LOS-P	Critical ▼	GFP CSF Link	Critical ▼
AN Link Status	Critical ▼	Fail	Critical ▼
Loss Of Sync	Critical ▼	GFP CSF Signal	Critical ▼
Auto		Fail	Critical ▼
Negotiation		GFP Core	Critical ▼
Error	Critical ▼	Header Error	Critical ▼
FCS Error	Critical ▼	GFP Ext Header	Critical ▼
		Error	Critical ▼

Refresh

7.3.4.3 Optical Alarm Severity

You can assign or modify the severity of optical alarms. The severity levels that you can assign are: Critical, Major, and Minor.

Optical Alarms

Input

IOP Failure	Critical ▼
Low IOP	Critical ▼
High IOP	Critical ▼

Output

OOP Failure	Critical ▼
Low OOP	Critical ▼
High OOP	Critical ▼

Temperature

High Temp	Critical ▼
Low Temp	Critical ▼

Bias

High Bias	Critical ▼
Low Bias	Critical ▼

Refresh

7.3.4.4 OTU2 Alarm Severity

You can assign or modify the severity of OTU2 alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OTU2 Alarms Severity

LOS-P	Critical ▼	OTU-BIAE	Critical ▼
LOF	Critical ▼	ODU-OCI	Critical ▼
LOM	Critical ▼	ODU-SD	Critical ▼
OOF	Critical ▼	ODU-BDI	Critical ▼
OOM	Critical ▼	ODU-LCK	Critical ▼
OTU-BDI	Critical ▼	ODU-TIM	Critical ▼
OTU-SD	Critical ▼	ODU-AIS	Critical ▼
OTU-TIM	Critical ▼	ODU-IAE	Critical ▼
OTU-AIS	Critical ▼	ODU-BIAE	Critical ▼
OTU-IAE	Critical ▼	PLM	Critical ▼

[Refresh](#)

7.4 OBA

The following options are available under OBA:

1. Alarms.
2. Settings.
3. Performance.
4. Alarm Severity.

Select OBA card on node view, then FCAPS buttons will be enabled.

7.4.1 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements.

Click on **Alarms** button, then alarms dialog will be displayed as shown below.

Alarms

IOP Failure		[Below -30.0 dBm]	Apply
Degraded IOP		[Below -8.0 dBm]	Apply
Low IOP		[Below -10.0 dBm]	Apply
IOP Out of Range		[Below -10.0 dBm]	Apply or Above 3.0 dBm Apply
Degraded OOP		[Below 16.0 dBm]	Apply
OOP Out of Range		[Below 14.5 dBm]	Apply or Above 20.5 dBm Apply
OOP Failure		[Below 5.0 dBm]	Apply

EDFA High Temperature  [Above +80 Degree]

Alarms Symbol Description

Critical : 
Major : 
Minor : 
No Alarm : 

[Threshold Refresh](#)
[Threshold Calculator](#)

It will automatically refresh every 10 sec.

7.4.2 Settings

Click on **Settings** button, then settings dialog will be displayed as shown below.

If you want to apply any settings

- Select required value and click on **Apply** button.

Card Details

Serial No [Refresh](#)
HW Version [Refresh](#)
SW Version [Refresh](#)
Card Reset [Soft Reset](#)

Operation Mode Settings

Operation Mode: [Apply](#) [Refresh](#)

LD Settings

LD Status: [Apply](#) [Refresh](#)

EVOA Settings

EVOA Settings db [Apply](#) [Refresh](#)

7.4.3 Performance

Click on **Performance** button, then performance dialog will be displayed as shown below. This screen collects optical performance data of the card.

Optical Performance

Input[dBm]
Output[dBm]
Gain
Mode
EDFA Temp
[Retrieves performance values] [Refresh](#)

7.4.4 Alarm Severity

Click on **Alarm Severity** button, then alarm severity dialog will be displayed as shown below. You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

OBA Alarm Severity

IOP Failure	Critical ▼
Degraded IOP	Major ▼
Low IOP	Major ▼
IOP Out of Range	Major ▼
Degraded OOP	Major ▼
OOP Out of Range	Major ▼
OOP Failure	Critical ▼
EDFA High Temperature	Major ▼

Refresh

Data Retrieved Successfully..

- Select the severity type and click on **Apply** button.

7.5 OPA

The following options are available under MSPP:

1. Alarms.
2. Settings.
3. Performance.
4. Alarm Severity.

Select OBA card on node view, then FCAPS buttons will be enabled.

7.5.1 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements.

Click on **Alarms** button, then alarms dialog will be displayed as shown below.

Stage-1 Alarms

IOP Failure		[Below	-30.0	dBm]	Apply	
Degraded IOP		[Below	-8.0	dBm]	Apply	
Low IOP		[Below	-10.0	dBm]	Apply	
IOP Out of Range		[Below	-10.0	dBm]	Apply	or Above 3.0 dBm] Apply
OOP Out of Range		[Below	8.0	dBm]	Apply	or Above 16.0 dBm] Apply
OOP Failure		[Below	5.0	dBm]	Apply	

Stage-2 Alarms

IOP Failure		[Below	-30.0	dBm]	Apply	
Degraded IOP		[Below	-12.0	dBm]	Apply	
Low IOP		[Below	-15.0	dBm]	Apply	
IOP Out of Range		[Below	-15.0	dBm]	Apply	or Above -2.0 dBm] Apply
Degraded OOP		[Below	5.0	dBm]	Apply	
OOP Out of Range		[Below	3.0	dBm]	Apply	or Above 11.0 dBm] Apply
OOP Failure		[Below	0.0	dBm]	Apply	

EDFA High Temperature  [Above +80 Degree]

Alarms Symbol Description

Critical :  Major :  Minor :  No Alarm : 

7.5.2 Settings

Click on **Settings** button, then settings dialog will be displayed as shown below.

Card Details

Serial No:

HW Version:

SW Version:

Card Reset:

TILT Settings

TILT :

[Value range: -3 db to +1 db]

Operation Mode Settings

Operation Mode:

LD Settings

LD Status:

Gain Settings

Gain[Stage-1]: db

Gain[Stage-2]: db

EVOA Settings

EVOA Settings: db

7.5.3 Performance

Click on **Performance** button, then performance dialog will be displayed as shown below.

This screen collects performance data of the card.

Optical Performance

Input(dBm)	no_input_power	Mid-In(dBm)	-57.00
Mid-Out(dBm)	-57.00	Output(dBm)	no_output_power
Gain[Stage-1]	11.00	Gain[Stage-2]	24.00
Mode[Stage-1]	AGC	Mode[Stage-2]	AGC
EDFA Temp	33.20		

[Refresh](#)

7.5.4 Alarm Severity

Click on **Severity** button, then severity dialog will be displayed as shown below. You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Stage-1 Alarm Severity

IOP Failure	Critical ▼
Degraded IOP	Major ▼
Low IOP	Major ▼
IOP Out of Range	Major ▼
OOP Out of Range	Major ▼
OOP Failure	Critical ▼

Stage-2 Alarm Severity

IOP Failure	Critical ▼
Degraded IOP	Major ▼
Low IOP	Major ▼
IOP Out of Range	Major ▼
Degraded OOP	Major ▼
OOP Out of Range	Major ▼
OOP Failure	Critical ▼

Temperature Alarm Severity

EDFA High Temp	Major ▼
----------------	---------

[Refresh](#)

Data Retrieved Successfully..

7.6 SCC

The following options are available under SCC:

- Alarms.
- Settings.

- Performance.
- Alarm Severity

Select SCC card on node view, then FCAPS buttons will be enabled.

7.6.1 Alarms

Click on **Alarms** button, then alarms dialog is displayed as shown below.

OSC-Alarms

	OSC-1	OSC-2	OSC-3	OSC-4		OSC-1	OSC-2	OSC-3	OSC-4
OutputPowerFailure					InputPowerFailure				
HighOutputPower					HighInputPower				
LowOutputPower					LowInputPower				
HighBiasCurrent					LowBiasCurrent				
HighTemperature					LowTemperature				
Link Failure									

SCC-Alarms

HighTemperature		LowTemperature	
-----------------	--	----------------	--

SFP Status

- OSC1 - SFP Absent - Port Configured
- OSC3 - SFP Absent - Port Configured

- OSC2 - SFP Present - Port Configured
- OSC4 - SFP Absent - Port UnConfigured

[SFP Status](#)

Alarms Symbol Description

Critical:
 Normal:
 Minor:
 Major:
 SFP Absent: (Configured ports only)

7.6.2 Settings

Click on **Settings** button, then settings dialog as shown in below.

7.6.2.1 IP Address

In SNMP, traps enable an agent to notify the management server of significant events by means of unsolicited SNMP messages called TRAP. You can add up to twenty trap IP Address receivers.

IP Address

	IP Address	NetMask	
Node IP	192.168.21.46	255.255.255.0	Apply Refresh
VOIP	192.168.1.77	255.255.255.252	Apply Delete Refresh
OSC1	192.168.1.81	255.255.255.252	Apply Delete Refresh
OSC2	192.168.1.114	255.255.255.252	Apply Delete Refresh
OSC3	192.168.1.70	255.255.255.252	Apply Delete Refresh
OSC4	0.0.0.0	0.0.0.0	Apply Delete Refresh

Trap Receiver IP:
[Apply](#)
[Delete](#)

[Refresh](#)

OSC1 - SFP Absent - Port Configured

OSC2 - SFP Present - Port Configured

OSC3 - SFP Absent - Port Configured

OSC4 - SFP Absent - Port UnConfigured

7.6.2.2 Card Details

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

Card Details

Serial No	UTL000000	Apply	Refresh
Hardware Version	1.0	Refresh	
Software Version	1.0.0	Refresh	
Card Reset	SoftReset		

Warning: The card software will reboot

7.6.2.3 ALS

OSC1 - ALSD Settings ALSD Status: Enable Apply Refresh	OSC2 - ALSD Settings ALSD Status: Enable Apply Refresh																
OSC3 - ALSD Settings ALSD Status: Enable Apply Refresh	OSC4 - ALSD Settings ALSD Status: Enable Apply Refresh																
ALS Pulse Repetition time ALS Time: 10 sec Apply Refresh Note:[Value range: 1 sec to 300 sec]	ALS For Osc Ports <table border="0"> <tr> <td>OSC1 ALS:</td> <td> On </td> <td> Apply </td> <td> Refresh </td> </tr> <tr> <td>OSC2 ALS:</td> <td> On </td> <td> Apply </td> <td> Refresh </td> </tr> <tr> <td>OSC3 ALS:</td> <td> On </td> <td> Apply </td> <td> Refresh </td> </tr> <tr> <td>OSC4 ALS:</td> <td> On </td> <td> Apply </td> <td> Refresh </td> </tr> </table>	OSC1 ALS:	On	Apply	Refresh	OSC2 ALS:	On	Apply	Refresh	OSC3 ALS:	On	Apply	Refresh	OSC4 ALS:	On	Apply	Refresh
OSC1 ALS:	On	Apply	Refresh														
OSC2 ALS:	On	Apply	Refresh														
OSC3 ALS:	On	Apply	Refresh														
OSC4 ALS:	On	Apply	Refresh														

7.6.2.4 Ethernet Port Access

EMS/LCT Ethernet Port Access Settings

EMS Port:	EMS	Enable	Apply	Refresh
LCT Port:	LCT	Enable	Apply	Refresh

7.6.2.5 Software Upload

The UTL MP-0072 EMS provides the system administrator with the means to easily upload and upgrade the software in any existing and functional SCC.

- SCC Software File Upload: Choose the file to transfer to the SCC using SFTP.
- Other Cards Software File Upload: Choose the file to transfer to the Card using SFTP.

The screenshot displays two distinct sections for software uploads. The top section, titled "SCC Software File Transfer", contains a label "SCC File:" followed by a "Choose File" button and the text "No file chosen". Below this is a blue button labeled "SCC SW Upload". The bottom section, titled "Card Software File Transfer", contains a label "Card File:" followed by a "Choose File" button and the text "No file chosen". Below this is a blue button labeled "Card SW Upload".

7.6.2.6 NTP Settings

The Network Time Protocol (**NTP**) is one way to provide clock accuracy in a network. User can configure NTP server IP using below screen.

The screenshot shows the "NTP Server IP Address" configuration screen. At the top, there is a header "NTP Server IP Address". Below it, there is a field labeled "NTPServerIP:" with a text input box containing three dots, and two buttons: "Apply" and "Delete". Below this is a larger section containing a label "NTPServerIP 1" and a text input box containing the IP address "192.168.21.234". At the bottom right of this section is a blue button labeled "Refresh".

7.6.3 Performance

Click on **Performance** button, then performance dialog will be displayed as shown below.

This screen collects performance data of the card.

OSC Performance

	OSC-1	OSC-2	OSC-3	OSC-4
Rx Input Power(dBm)	sfp_absent_for	no_input_powe	sfp_absent_for	sfp_absent_for
Tx Output Power(dBm)	sfp_absent_for	2.52	sfp_absent_for	sfp_absent_for
Temperature	sfp_absent_for	34.88	sfp_absent_for	sfp_absent_for
Bias Current	sfp_absent_for	19.40	sfp_absent_for	sfp_absent_for
	Refresh	Refresh	Refresh	Refresh

SCC

Card Temperature	45.0	Refresh
------------------	------	---------

7.6.4 Alarm Severity

Click on **Alarm Severity** button, then alarm severity dialog will be displayed as shown below.

You can assign or modify the severity of OSC alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OSC Alarm Severity

OutputPowerFailure	Critical ▼	InputPowerFailure	Critical ▼
HighOutputPower	Major ▼	HighInputPower	Major ▼
LowOutputPower	Major ▼	LowInputPower	Major ▼
HighBiasCurrent	Major ▼	LowBiasCurrent	Major ▼
HighTemperature	Major ▼	LowTemperature	Major ▼
Link Failure	Critical ▼		

SCC Alarm Severity

HighTemperature	Critical ▼	LowTemperature	Critical ▼
-----------------	-------------------	----------------	-------------------

[Refresh](#)

Data Retrieved Successfully..

7.7 SCU

Sub rack Control Unit (SCU) receives information data from BMU and after analyzing its destination addresses it passes on to corresponding slot unit.

Select SCU card on node view, then Settings button will be enabled.

7.7.1 Settings

Click on **Settings** button, then settings dialog will be displayed as shown below.

Card Details

Serial No.		Refresh
HW Version		Refresh
SW Version		Refresh
Card Reset	Soft Reset Warning: The card software will reboot	

If you want to reset the card click the **Soft Reset** button and click on yes.

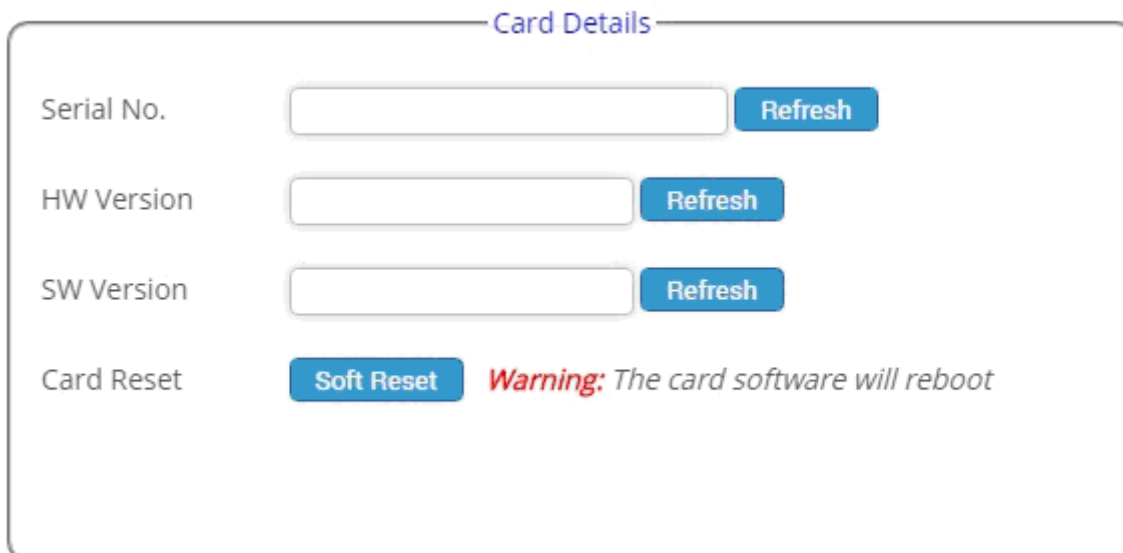
7.8 BMU

Bay Monitoring Unit (BMU) receives information data from SCC and after analyzing its destination addresses it passes on to corresponding sub rack.

Select BMU card on node view, then Settings button will be enabled.

7.8.1 Settings

Click on **Settings** button, then settings dialog will be displayed as shown below.



The 'Card Details' dialog box contains the following fields and controls:

- Serial No.:** A text input field with a 'Refresh' button to its right.
- HW Version:** A text input field with a 'Refresh' button to its right.
- SW Version:** A text input field with a 'Refresh' button to its right.
- Card Reset:** A 'Soft Reset' button followed by a red warning message: *Warning: The card software will reboot*.

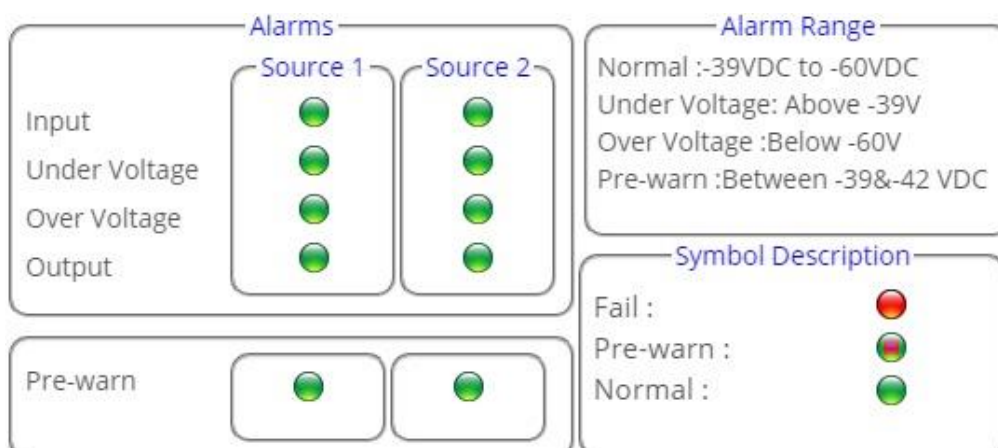
If you want to reset the card click the **Soft Reset** button and click on yes.

7.9 PS1 & PS2

A power supply (PS) is a component that supplies power to DWDM systems. PSU monitors continuously the Input supply. When the input supply reaches under voltage or Under Voltage Pre-warn (EMS only) or Over voltage it gives an LED indication and this information displayed in the EMS.

7.9.1 Alarms

Select PS1 & PS2 card on node view, click on **Alarms** button, then alarms dialog will be displayed as shown below.



The 'Alarms' dialog box displays the following information:

- Alarm Range:**
 - Normal :-39VDC to -60VDC
 - Under Voltage: Above -39V
 - Over Voltage :Below -60V
 - Pre-warn :Between -39&-42 VDC
- Symbol Description:**
 - Fail : (Red LED)
 - Pre-warn : (Yellow LED)
 - Normal : (Green LED)
- Alarms Table:**

	Source 1	Source 2
Input	Green LED	Green LED
Under Voltage	Green LED	Green LED
Over Voltage	Green LED	Green LED
Output	Green LED	Green LED
Pre-warn	Green LED	Green LED

It will automatically refresh every 10 sec.

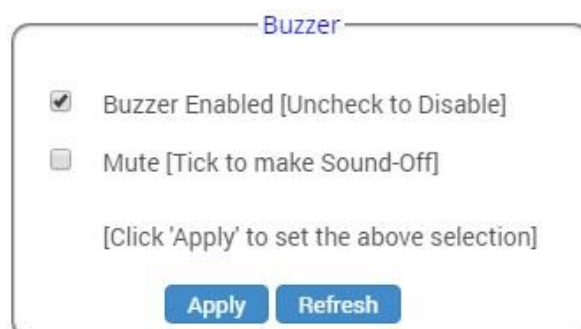
7.10 Buzzer

Buzzer card is used for extending Potential Free Contact (PFC) to the users Alarm Display Board, for global alarm monitoring. Select Buzzer card on node

view, then Settings buttons will be enabled.

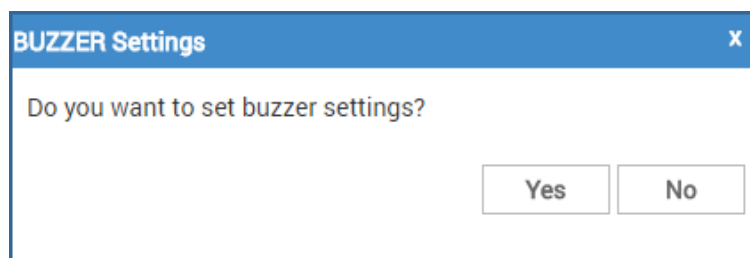
7.10.1 Settings

Click on **Settings** button, then settings dialog is displayed as shown below.



If you want to apply any settings, select required value and click on **Apply** button.

After clicking the **Apply** button, a confirmation prompt will be displayed as below.



Click **Yes** button to apply Settings.

7.11 Fan

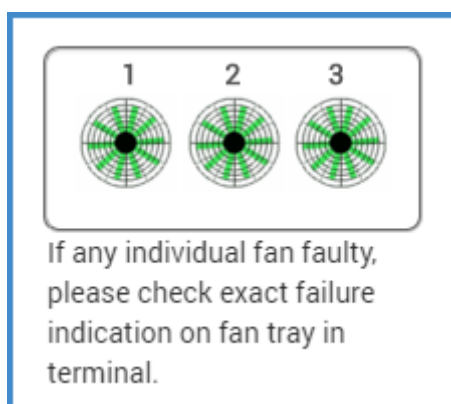
The fan tray assembly implements heat dissipation on a sub rack, which allows the sub rack to operate normally within the designated temperature range.

The following options are available under Fan:

1. Alarms.

7.11.1 Alarms

Click on **Alarms** button, then alarms dialog as shown in below.



7.12 OMTP11

The following options are available under OMTP11:

1. STM 64.
2. 10G WAN.
3. 10G LAN.
4. OTU2.

Select OMTP11 card on node view, then FCAPS buttons will be enabled.

7.12.1 Configuration Settings

7.12.1.1 Client Traffic

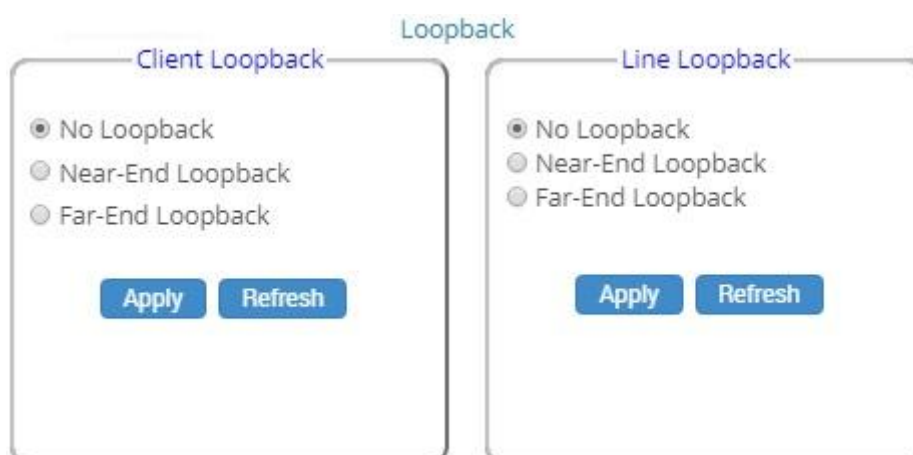
Click on **Settings** button, then client traffic dialog will be displayed as shown in below.



7.12.1.2 Loopback

Fiber optic loopback is known as the routing of electronic signals, digital data streams, or flows of items back to their source without intentional processing or modification. In terms of telecommunication, loopback is a hardware or software method to feed a received signal or data back to the sender. It is very useful for solving physical connection problems.

If you want to set the loopback, Select loopback type and click on **Apply** button.



7.12.1.3 ALS & FEC

User can set the Laser ON/OFF, Automatic Laser Shutdown (ALS) and forward error correction (FEC) to correct bit errors in the received data using below mentioned dialog.

- Select configuration type and click on **Apply** button.

ALS & FEC

Laser Configuration

Client Laser

☐ LASER OFF
☐ LASER ON

Apply **Refresh**

Line Laser

☐ LASER OFF
☐ LASER ON

Apply **Refresh**

ALS Configuration

Client ALS

☒ ALS Disable
☐ ALS Enable

Apply **Refresh**

Line ALS

☒ ALS Disable
☐ ALS Enable

Apply **Refresh**

FEC Configuration

Line FEC

☐ No FEC
☒ Super FEC
☐ Normal FEC

Apply **Refresh**

7.12.1.4 Threshold Settings

You can also configure the BER thresholds that raise or clear a signal degrade alarm and the time interval for the thresholds. If the BER thresholds and interval are not configured, the default values are used.

- Select threshold settings and click on **Apply** button.

Threshold

Client SDH Threshold Settings

B2 SF: **Apply** **Refresh**

B2 SD: **Apply** **Refresh**

Line Performance Threshold Settings

OTU SD: **Apply** **Refresh**

ODU SD: **Apply** **Refresh**

7.12.1.5 TTI Settings

The Trail trace identifier (TTI) is used to identify the signal from the source to the destination within the network. Please refer below screens for OTU/ODU TTI message settings.

I. Line OTU TTI Message

- Fill the required configuration and click on **Apply** button.

TTI Message
Line OTU TTI Message Settings

☒ OTU TTI ☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear
OTU Expected	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear

II. Line ODU TTI Message

Fill the required configuration and click on **Apply** button.

TTI Message
Line ODU TTI Message Settings

☐ OTU TTI ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear
ODU Expected	UTL	UTL	UTL
	Apply Refresh Clear	Apply Refresh Clear	Apply Refresh Clear

7.12.1.6 Wavelength Settings

UTL DWDM transport systems include pluggable and software-tunable transceiver modules capable of operating on 40 or 80 channels. UTL DWDM tunable transceivers are able to function on various wavelengths and to adjust wavelengths according to user needs, making them prevalent among DWDM systems.

- Select the required channel and click on **Apply** button.
- If tunable card then only the **Apply** button will be enabled.

Wavelength

☒ 40 Channel ☐ 80 Channel

40 Channel Wavelength Selection

Line Channel	1 ▼	Wavelength		Apply	Refresh
--------------	-----	------------	--	-------	---------

7.12.1.7 Card Inventory

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

Card Details

Serial No.

Refresh

HW Version

Refresh

SW Version

Refresh

Card Reset

Soft Reset

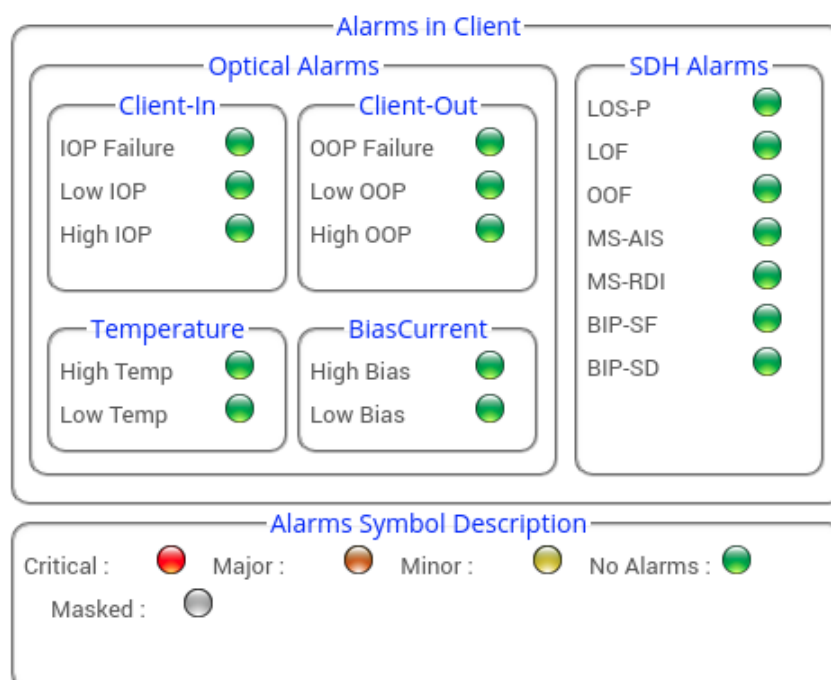
Warning: The card software will reboot

7.12.2 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements. Therefore, alarms need to be obtained to monitor and manage alarms at the card, node, or network level. In the card, node, or network view, click the Alarms tab to display the alarms for that card, node, or network.

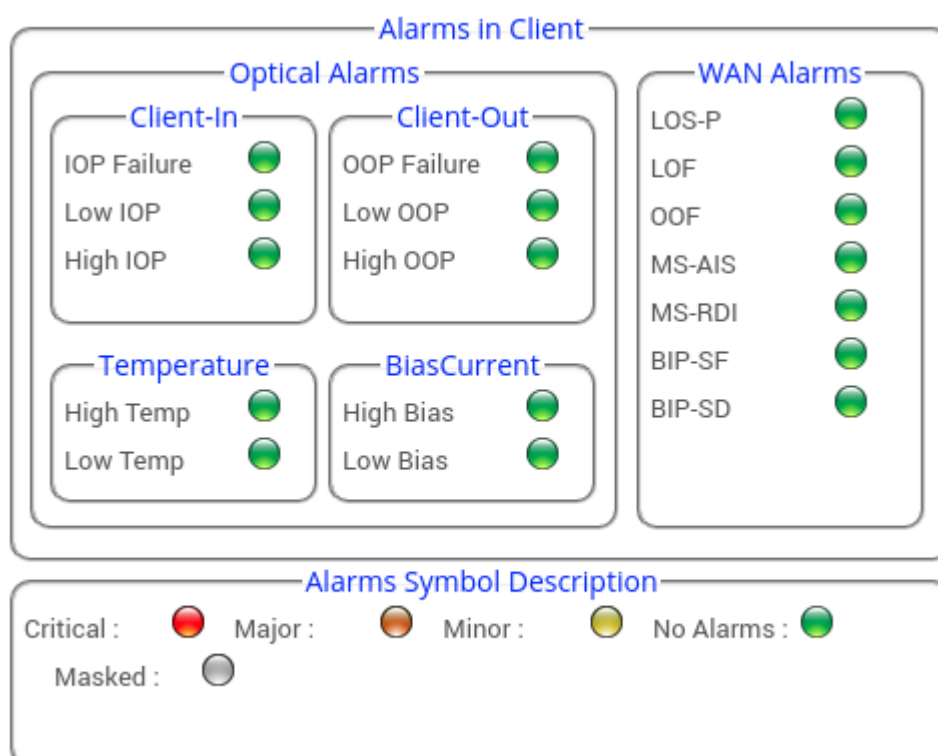
7.12.2.1 STM64 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the STM64 alarms.



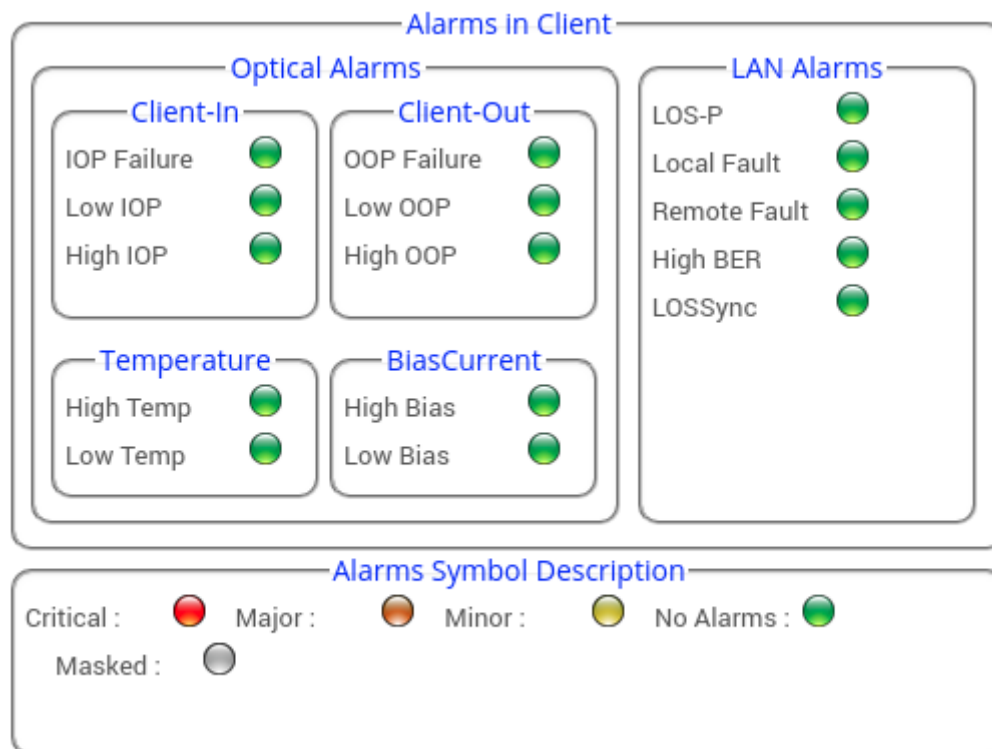
7.12.2.2 10G WAN Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the WAN alarms.



7.12.2.3 10G LAN Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the LAN alarms.

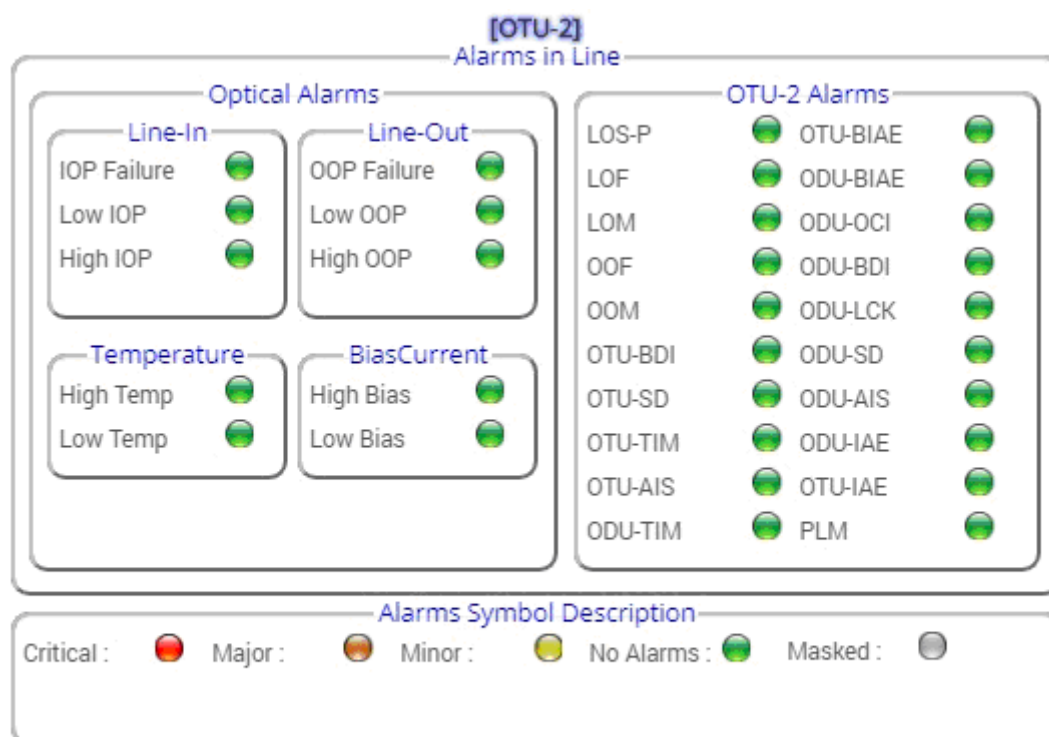


7.12.2.4 OTU2 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the OTU2 alarms.

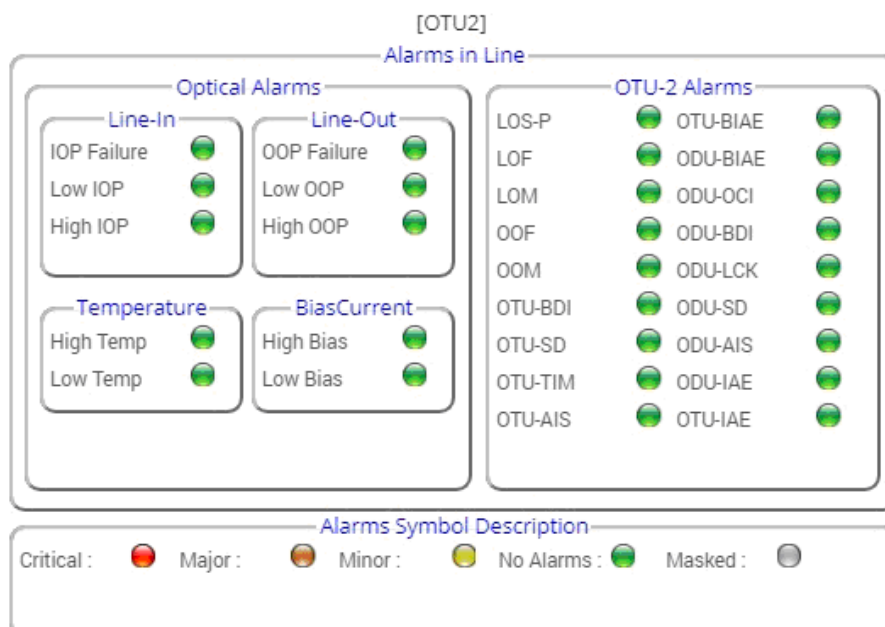
I. Client Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the client OTU2 alarms.



II. Line Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the Line OTU2 alarms.



7.12.3 Performance

Click on **Performance** button, then performance dialog will be displayed as shown below.

UTL DWDM products contain fiber optic components, including fiber optic transmitters and receivers. These components are extremely susceptible to contamination by particles of dirt or dust, which can obstruct the optic path and cause performance degradation. To ensure optimum product performance, it is important that all optics and connector ferrules be kept clean.

7.12.3.1 Optical Performance

This screen collects optical performance data of the card.

	Client	Line
Rx Power(dBm)	no_input_power	no_input_power
Tx Power(dBm)	1.00	0.90
TxBias(mA)	47.59	64.80
TxTemp(Degree)	35.43	36.25
TxOSNR(dB)		43.95
	<button>Refresh</button>	<button>Refresh</button>

7.12.3.2 Client OTU Performance

I. OTU Performance

This screen collects OTU performance data of the card.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Background Block Errors	0	Background Block Errors	0
Error Free Seconds	0	Error Free Seconds	593
Unavailable Seconds	593	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
<button>Refresh</button>		<button>Re-initialize</button>	
[Re-initializes all the Client port counters]			

II. ODU Performance

This screen collects ODU performance data of the card.

ODU-Near End-Performance		ODU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Error Free Seconds	640	Error Free Seconds	641
Background Block Errors	0	Background Block Errors	0
Unavailable Seconds	0	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Error	0		
TCM-2 BIP Error	0		

[Re-initializes all the Client port counters]

III. FEC Performance

This screen collects FEC performance data of the card.

FEC-Performance	
Zeros Corrected Bits	0
Ones Corrected Bits	0
Uncorrected Words	0
Corrected Bits	0

[Re-initializes all the Client port counters]

7.12.3.3 Line Performance

I. OTU Performance

This screen collects OTU performance data of the card.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Error Free Seconds	0	Error Free Seconds	464
Background Block Errors	0	Background Block Errors	0
Unavailable Seconds	464	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0

[Re-initializes all the Line port counters]

II. ODU Performance

This screen collects ODU performance data of the card.

ODU-Near End-Performance		ODU-Far End-Performance	
Bit Interleaved Parity	0	Error Seconds	0
Error Seconds	0	Severe Error Seconds	0
Severe Error Seconds	0	Error Free Seconds	524
Error Free Seconds	524	Background Block Errors	0
Background Block Errors	0	Unavailable Seconds	0
Unavailable Seconds	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Error	0		
TCM-2 BIP Error	0		

[Re-initializes all the Line port counters]

III. FEC Performance

This screen collects FEC performance data of the card.

FEC-Performance	
Zeros Corrected Bits	0
Ones Corrected Bits	0
Uncorrected Words	0
Corrected Bits	0

[Re-initializes all the Line port counters]

7.12.3.4 STM64 Performance

This screen collects STM64 performance data of the card.

SDH Performance-Client

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	66
B1 Error Count	3696000
B2 Error Count	407616000
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.12.3.5 10G WAN Performance

This screen collects WAN performance data of the card.

WAN Performance-Client

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	106
B1 Error Count	5936000
B2 Error Count	654656000
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.12.3.6 10G LAN Performance

This screen collects LAN performance data of the card.

10G LAN Performance(Client)

UnderSize Frames Received	0	Pause Frames Received	0
Fragment Frames Received	0	Broadcast Frames Received	0
Over Size Frames Received	0	Multicast Frames Received	0
Frames Received	0	Unicast Frames Received	0
Frames Transmitted	0	Unicast Frames Transmitted	0
FrameCheck Seq. Errors Received	0	Broadcast Frames Transmitted	0
		Multicast Frames Transmitted	0
		Refresh	
		Re-initialize	

7.12.4 Alarm Severity

You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Some alarms with default severity are displayed on EMS as follows

Loss of I/P power - Critical, Loss of signal – Critical, Loss of frame - Critical ,
Input High Power – Major, OTU/ODU Signal Degrade – Minor etc.

7.12.4.1 Optical Alarm Severity

You can assign or modify the severity of optical alarms. The severity levels that you can assign are: Critical, Major, and Minor.

Optical Alarms

Input

IOP Failure	Critical ▼
Low IOP	Critical ▼
High IOP	Critical ▼

Output

OOP Failure	Critical ▼
Low OOP	Critical ▼
High OOP	Critical ▼

Temperature

High Temp	Critical ▼
Low Temp	Critical ▼

Bias

High Bias	Critical ▼
Low Bias	Critical ▼

[Refresh](#)

7.12.4.2 OTU2 Alarm Severity

You can assign or modify the severity of OTU2 alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OTU2 Alarms Severity

LOS-P	Critical ▼	OTU-BIAE	Critical ▼
LOF	Critical ▼	ODU-OCI	Critical ▼
LOM	Critical ▼	ODU-SD	Critical ▼
OOF	Critical ▼	ODU-BDI	Critical ▼
OOM	Critical ▼	ODU-LCK	Critical ▼
OTU-BDI	Critical ▼	ODU-TIM	Critical ▼
OTU-SD	Critical ▼	ODU-AIS	Critical ▼
OTU-TIM	Critical ▼	ODU-IAE	Critical ▼
OTU-AIS	Critical ▼	ODU-BIAE	Critical ▼
OTU-IAE	Critical ▼	PLM	Critical ▼

[Refresh](#)

7.12.4.3 STM64 Alarm Severity

You can assign or modify the severity of STM64 alarms. The severity levels that you can assign are: Critical, Major, and Minor.

SDH Alarms Severity

LOS-P	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SF	Critical ▼
OOF	Critical ▼	BIP-SD	Critical ▼
MS-AIS	Critical ▼		

[Refresh](#)

7.12.4.4 10G WAN Alarm Severity

You can assign or modify the severity of WAN alarms. The severity levels that you can assign are: Critical, Major, and Minor.

WAN Alarms Severity

LOS-P	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SF	Critical ▼
OOF	Critical ▼	BIP-SD	Critical ▼
MS-AIS	Critical ▼		

[Refresh](#)

7.12.4.5 10G LAN Alarm Severity

You can assign or modify the severity of LAN alarms. The severity levels that you can assign are: Critical, Major, and Minor.

LAN Alarms Severity

LOS-P	Critical ▼
LOSSync	Critical ▼
High Bit Errors	Critical ▼
Local Fault	Critical ▼
Remote Fault	Critical ▼

[Refresh](#)

7.13 OMMP81

The following data traffic options are available under OMMP81:

1. STM16.
2. STM4.

3. STM1.
4. GigE.
5. OTU1.

Select OMMP81 card on node view, then FCAPS buttons will be enabled.

7.13.1 Configuration Settings

7.13.1.1 Client Traffic

Click on **Settings** button, then traffic configuration dialog will be displayed as shown below. The user can set any traffic like STM16, STM4, STM1, 1GigE and OTU2 for this card.

Client Traffic Configuration

Port-1	STM16 ▼	Apply
Port-2	STM16 ▼	Apply
Port-3	STM16 ▼	Apply
Port-4	Free ▼	Apply
Port-5	Free ▼	Apply
Port-6	Free ▼	Apply
Port-7	Free ▼	Apply
Port-8	Free ▼	Apply

Refresh All

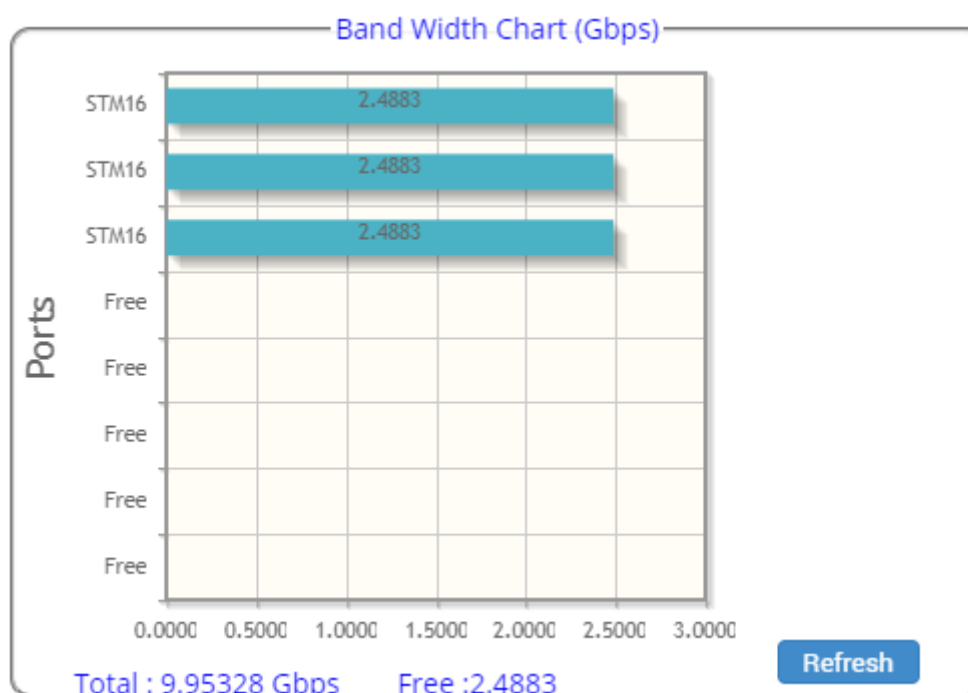
SFP Status:

Port1 - SFP(2.50 Gbps) Present - Port Configured
Port2 - SFP(2.50 Gbps) Present - Port Configured
Port3 - SFP(2.50 Gbps) Present - Port Configured
Port4 - SFP Absent - Port UnConfigured

Port5 - SFP Absent - Port UnConfigured
Port6 - SFP Absent - Port UnConfigured
Port7 - SFP Absent - Port UnConfigured
Port8 - SFP Absent - Port UnConfigured

7.13.1.2 Bandwidth Chart

The bandwidth of all client ports is displayed as shown below.



7.13.1.3 ALS & FEC

User can set the Laser ON/OFF, Automatic Laser Shutdown (ALS) and forward error correction (FEC) to correct bit errors in the received data using below mentioned dialog.

- Select port settings type and click on **Apply** button.

Client-SFP Laser Configuration

Port-1	Apply	Refresh
Port-2	Apply	Refresh
Port-3	Apply	Refresh
Port-4	Apply	Refresh
Port-5	Apply	Refresh
Port-6	Apply	Refresh
Port-7	Apply	Refresh
Port-8	Apply	Refresh

ALS Configuration

Port-1	Apply	Refresh
Port-2	Apply	Refresh
Port-3	Apply	Refresh
Port-4	Apply	Refresh
Port-5	Apply	Refresh
Port-6	Apply	Refresh
Port-7	Apply	Refresh
Port-8	Apply	Refresh

Client-OTU1 FEC Configuration

Port-5	Apply	Refresh
--------	--------------------------------------	--

Line-XFP Laser Configuration

Line :	Apply	Refresh
--------	--------------------------------------	--

Line ALS Configuration

Line :	Apply	Refresh
--------	--------------------------------------	--

Line-OTU2 FEC Configuration

Line-1	Apply	Refresh
--------	--------------------------------------	--

7.13.1.4 Loopback

Fiber optic loopback is known as the routing of electronic signals, digital data streams, or flows of items back to their source without intentional processing or modification. In terms of telecommunication, loopback is a hardware or software

method to feed a received signal or data back to the sender. It is very useful for solving physical connection problems.

If you want to set the loopback, Select loopback type and click on **Apply** button.

Loopback

Client Diagnostic Loopback

Port-1	Clear Loopback ▼	Apply	Refresh
Port-2	Clear Loopback ▼	Apply	Refresh
Port-3	Clear Loopback ▼	Apply	Refresh
Port-4	Clear Loopback ▼	Apply	Refresh
Port-5	Clear Loopback ▼	Apply	Refresh
Port-6	Clear Loopback ▼	Apply	Refresh
Port-7	Clear Loopback ▼	Apply	Refresh
Port-8	Clear Loopback ▼	Apply	Refresh

Refresh All

Line Loopback

Line: Clear Loopback ▼

Apply Refresh

7.13.1.5 Auto Negotiation

If you want to set the Auto-Negotiation Configuration

- Select port settings type and click on **Apply** button.

Client Port Auto-Negotiation Settings

Port-4: Enable ▼

Apply Refresh

Refresh All

7.13.1.6 Threshold Settings

You can also configure the BER thresholds that raise or clear a signal degrade alarm and the time interval for the thresholds. If the BER thresholds and interval are not configured, the default values are used.

- Select threshold settings and click on **Apply** button.

Client SDH B2 SF Threshold

Select Port: Client-1 ▼

B2 SF: 1e-5 ▼

Apply Refresh

Client OTU1 Threshold Settings

Select Port: Client-5 ▼

OTU SD: 1e-7 ▼

ODU SD: 1e-7 ▼

Apply Refresh

Client SDH B2 SD Threshold

Select Port: Client-1 ▼

B2 SD: 1e-7 ▼

Apply Refresh

Line OTU2 Threshold Settings

Select Port: Line-1 ▼

OTU SD: 1e-7 ▼

ODU SD: 1e-7 ▼

Apply Refresh

7.13.1.7 TTI Settings

The Trail trace identifier (TTI) is used to identify the signal from the source to the destination within the network. Please refer below Line-1 and Line-2 screens for OTU/ODU TTI message settings.

I. Line-1 OTU TTI Message

- Fill the required configuration and click on **Apply** button.

Line OTU TTI Message Settings

☒ OTU TTI ☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

II. Line-1 ODU TTI Message

- Fill the required configuration and click on **Apply** button.

Line ODU TTI Message Settings

☐ OTU TTI ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

V. Line-2 OTU TTI Message

- Fill the required configuration and click on **Apply** button.

TTI Message

☐ Line-1
 ☒ Line-2

Line2 OTU TTI Message Settings

☒ OTU TTI
 ☐ ODU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

VI. Line-2 ODU TTI Message

- Fill the required configuration and click on **Apply** button.

TTI Message

☐ Line-1
 ☒ Line-2

Line2 ODU TTI Message Settings

☐ OTU TTI
 ☒ ODU TTI

	SAPI	DAPI	Operator
ODU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
ODU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

7.13.1.8 Wavelength Settings

UTL DWDM transport systems include pluggable and software-tunable transceiver modules capable of operating on 40 or 80 channels. UTL DWDM tunable transceivers are able to function on various wavelengths and to adjust wavelengths according to user needs, making them prevalent among DWDM systems.

- Select the required channel and click on **Apply** button.
- If tunable card then only the **Apply** button will be enabled.

☒ 40 Channel ☐ 80 Channel

40 Channel Wavelength Selection

Line-1 Channel: Wavelength:

7.13.1.9 Card Inventory

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

Card Details

Serial No.

HW Version

SW Version

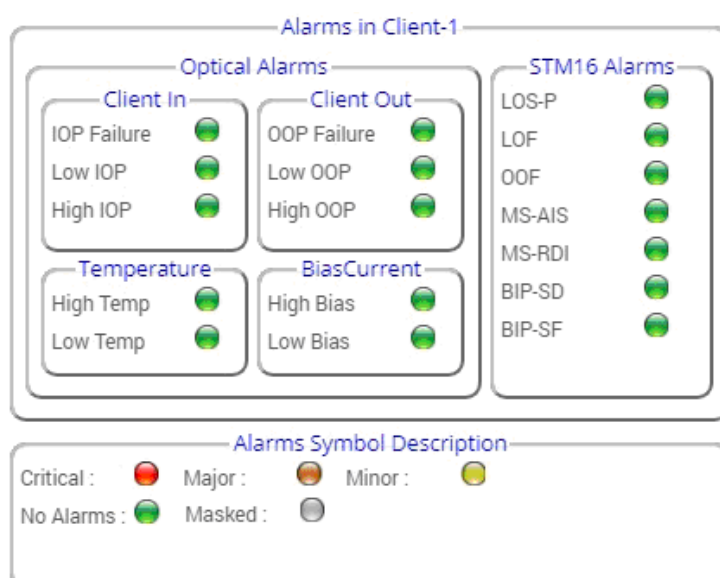
Card Reset **Warning:** The card software will reboot

7.13.2 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements. Therefore, alarms need to be obtained to monitor and manage alarms at the card, node, or network level. In the card, node, or network view, click the Alarms tab to display the alarms for that card, node, or network.

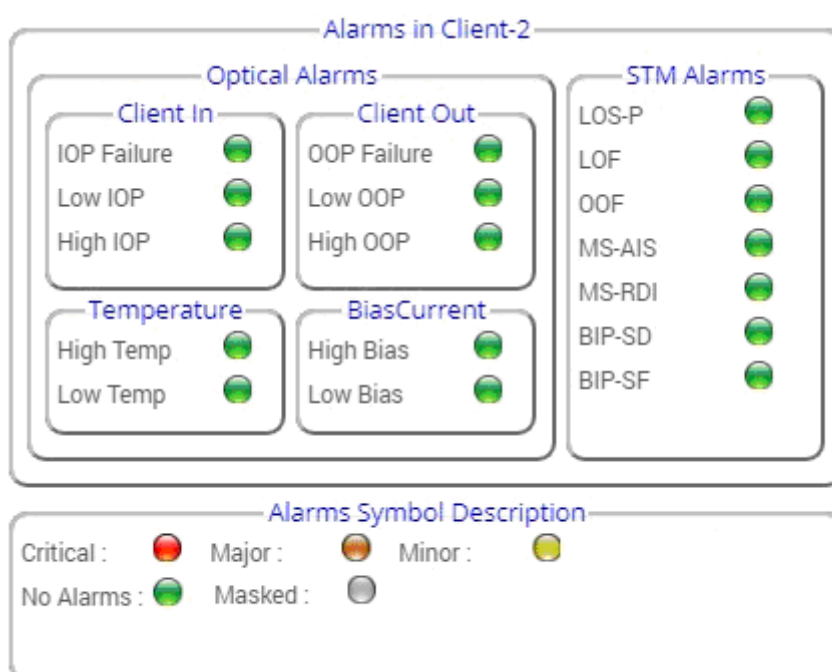
7.13.2.1 STM16 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the STM16 alarms.



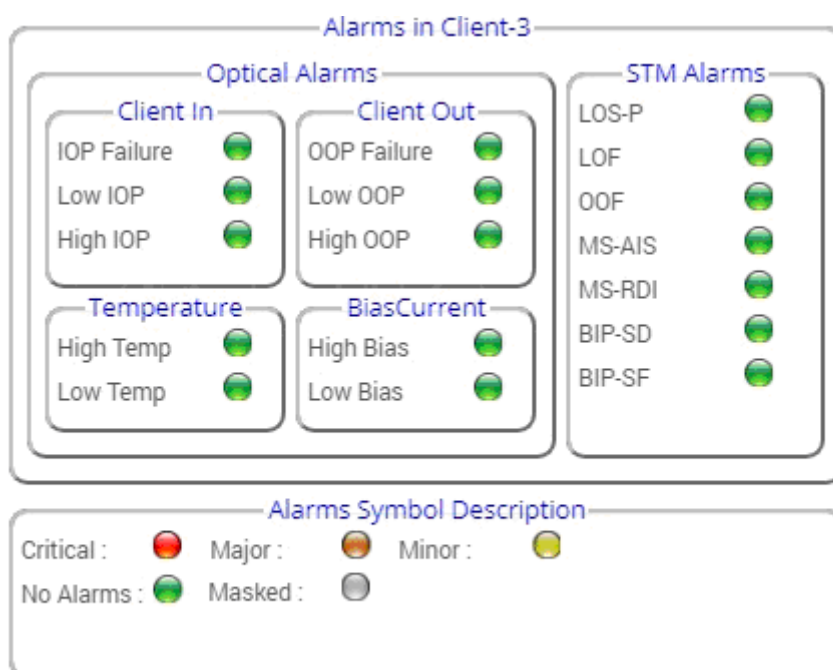
7.13.2.2 STM4 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the STM4 alarms.



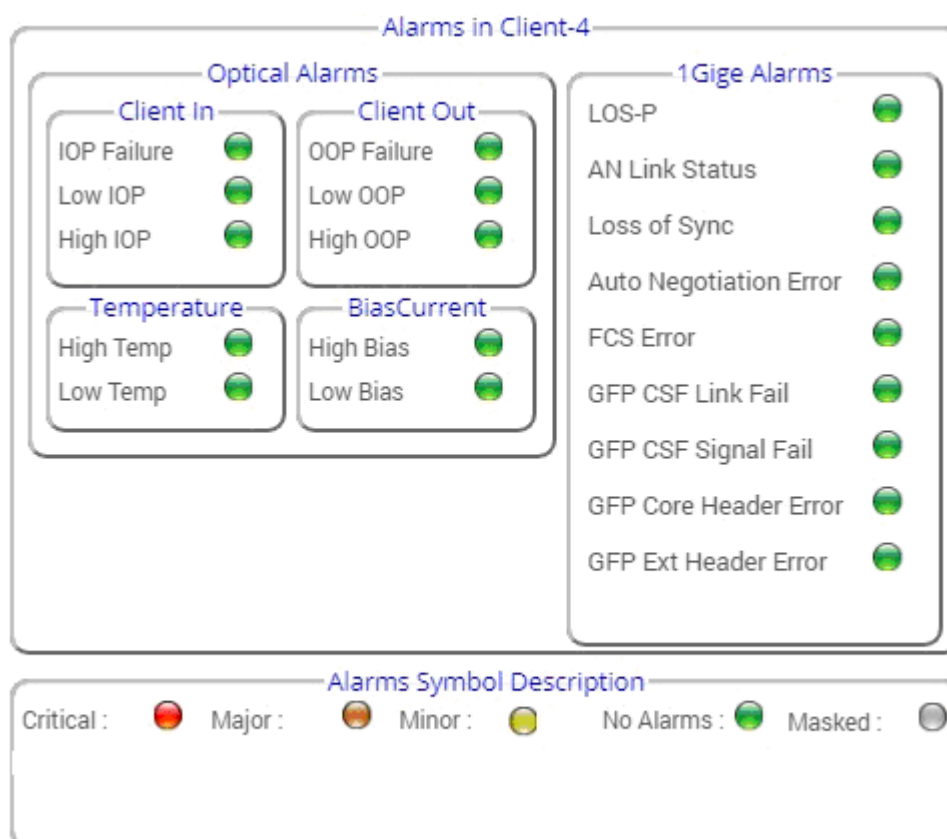
7.13.2.3 STM1 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the STM1 alarms.



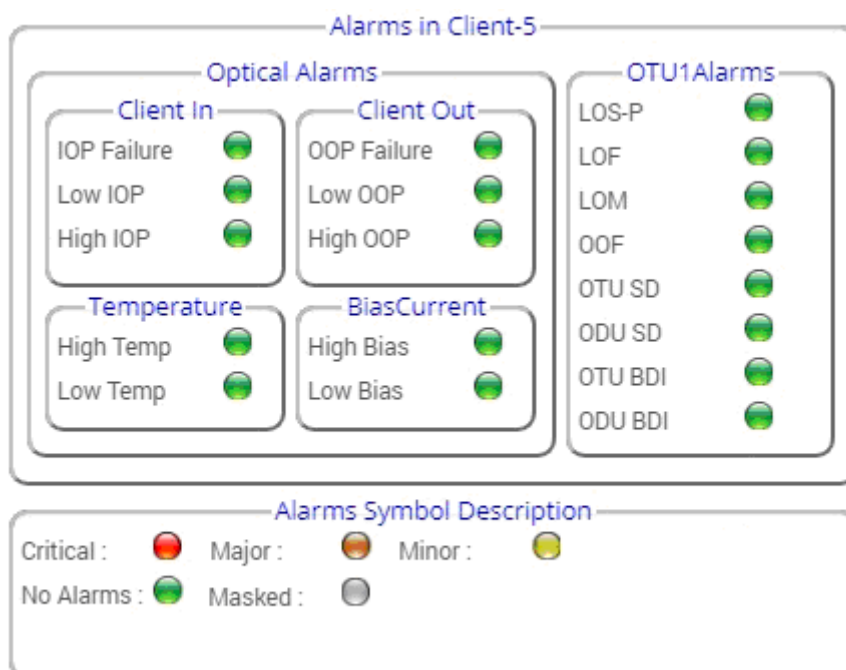
7.13.2.4 GigE Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the GigE alarms.



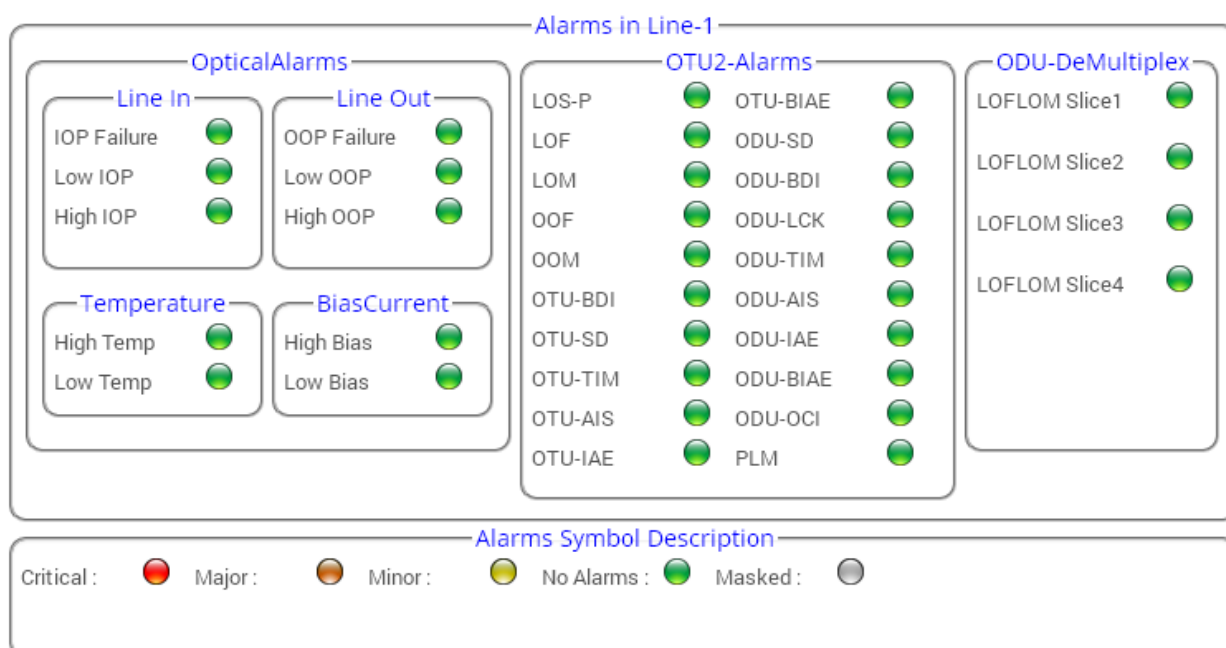
7.13.2.5 OTU1 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the OTU1 alarms.



7.13.2.6 OTU2 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the OTU2 alarms.



7.13.3 Performance

Click on **Performance** button, then performance dialog will be displayed as shown below.

7.13.3.1 STM16 Performance

This screen collects STM16 performance data of the card.

STM16-Performance in Client-1

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	713
B1 Error Count	5730302
B2 Error Count	1117408889
MS-REI Error Count	183369664
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.13.3.2 STM4 Performance

This screen collects STM4 performance data of the card.

STM-4 Performance in Client-2

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	747
B1 Error Count	17969700
B2 Error Count	275898984
MS-REI Error Count	341874828
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.13.3.3 STM1 Performance

This screen collects STM1 performance data of the card.

STM-1 Performance in Client-3

Error Seconds	0
Severely Errored Seconds	0
Background Block Errors	0
Error Free Seconds	0
Unavailable Seconds	774
B1 Error Count	37169739
B2 Error Count	62092871
MS-REI Error Count	142813601
[Retrieves counter values]	Refresh
[Re-initializes all the counters]	Re-initialize

7.13.3.4 GigE Performance

This screen collects GigE performance data of the card.

1Gige-Performance in Client-4

Frames Transmitted	0	OverSize Frames Received	0
Frames Received	0	Vlan Tagged Frames Transmitted	0
GfpExtension Header ErrorCount	0	Vlan Tagged Frames Received	0
Pause Frames Received	0	Fragment Frames Received	0
Unicast Frames Received	0	FrameCheck Seq Errors Received	0
Multicast Frames Received	0	Frames Received With NoDefect	0
Broadcast Frames Received	0	GfpCore Header ErrorCount	0
UnderSize Frames Received	0	OverSize Frames Transmitted	0
GfpFrameCount With SingleBitError	0		

[Re-initializes all the counters]

7.13.3.5 OTU1 Performance

This screen collects OTU1 performance data of the card.

OTU1-Performance in Client-5

Error Seconds	0	Ones Corrected Errors	35
Severe Error Seconds	0	Uncorrected Words	1088545236
Background Block Errors	0	Section Monitoring BIP	17008520
Error Free Seconds	0	Path Monitoring BIP	17008519
Unavailable Seconds	833	SM Backward Error Counter	1
TCM-1 Counter	17008519	PM Backward Error Counter	0
TCM-2 Counter	17008519	Corrected Bit Counter	70
Zeros Corrected Errors	35		

[Retrieves counter values] **Refresh** [Re-initializes all the counters] **Re-initialize**

7.13.3.6 Line Performance

I. Line-1 OTU Performance

This screen collects Line-1 OTU performance data of the card.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Background Block Errors	0	Background Block Errors	0
Error Free Seconds	0	Error Free Seconds	0
Unavailable Seconds	492	Unavailable Seconds	492
Bit Interleaved Parity	32243220	Backward Error Indication	27245718
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0

Refresh

[Re-initializes all the counters] **Re-initialize**

II. Line-1 ODU Performance

This screen collects Line-1 ODU performance data of the card.

ODU-Near End-Performance		ODU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Background Block Errors	0	Background Block Errors	0
Error Free Seconds	0	Error Free Seconds	0
Unavailable Seconds	546	Unavailable Seconds	546
Bit Interleaved Parity	35782110	Backward Error Indication	16429744
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Errors	35782110		
TCM-2 BIP Errors	35782110		

[Re-initializes all the counters]

III. Line-1 FEC Performance

This screen collects Line-1 FEC performance data of the card.

FEC-Performance	
Zeros Corrected Bits	1542399
Ones Corrected Bits	591794
Uncorrected Words	3178213538
Corrected Bits	2134193

[Re-initializes all the counters]

7.13.3.7 Optical Performance

This screen collects optical performance data of the card.

Optical Performance

Client

	Client-1	Client-2	Client-3	Client-4	Client-5	Client-6	Client-7	Client-8
Rx Power(dBm)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
Tx Power(dBm)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
TxBias (mA)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
TxTemp(Degree)	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f	sfp_absent_f
	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh	Refresh

Line1

Rx Power(dBm)	no_input_power	TxBias(mA)	43.69	TXOSNR	43.53
Tx Power(dBm)	0.34	TxTemp(Degree)	35.56	Refresh	

7.13.4 Alarm Severity

Click on **Alarm Severity** button, then alarm severity dialog will be displayed as shown below. You can assign or modify the severity of alarms. The severity levels that you can assign are: Critical, Major, and Minor.

Some alarms with default severity are displayed on EMS as follows

Loss of I/P power - Critical, Loss of signal – Critical, Loss of frame - Critical ,
Input High Power – Major, OTU/ODU Signal Degrade – Minor etc.

7.13.4.1 SDH Alarm Severity

You can assign or modify the severity of SDH alarms. The severity levels that you can assign are: Critical, Major, and Minor.

SDH-Alarms Severity

LOS	Critical ▼	MS-RDI	Critical ▼
LOF	Critical ▼	BIP-SD	Critical ▼
OOF	Critical ▼	BIP-SF	Critical ▼
MS-AIS	Critical ▼		
Refresh			

7.13.4.2 GigE Alarm Severity

You can assign or modify the severity of GigE alarms. The severity levels that you can assign are: Critical, Major, and Minor.

1GIGE-Alarms Severity

LOS-P	Critical ▼	GFP CSF Link Fail	Critical ▼
AN Link Status	Critical ▼	GFP CSF Signal Fail	Critical ▼
Loss Of Sync	Critical ▼	GFP Core Header Error	Critical ▼
Auto Negotiation Error	Critical ▼	GFP Ext Header Error	Critical ▼
FCS Error	Critical ▼		

[Refresh](#)

7.13.4.3 OTU1 Alarm Severity

You can assign or modify the severity of OTU1 alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OTN-Alarm Severity

LOS-P	Critical ▼	OTU-BIAE	Critical ▼
LOF	Critical ▼	ODU-OCI	Critical ▼
LOM	Critical ▼	ODU-SD	Critical ▼
OOF	Critical ▼	ODU-BDI	Critical ▼
OOM	Critical ▼	ODU-LCK	Critical ▼
OTU-BDI	Critical ▼	ODU-TIM	Critical ▼
OTU-SD	Critical ▼	ODU-AIS	Critical ▼
OTU-TIM	Critical ▼	ODU-IAE	Critical ▼
OTU-AIS	Critical ▼	ODU-BIAE	Critical ▼
OTU-IAE	Critical ▼	PLM	Critical ▼

[Refresh](#)

7.13.4.4 ODU-Demultiplex Alarm Severity

You can assign or modify the severity of ODU-Demultiplex alarms. The severity levels that you can assign are: Critical, Major, and Minor.

ODU-DeMultiplex

ODU LOFLOM	Critical ▼
------------	-------------------

[Refresh](#)

7.13.4.5 Optical Alarm Severity

You can assign or modify the severity of optical alarms. The severity levels that you can assign are: Critical, Major, and Minor.

Optical Alarms

Input

IOP FailureCritical ▾

Low IOPCritical ▾

High IOPCritical ▾

Output

OOP FailureCritical ▾

Low OOPCritical ▾

High OOPCritical ▾

Temperature

High TempCritical ▾

Low TempCritical ▾

Bias

High BiasCritical ▾

Low BiasCritical ▾

Refresh

- Select the severity type and click on **Apply** button.

7.14 OMTP11T

7.14.1 Configuration Settings

7.14.1.1 ALS & FEC

User can set the Laser ON/OFF, Automatic Laser Shutdown (ALS) and forward error correction (FEC) to correct bit errors in the received data using below mentioned dialog.

The screenshot shows the 'ALS & FEC' configuration page. It features a top navigation bar with tabs: 'ALS & FEC', 'Loopback', 'Threshold', 'TTI Message', 'Wavelength', and 'Card Inventory'. The 'ALS & FEC' tab is selected. Below the tabs, there are three main configuration panels:

- Laser Configuration:** Contains two sections, 'Line-1 Laser' and 'Line-2 Laser'. Each section has radio buttons for 'LASER OFF' and 'LASER ON' (selected). Below each section are 'Apply' and 'Refresh' buttons.
- ALS Configuration:** Contains two sections, 'Line-1 ALS' and 'Line-2 ALS'. Each section has radio buttons for 'ALS Disable' (selected) and 'ALS Enable'. Below each section are 'Apply' and 'Refresh' buttons.
- OTU2 FEC Configuration:** Contains two sections, 'Line-1 FEC' and 'Line-2 FEC'. Each section has radio buttons for 'No FEC', 'Normal FEC' (selected), and 'Super FEC'. Below each section are 'Apply' and 'Refresh' buttons.

7.14.1.2 Card Inventory

Click on **Settings->Card Inventory**, then card details dialog will be displayed as shown below. This screen displays the inventory information like serial number, software version and hardware version for the card in that slot. Use the soft reset button to initiate software reboot on the selected card.

The screenshot shows the 'Card Details' dialog box. It has a title bar 'Card Details'. Inside, there are three rows of information:

- Serial No.:** A text input field followed by a 'Refresh' button.
- HW Version:** A text input field followed by a 'Refresh' button.
- SW Version:** A text input field followed by a 'Refresh' button.

At the bottom, there is a 'Card Reset' section containing a 'Soft Reset' button and a warning message: *Warning: The card software will reboot*.

7.14.1.3 Loopback

Fiber optic loopback is known as the routing of electronic signals, digital data streams, or flows of items back to their source without intentional processing or modification. In terms of telecommunication, loopback is a hardware or software method to feed a received signal or data back to the sender. It is very useful for solving physical connection problems.

If you want to set the loopback, Select loopback type and click on **Apply** button.

Loopback

Line-1 Loopback

☒ No Loopback
☐ Near-End Loopback
☐ Far-End Loopback

Apply **Refresh**

Line-2 Loopback

☒ No Loopback
☐ Near-End Loopback
☐ Far-End Loopback

Apply **Refresh**

7.14.1.4 Threshold Settings

You can also configure the BER thresholds that raise or clear a signal degrade alarm and the time interval for the thresholds. If the BER thresholds and interval are not configured, the default values are used.

Threshold

Line-1 Threshold Settings

OTU SD: 1e-7 **Apply** **Refresh**
ODU SD: 1e-7 **Apply** **Refresh**

Line-2 Performance Threshold Settings

OTU SD: 1e-7 **Apply** **Refresh**
ODU SD: 1e-7 **Apply** **Refresh**

7.14.1.5 TTI Message

TTI Message

Line1 OTU TTI Message Settings

☒ OTU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

Line2 OTU TTI Message Settings

☒ OTU TTI

	SAPI	DAPI	Operator
OTU Transmit	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear
OTU Expected	UTL Apply Refresh Clear	UTL Apply Refresh Clear	UTL Apply Refresh Clear

7.14.1.6 Wavelength

ALS & FEC

Loopback

Threshold

TTI Message

Wavelength

Card Inventory

☒ 40 Channel

☐ 80 Channel

40 Channel Wavelength Selection

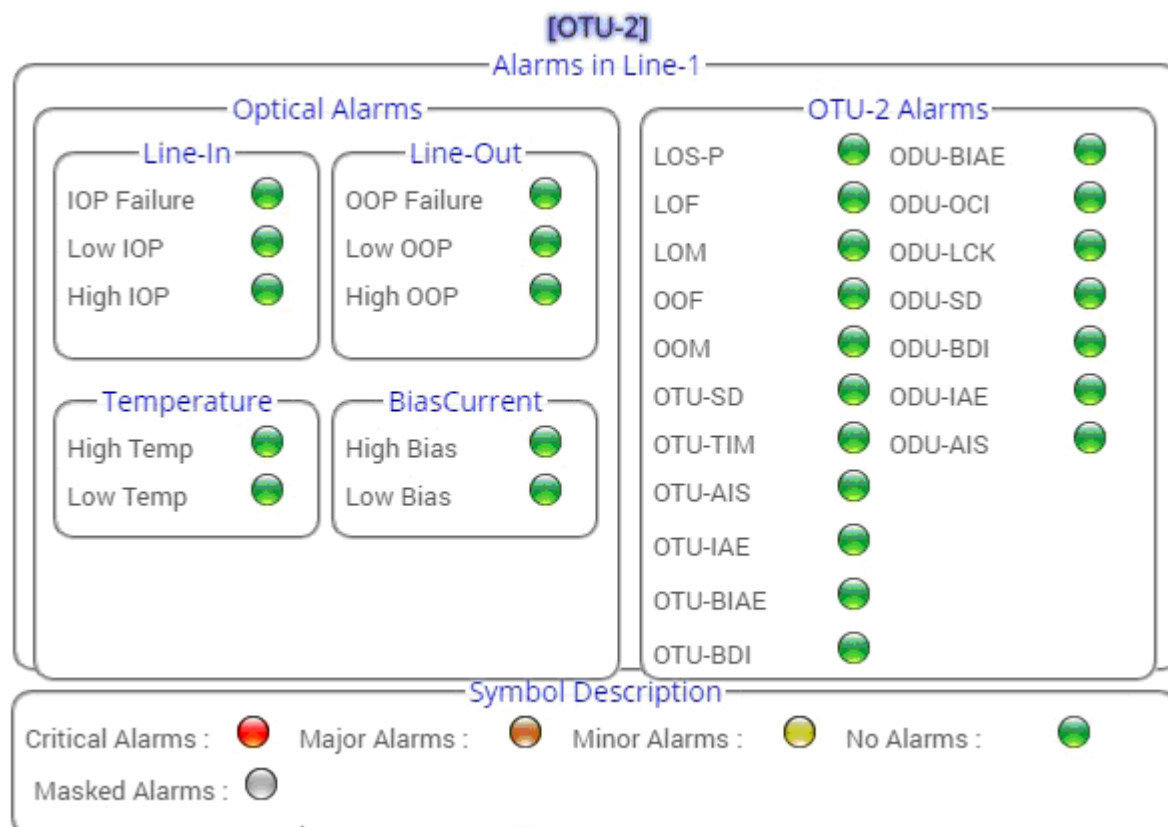
Line-1 Channel	1	Wavelength	1530.33 Apply Refresh
Line-2 Channel	1	Wavelength	1530.33 Apply Refresh

7.14.2 Alarms

UTL MP-0072 Spectrum EMS detects and reports alarms generated by the UTL DWDM MP-0072 network elements. Therefore, alarms need to be obtained to monitor and manage alarms at the card, node, or network level. In the card, node, or network view, click the Alarms tab to display the alarms for that card, node, or network.

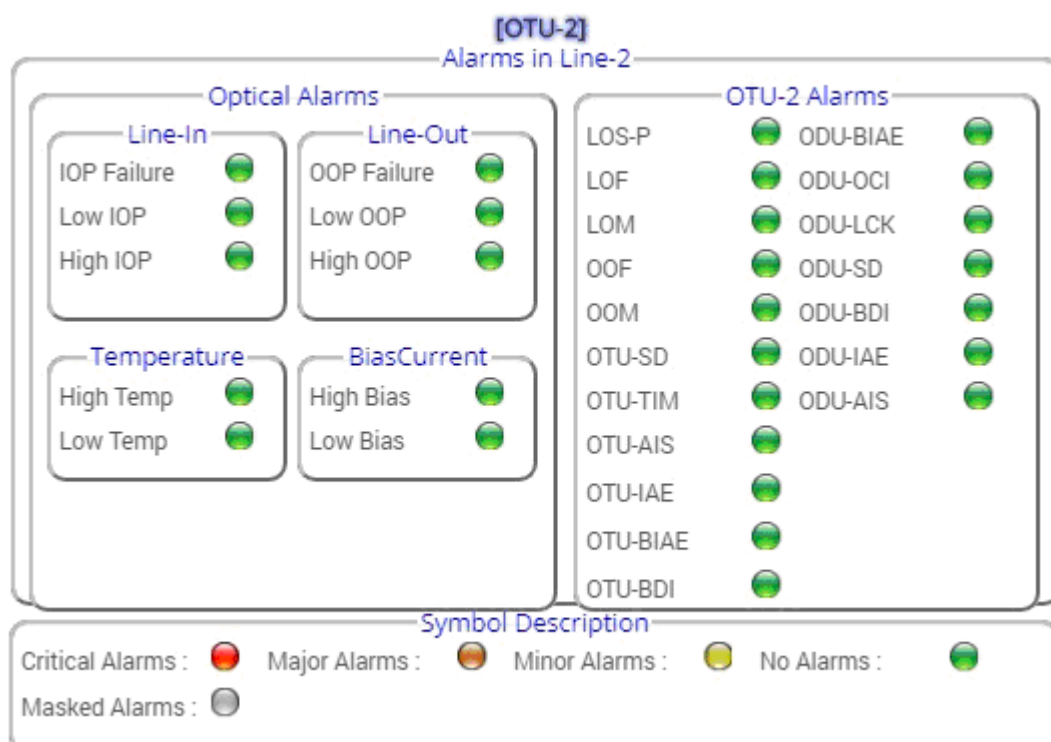
7.14.2.1 Line1 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the Line1 alarms.



7.14.2.2 Line2 Alarms

The screen uses evident colour changes on the GUI to help you quickly locate the Line2 alarms.



7.14.3 Performance

Click on Performance button, then performance dialog is displayed as shown below. UTL DWDM products contains fiber optic components, including fiber optic transmitters and receivers.

These components are extremely susceptible to contamination by particles of dirt or dust, which can obstruct the optic path and cause performance degradation. To ensure optimum product performance, it is important that all optics and connector ferrules be kept clean.

7.14.3.1 Optical Performance

This screen collects optical performance data of the card.

Optical Performance

	Line-1	Line-2
Rx Power(dBm)	no_input_power	no_input_power
Tx Power(dBm)	0.60	0.61
TxBias(mA)	25.95	22.35
TxTemp(Degree)	52.06	50.98
TxOSNR(dB)	43.78	43.79

[Refresh](#)

7.14.3.2 OTU2 Performance

This screen collects OTU2 performance data of the card.

[OTU-2]

OTU
ODU
FEC

OTU-Near End-Performance

Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	0
Background Block Errors	0
Unavailable Seconds	1627
Bit Interleaved Parity	0
Incoming Alignment Error Seconds	0

OTU-Far End-Performance

Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	1627
Background Block Errors	0
Unavailable Seconds	0
Backward Error Indication	0
Backward Incoming Alignment Error Seconds	0

[Refresh](#)
 [Re-initializes all the Line port counters] [Re-initialize](#)

7.14.3.3 Line-1 OTU Performance

This screen collects Line-1 OTU performance data of the card.

OTU-Near End-Performance		OTU-Far End-Performance	
Error Seconds	0	Error Seconds	0
Severe Error Seconds	0	Severe Error Seconds	0
Error Free Seconds	0	Error Free Seconds	158602
Background Block Errors	0	Background Block Errors	0
Unavailable Seconds	158602	Unavailable Seconds	0
Bit Interleaved Parity	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0

[Re-initializes all the Line port counters]

Refresh Re-initialize

7.14.3.4 Line-1 ODU Performance

This screen collects Line-1 ODU performance data of the card.

ODU-Near End-Performance		ODU-Far End-Performance	
Bit Interleaved Parity	0	Error Seconds	0
Error Seconds	0	Severe Error Seconds	0
Severe Error Seconds	0	Error Free Seconds	158656
Error Free Seconds	158656	Background Block Errors	0
Background Block Errors	0	Unavailable Seconds	0
Unavailable Seconds	0	Backward Error Indication	0
Incoming Alignment Error Seconds	0	Backward Incoming Alignment Error Seconds	0
TCM-1 BIP Error	0		
TCM-2 BIP Error	0		

[Re-initializes all the Line port counters]

Refresh Re-initialize

7.14.3.5 Line-1 FEC Performance

This screen collects Line-1 FEC performance data of the card.

FEC-Performance

Zeros Corrected Bits	0
Ones Corrected Bits	0
Uncorrected Words	0
Corrected Bits	0

[Refresh](#)

[Re-initializes all the Line port counters] [Re-initialize](#)

7.14.3.6 Line-2 OTU Performance

This screen collects Line-2 OTU performance data of the card.

OTU-Near End-Performance

Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	0
Background Block Errors	0
Unavailable Seconds	158725
Bit Interleaved Parity	0
Incoming Alignment Error Seconds	0

OTU-Far End-Performance

Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	158725
Background Block Errors	0
Unavailable Seconds	0
Backward Error Indication	0
Backward Incoming Alignment Error Seconds	0

[Re-initializes all the Line port counters] [Refresh](#)
[Re-initialize](#)

7.14.3.7 Line-2 ODU Performance

This screen collects Line-2 ODU performance data of the card.

ODU-Near End-Performance

Bit Interleaved Parity	0
Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	158757
Background Block Errors	0
Unavailable Seconds	0
Incoming Alignment Error Seconds	0
TCM-1 BIP Error	0
TCM-2 BIP Error	0

ODU-Far End-Performance

Error Seconds	0
Severe Error Seconds	0
Error Free Seconds	158757
Background Block Errors	0
Unavailable Seconds	0
Backward Error Indication	0
Backward Incoming Alignment Error Seconds	0

[Re-initializes all the Line port counters] [Refresh](#)
[Re-initialize](#)

7.14.3.8 Line2 FEC Performance

This screen collects Line-2 FEC performance data of the card.

FEC-Performance

Zeros Corrected Bits	0
Ones Corrected Bits	0
Uncorrected Words	0
Corrected Bits	0

Refresh

[Re-initializes all the Line port counters] **Re-initialize**

7.14.4 Alarm Severity

You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Some alarms with default severity are displayed on EMS as follows

Loss of I/P power - Critical, Loss of signal – Critical, Loss of frame - Critical ,
Input High Power – Major, OTU/ODU Signal Degrade – Minor etc.

7.14.4.1 OTU Alarm Severity

You can assign or modify the severity of OTU alarms. The severity levels that you can assign are: Critical, Major, and Minor.

OTN Alarms Severity

LOS-P	Critical ▼	OTU-BIAE	Minor ▼
LOF	Critical ▼	ODU-OCI	Minor ▼
LOM	Major ▼	ODU-SD	Minor ▼
OOF	Major ▼	ODU-BDI	Minor ▼
OOM	Major ▼	ODU-LCK	Minor ▼
OTU-BDI	Minor ▼	ODU-BIAE	Minor ▼
OTU-SD	Minor ▼	ODU-AIS	Minor ▼
OTU-TIM	Major ▼	ODU-IAE	Minor ▼
OTU-AIS	Minor ▼		
OTU-IAE	Minor ▼		

Refresh

Data Retrieved Successfully..

7.14.4.2 Optical Alarm Severity

You can assign or modify the severity of optical alarms. The severity levels that you can assign are: Critical, Major, and Minor.

Optical Alarms	
Input	
IOP Failure	Critical ▼
Low IOP	Critical ▼
High IOP	Critical ▼
Output	
OOP Failure	Critical ▼
Low OOP	Critical ▼
High OOP	Critical ▼
Temperature	
High Temp	Critical ▼
Low Temp	Critical ▼
Bias	
High Bias	Critical ▼
Low Bias	Critical ▼
Refresh	

7.15 OLA

7.15.1 Alarms

By viewing alarms, you can learn network and NE status, and locate faults. Therefore, alarms need to be obtained and managed effectively. The screen uses evident colour changes on the GUI to help you quickly locate the amplifier alarms.

Stage-1 Alarms

IOP Failure		[Below -30.0 dBm]	Apply	
Degraded IOP		[Below -8.0 dBm]	Apply	
Low IOP		[Below -10.0 dBm]	Apply	
IOP Out of Range		[Below -10.0 dBm]	Apply	or Above 3.0 dBm Apply
OOP Out of Range		[Below 8.0 dBm]	Apply	or Above 16.0 dBm Apply
OOP Failure		[Below 5.0 dBm]	Apply	

Stage-2 Alarms

IOP Failure		[Below -30.0 dBm]	Apply	
Degraded IOP		[Below -12.0 dBm]	Apply	
Low IOP		[Below -15.0 dBm]	Apply	
IOP Out of Range		[Below -15.0 dBm]	Apply	or Above -2.0 dBm Apply
Degraded OOP		[Below 5.0 dBm]	Apply	
OOP Out of Range		[Below 3.0 dBm]	Apply	or Above 11.0 dBm Apply
OOP Failure		[Below 0.0 dBm]	Apply	

EDFA High Temperature  [Above +80 Degree]

[Threshold Refresh](#)

[Threshold Calculator](#)

Alarms Symbol Description

Critical : 
 Major : 
 Minor : 
 No Alarm : 

7.15.2 Settings

In this dialog, you can modify Gain, LD and TILT of the amplifiers.

Card Details

Serial No: [Refresh](#)

HW Version: [Refresh](#)

SW Version: [Refresh](#)

Card Reset: [Soft Reset](#)

TILT Settings

TILT: [Apply](#) [Refresh](#)

[Value range: -3 db to +1 db]

Operation Mode Settings

Operation Mode: [Apply](#) [Refresh](#)

LD Settings

LD Status: [Apply](#) [Refresh](#)

OLA Module Type

OLA Type: [Refresh](#)

Gain Settings

Gain[Stage-1]: db [Apply](#) [Refresh](#)

Gain[Stage-2]: db [Refresh](#)

EVOA Settings

EVOA Settings: db [Apply](#) [Refresh](#)

7.15.3 Performance

This screen collects optical performance data of the card.

Optical Performance

Input(dBm)	no_input_power	Mid-In(dBm)	-57.00
Mid-Out(dBm)	-57.00	Output(dBm)	no_output_power
Gain[Stage-1]	11.00	Gain[Stage-2]	24.00
Mode[Stage-1]	AGC	Mode[Stage-2]	AGC
EDFA Temp	32.50		

Refresh

7.15.4 Alarm Severity

You can assign or modify the severity of an alarm. The severity levels that you can assign are: Critical, Major, and Minor.

Stage-1 Alarm Severity

IOP Failure	Critical ▼
Degraded IOP	Major ▼
Low IOP	Major ▼
IOP Out of Range	Major ▼
OOP Out of Range	Major ▼
OOP Failure	Critical ▼

Stage-2 Alarm Severity

IOP Failure	Critical ▼
Degraded IOP	Major ▼
Low IOP	Major ▼
IOP Out of Range	Major ▼
Degraded OOP	Major ▼
OOP Out of Range	Major ▼
OOP Failure	Critical ▼

Temperature Alarm Severity

EDFA High Temp	Major ▼
----------------	----------------

Refresh

Data Retrieved Successfully..

7.16 OPS

OPS (Optical Protection Switch), The optical Protection Switch Module (OPS) provides protection against a fiber cut by optically splitting an optical signal over two diverse paths. Under DWDM Node we have three popup menu items.



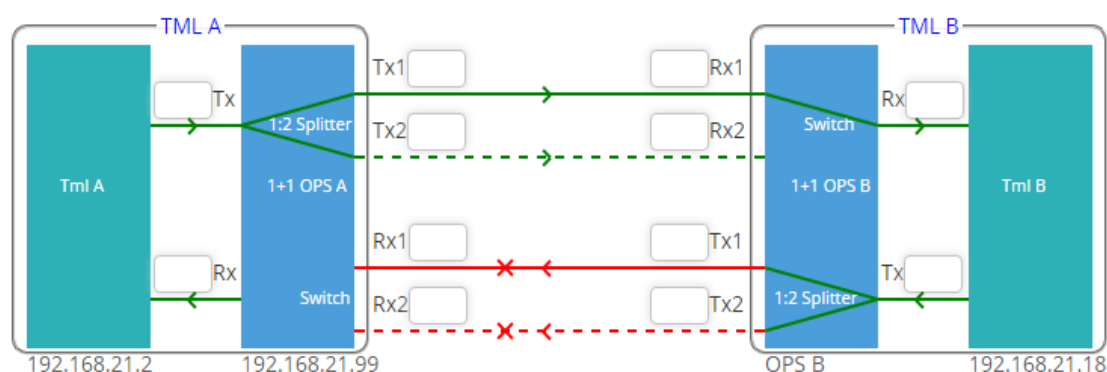
OPS having the maximum of 4 OLP, if you are select any OPL as shown in below.



Selected OLP 1 Details

Alarm Settings Performance

Link Status Of OPS A And OPS B

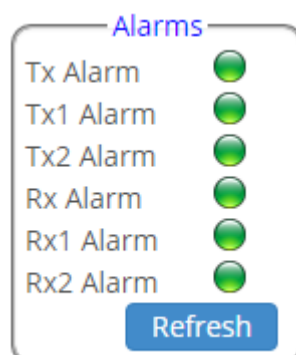


Refresh

1. Alarms
2. Settings
3. Performance

7.16.1 Alarms

Click on **Alarms** button, then alarms dialog as shown in below.



It will automatically refresh every 10 sec.

7.16.2 Settings

Click on **Settings** button, then settings dialog as shown in below.

OLP Details

Type		
Serial Number		
FirmWare Version		
Mode		Apply
Link		Apply
Return Time		Apply
Switch Delay		Apply
Refresh		

Threshold

Tx Alarm		Apply	Rx Alarm		Apply
Tx1 Alarm		Apply	Rx1 Alarm		Apply
Tx2 Alarm		Apply	Rx2 Alarm		Apply
Refresh					

If you want to apply any value

- Select required value and click on **Apply** button.

7.16.3 Performance

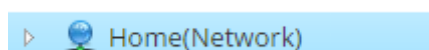
Click on **Performance** button, then performance dialog as shown in below. This screen collects optical performance data of the card. To ensure optimum product performance, it is important that all optics and connector ferrules be kept clean.

Performance in dBm

Tx Power	
Tx1 Power	
Tx2 Power	
Rx Power	
Rx1 Power	
Rx2 Power	
Refresh	

8. Tree Management

The basic tree is shown below.



The tree contains one root that is Network. This is the default root.

9.1 Zone

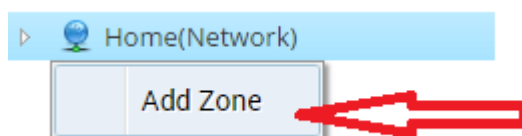
Under zone we have two Popup menu items.

- Add Zone

- Remove Zone

8.1.1 Add Zone

Right click on Network tree node, then the Add Zone popup will be displayed as shown below:



Click on the **Add Zone** popup menu option, then Add zone dialog will be displayed as shown below.

A screenshot of the 'Add Zone' dialog box. It has a blue header bar with the text 'Add Zone'. Below the header, there is a label 'Zone Name:' followed by a dropdown menu currently showing 'Select'. At the bottom right, there are two buttons: 'Add' and 'Close'.

From the above drop down list, select a zone name and click **Add** button. After clicking the add button, a confirmation prompt will be displayed as shown below.

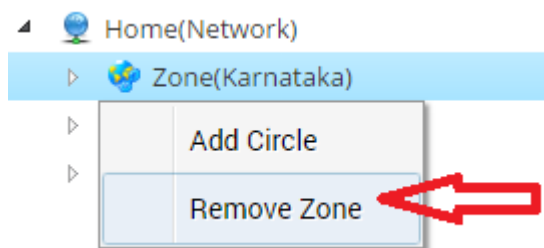
A screenshot of a confirmation dialog box. It has a blue header bar with the text 'Add Zone' and a close button 'x'. The main area contains the text 'Do you want to add zone ?'. At the bottom right, there are two buttons: 'Yes' and 'No'.

Click add button to add new zone, and then response dialog will be displayed as shown below.

A screenshot of the 'Add Zone' dialog box after a successful operation. It has a blue header bar with the text 'Add Zone'. Below the header, there is a label 'Zone Name:' followed by a dropdown menu currently showing 'Select'. Below the dropdown, there is a blue text message: 'Zone Added Successfully.'. At the bottom right, there are two buttons: 'Add' and 'Close'.

8.1.2 Remove Zone

Right click on already existing Zone, then the Remove Zone popup will be displayed as shown below:



Click on the **Remove Zone** menu option, then Remove zone dialog will be displayed as shown below.

A screenshot of a 'Remove Zone' dialog box. The title bar is blue with the text 'Remove Zone'. Below the title bar, there is a label 'Zone Name:' followed by a dropdown menu that currently shows 'Karnataka'. At the bottom of the dialog, there are two buttons: 'Remove' and 'Close'.

From the above drop down list, select a zone name and click **Remove** button. After clicking the remove button, a confirmation prompt will be displayed as shown below:

A screenshot of a confirmation dialog box titled 'Remove Zone'. The main text inside the dialog asks 'Do you want to remove zone:Karnataka from network?'. At the bottom right, there are two buttons: 'Yes' and 'No'.

Click **Yes** button to Remove zone, and then response dialog will be displayed as shown below.

A screenshot of the 'Remove Zone' dialog box after a successful operation. The title bar is 'Remove Zone'. The 'Zone Name:' dropdown menu now shows 'Select'. Below this, the text 'Zone Removed Successfully.' is displayed in blue. At the bottom, there are 'Remove' and 'Close' buttons.

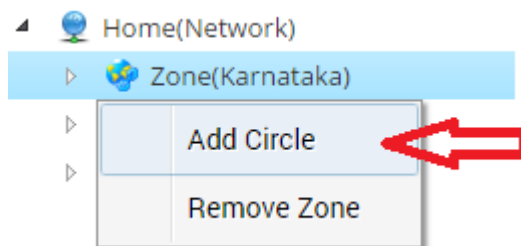
9.2 Circle

Under circle we have two popup menu items.

- Add Circle
- Remove Circle

8.2.1 Add Circle

Right click on already added Zone, then the Add circle popup will be displayed as shown below:



Click on the **Add Circle** menu option, then Add circle dialog will be displayed as below.

Add Circle

Zone Name:

Circle Name:

Topology Type:

- Enter circle name.
- Select topology type (Linear or Ring or Mesh).

Click **Add Circle** button. After clicking the add button, a confirmation prompt will be displayed as below.

Add Circle x

Do you want to add circle?

Click **Yes** button to add new circle, and then response dialog will be displayed as shown below.

Add Circle

Zone Name:

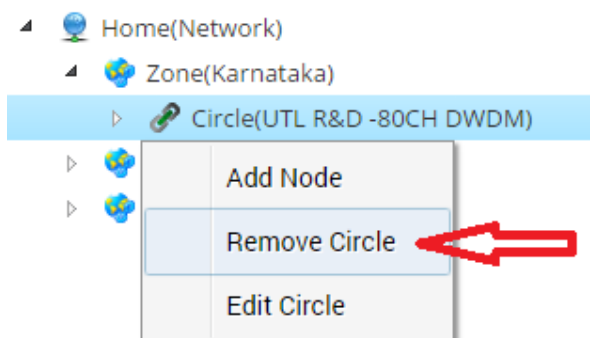
Circle Name:

Topology Type:

circle added successfully !

8.2.2 Remove Circle

Right click on already existing circle, then Remove Zone popup will be displayed as shown below:



Click on the **Remove Circle** menu option, then Remove circle dialog will be displayed as shown below.

Remove Circle

Zone Name:

Circle Name:

Click **Remove** button. After clicking the remove button, a confirmation prompt is displayed as below.

Remove Circle ✕

Do you want to remove circle?

Click **Yes** button to Remove circle, then response dialog will be displayed shown below:

Remove Circle

Zone Name:

Circle Name:

circle removed successfully !

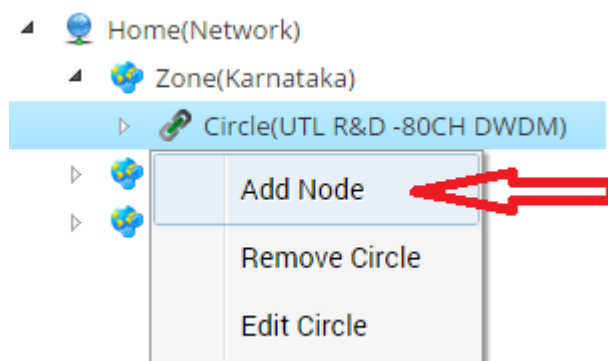
9.3 Node

Under Node we have three popup menu items.

- Add Node
- Edit Node
- Remove Node

8.3.1 Add Node

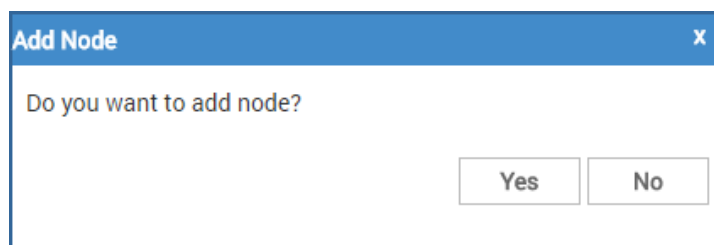
Right click on already existing circle, then the Add Node popup will be displayed as shown below:



Click on the **Add Node** menu option, then Add Node dialog is displayed as shown below:

- Select required zone name.
- Select required circle name.
- Select DWDM type (Terminal or ROADM or Amplifier or Regenerator).
- Select option Medium Node or Small Node (For ROADM type only)
- Select adjacent NE, if there (for Linear Topology only).
- Enter IP Address.
- Enter description (minimum 1, maximum 20).
- Enter name (minimum 1, maximum 20).
- Enter contact person name (minimum 1, maximum 20).
- Enter contact number (must be 10 digits).

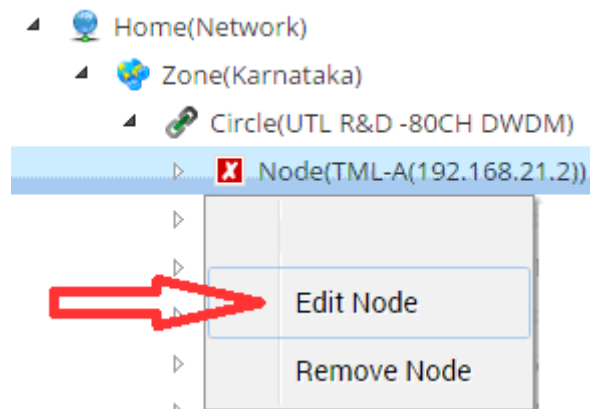
Click **Add Node** button. After clicking the add button, a confirmation prompt is displayed as below:



Click **Yes** button to add new node.

8.3.2 Edit Node

Right click on already existing node, then the Edit node popup is displayed as shown below:



Click on the **Edit Node** menu option, then Edit Node dialog is displayed as shown below:

Edit/Deconfigure Node

Zone Name:	India
Circle Name:	South
IP Address:	192.168.1.1
DWDM Type:	TERMINAL
Description:	TMLA
Location Name:	LAB
Contact Person:	SDE
Contact Number:	1111111111

De-Configure

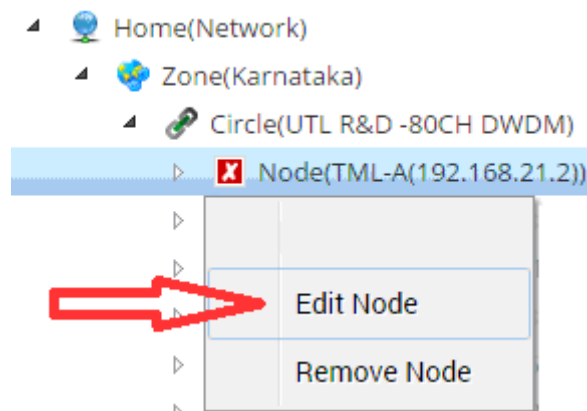
Modify

Close

1. De-Configure.
2. Modify.

8.3.2.1 De-Configure

Right click on already existing node, then the Edit node popup is displayed as shown below:



Edit node de-configure dialog is displayed as shown below.

Edit/Deconfigure Node

Zone Name:	India
Circle Name:	South
IP Address:	192.168.1.1
DWDM Type:	TERMINAL
Description:	TMLA
Location Name:	LAB
Contact Person:	SDE
Contact Number:	1111111111

De-Configure

Modify

Close

Click on **De-Configure** button. After clicking the de-configure button, a confirmation prompt is displayed as below:

Node De-Configuration x

Do you want to de-configure node?

Yes

No

Click **Yes** button to de-configure node.

Edit Node

Zone Name: [🚩]	Select ▼
Circle Name: [🚩]	Select ▼
IP Address: [🚩]	Select ▼
Node Type: [🚩]	DWDM ▼
Description: [🚩]	
Location Name: [🚩]	
Contact Person: [🚩]	
Contact Number: [🚩]	

Node De-Configured successfully.

De-Configure

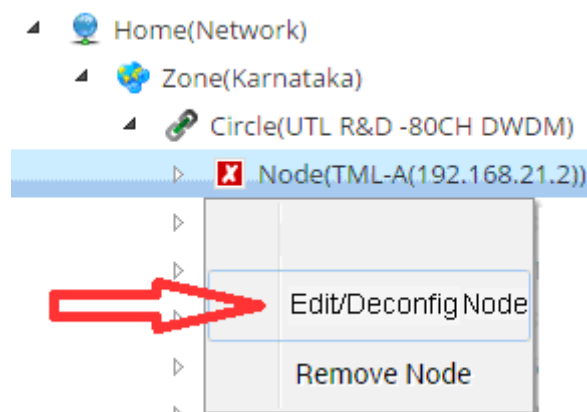
Configure

Modify

Close

8.3.2.2 Modify

Right click on already existing node, then the Edit node popup will be displayed as below:



Edit/Deconfigure Node

Zone Name:	India
Circle Name:	South
IP Address:	192.168.1.1
DWDM Type:	TERMINAL
Description:	TMLA
Location Name:	LAB
Contact Person:	SDE
Contact Number:	1111111111

De-Configure

Modify

Close

Click on **Modify** button. After clicking the modify button, a confirmation prompt is displayed as below.

Modify Node x

Do you want to modify node?

Yes

No

Click **Yes** button to modify node.

Edit Node

Zone Name:	Select
Circle Name:	Select
IP Address:	Select
Node Type:	DWDM
Description:	
Location Name:	
Contact Person:	
Contact Number:	

Node Modified Successfully

De-Configure

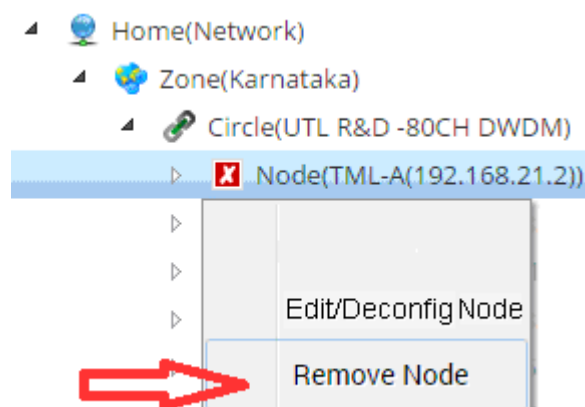
Configure

Modify

Close

8.3.3 Remove Node

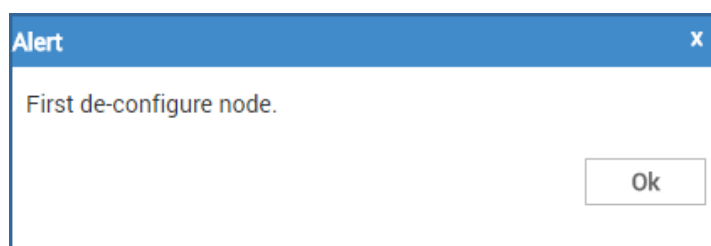
Right click on already existing node, then the **Remove Node** popup will be displayed as shown below:



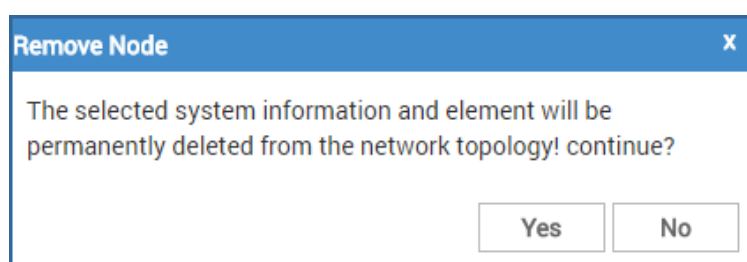
Click on the **Remove Node** menu option, then the below Remove Node dialog will be displayed.

- Select zone name.
- Select circle name.
- Select node IP to remove node (node IP must be 0.0.0.0), otherwise de-configure the node first and then only it will allow to delete.

If node is not de-configured, the below alert is displayed.



After de-configuring node, click **Remove Node** button. After clicking the remove button, a confirmation prompt is displayed as below.

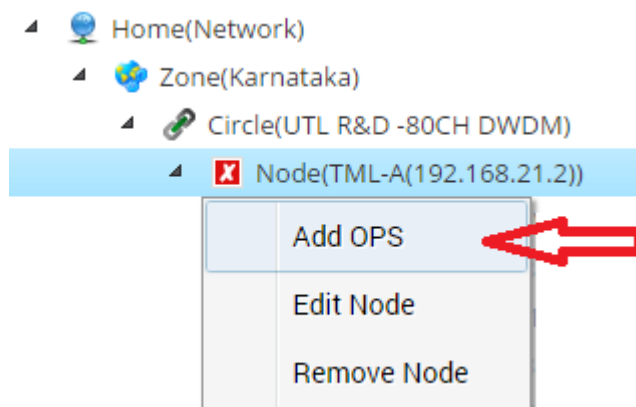


Click **Yes** button to Remove node.

8.3.4 Add OPS (Optical Protection Switch)

The optical Protection Switch Module (OPS) provides protection against a fiber cut by optically splitting an optical signal over two diverse paths.

Right click on already existing node, then the Add OPS popup as shown in below



Click on the **Add OPS** menu option, then Add OPS dialog as shown in below.

The 'Add OPS' dialog box contains the following elements:

- Add OPS** (Title bar)
- IP Address:
- Description:
- Add** button
- Close** button

- Enter IP Address.
- Enter description (minimum 1, maximum 20)

Click **Add** button. After clicking the add button and a confirmation prompt is displayed.

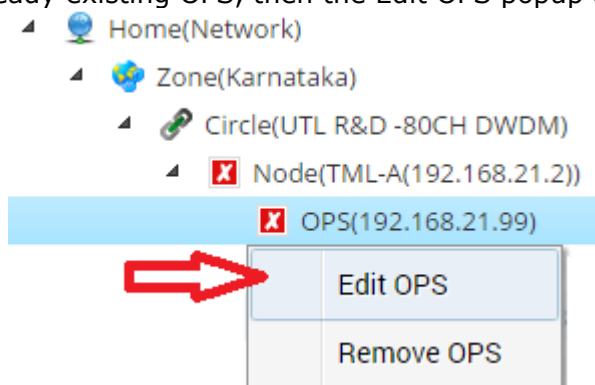
The confirmation dialog box contains the following elements:

- Add OPS** (Title bar)
- Do you want to add optical protection switch?
- Yes** button
- No** button

Click **Yes** button to add OPS.

8.3.5 Edit OPS

Right click on already existing OPS, then the Edit OPS popup as shown in below

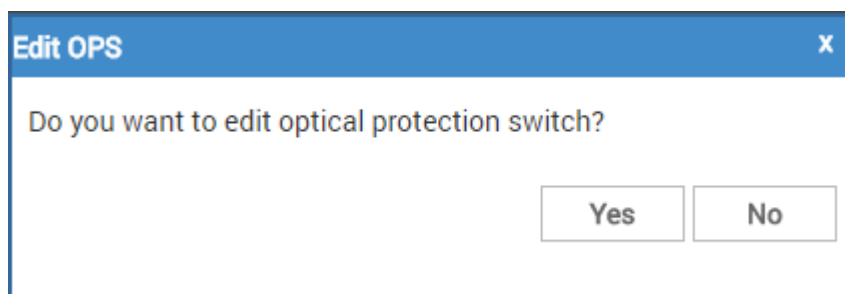


Click on the **Edit OPS** menu option, then Edit OPS dialog as shown in below.

The image shows a dialog box titled "Edit OPS". It has a blue header bar with the title. Below the header, there are two input fields: "IP Address:" followed by a text box containing three dots, and "Description:" followed by an empty text box. At the bottom right of the dialog, there are two buttons: "Edit" and "Close", both in blue.

- Enter IP Address.
- Enter description (minimum 1, maximum 20)

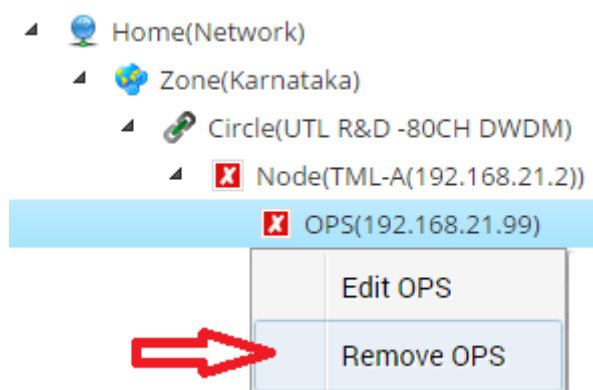
Click **Edit** button. After clicking the add button and a confirmation prompt is displayed.

The image shows a confirmation dialog box titled "Edit OPS" with a close button (X) in the top right corner. The main text of the dialog asks "Do you want to edit optical protection switch?". At the bottom right, there are two buttons: "Yes" and "No", both in white with blue borders.

Click **Yes** button to update OPS.

8.3.6 Remove OPS

Right click on already existing OPS, then the Remove OPS popup as shown in below.



Click on the **Remove OPS** menu option, then Remove OPS dialog as shown in below.

Remove OPS

IP Address:

Description:

Remove **Close**

Click **Remove** button. After clicking the add button and a confirmation prompt is displayed.

Remove OPS x

Do you want to remove optical protection switch?

Yes **No**

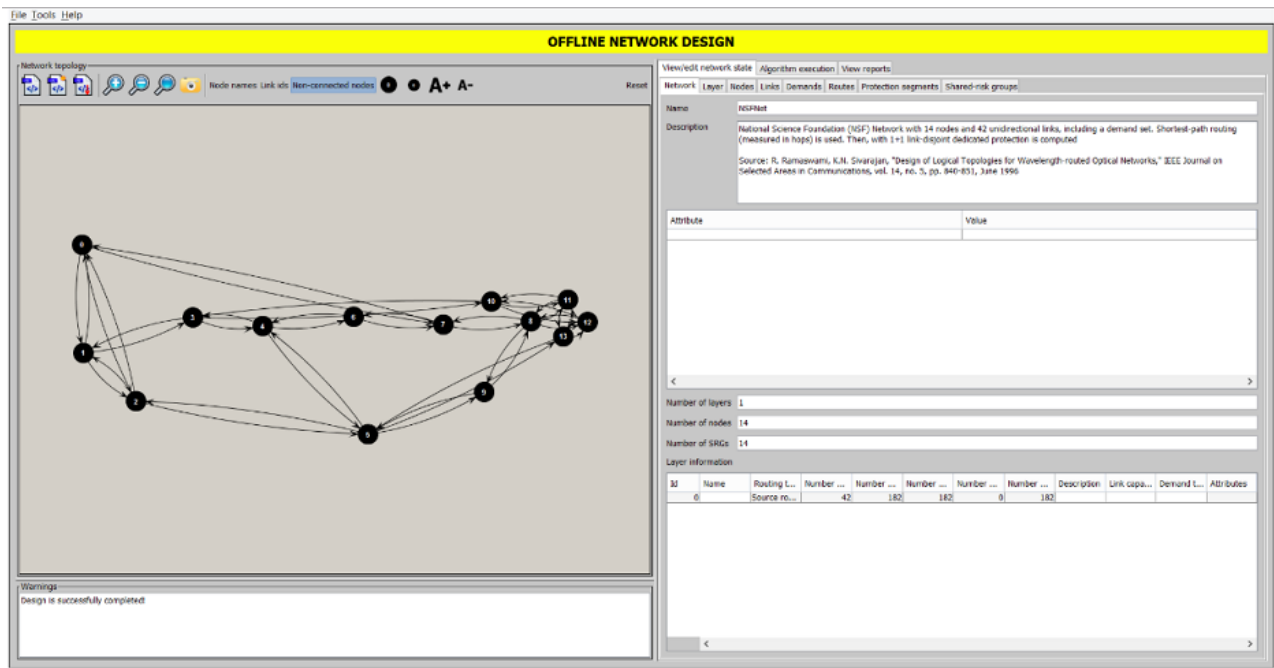
Click **Yes** button to Remove OPS.

9. Network Planning Tool

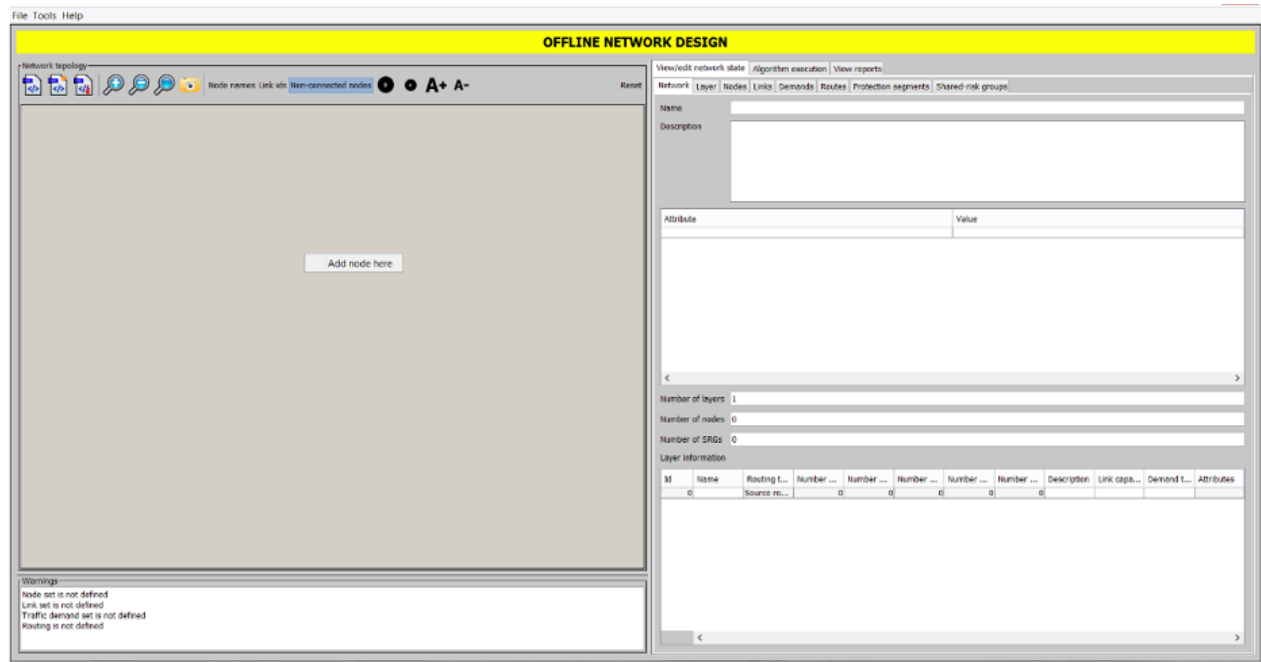
The Network Planning Tool is a standalone tool which assists users in the process of offline network design and planning. In practical network design different variables can be involved: network nodes, network links, link capacities, the traffic routing, the traffic demand etc.

Usually, offline network design problems receive some of this information as input parameters (e.g. traffic demand, and network topology) and try to optimize others (e.g. capacities in the links and traffic routing).

In the UTL Network Planning Tool, network designs can be completed incrementally, modifying the current network design, that the user can inspect visually in the “Network Topology” panel, and the “Edit network plan” tab. When the offline design tool starts, the current network design is empty. Then, the user can modify it (i) using manual editions in the “Network Topology” panel or the “View/edit network state” tab, (ii) loading existing designs and/or traffics demands, (iii) applying offline design algorithms (that take the current network design, and return a modified design which becomes the current design).



9.1 Inserting a node.



The screenshot displays the 'OFFLINE NETWORK DESIGN' application. The central canvas shows a network topology with two nodes, Node 0 and Node 1. A context menu is open for Node 0, providing options to 'Remove node', 'Create unidirectional link', or 'Create bidirectional link'. The right-hand sidebar features a 'View/edit network state' panel with multiple tabs: 'Network', 'Layer', 'Nodes', 'Links', 'Demands', 'Routes', 'Protection segments', and 'Shared risk groups'. The 'Network' tab is active, showing input fields for 'Name' and 'Description', and a table with 'Attribute' and 'Value' columns. Below this, the 'Layer information' section contains a table with various network parameters, including 'Number of layers', 'Number of nodes', 'Number of SRGs', and 'Routing is not defined'.

File Tools Help

OFFLINE NETWORK DESIGN

Network topology

Node names Link ids **Non-connected nodes** **● ● ● A+ A-**

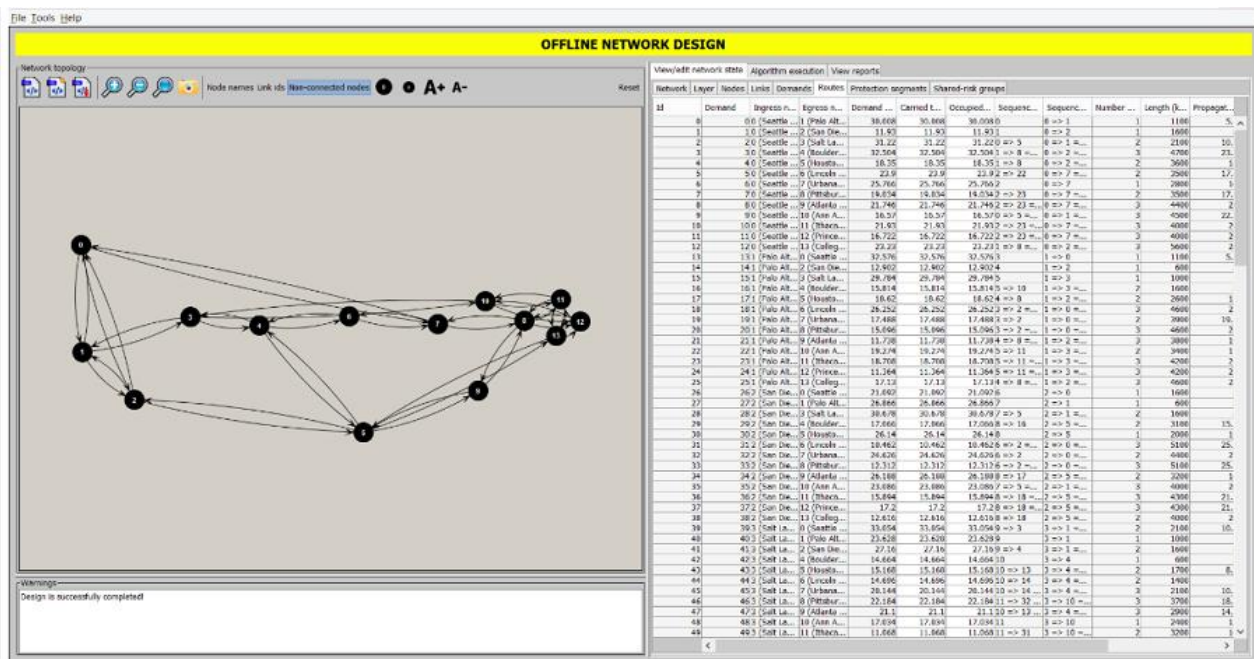
Reset

View/edit network state | Algorithms execution | View reports

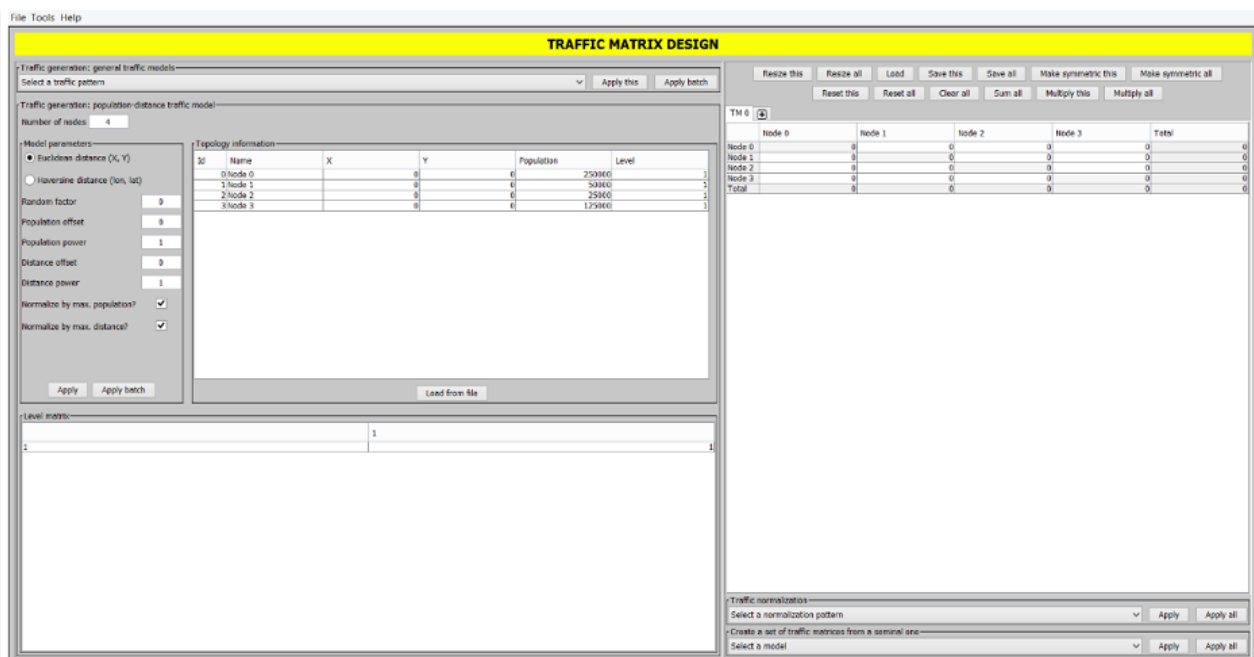
Network | Layer | Nodes | Links | Demands | Routes | Protection segments | Shared risk groups

ID	Show/hide	Origin no.	Destination	State	Coupled	Capacity	Carried	Reserved	Utilization	Utilization	Is border
0	☒	0	Seattle ... 1	Prato Alb...	✓	2000	191.68	745	0.108	0.051	
1	☒	0	Seattle ... 2	San Die...	✓	2000	238.740	482.280	0.205	0.094	
2	☒	0	Seattle ... 2	Urbanas...	✓	2000	225.224	240.922	0.228	0.118	
3	☒	1	Prato Alb...	0	Seattle ...	2000	164.944	256.126	0.261	0.077	
4	☒	1	Prato Alb...	0	San Die...	2000	138.616	168.314	0.136	0.052	
5	☒	1	Prato Alb...	0	Seattle ...	2000	126.198	167.442	0.302	0.098	
6	☒	2	San Die...	0	Seattle ...	2000	123.240	177.275	0.154	0.065	
7	☒	2	San Die...	1	Prato Alb...	2000	128.656	212.516	0.171	0.064	
8	☒	2	San Die...	0	Urbanas...	2000	229.676	254.982	0.376	0.118	
9	☒	3	Prato Alb...	1	Prato Alb...	2000	182.516	261.346	0.217	0.091	
10	☒	3	Prato Alb...	4	Boulder...	2000	259.862	346.352	0.108	0.079	
11	☒	3	Prato Alb...	10	Ann Ar...	2000	241.888	291.838	0.267	0.121	
12	☒	4	Boulder...	0	Seattle ...	2000	201.778	324.611	0.163	0.101	
13	☒	4	Boulder...	5	Urbanas...	2000	290.796	468.190	0.379	0.145	
14	☒	4	Boulder...	6	Urbanas...	2000	427.254	372.890	0.297	0.11	
15	☒	5	Urbanas...	2	San Die...	2000	238.240	271.56	0.275	0.116	
16	☒	5	Urbanas...	4	Boulder...	2000	385.174	587.556	0.434	0.14	
17	☒	5	Urbanas...	0	Adelphi...	2000	159.822	449.348	0.13	0.085	
18	☒	5	Urbanas...	11	College...	2000	262.794	408.676	0.286	0.111	
19	☒	6	Urbanas...	4	Boulder...	2000	377.4	287.426	0.382	0.136	
20	☒	6	Urbanas...	7	Urbanas...	2000	222.892	265.228	0.304	0.112	
21	☒	7	Urbanas...	0	Seattle ...	2000	151.664	543.976	0.168	0.076	
22	☒	7	Urbanas...	6	Urbanas...	2000	265.972	284.88	0.375	0.133	
23	☒	7	Urbanas...	0	Prato Alb...	2000	282.458	466.822	0.326	0.181	
24	☒	8	Prato Alb...	7	Urbanas...	2000	274.998	426.622	0.291	0.117	
25	☒	8	Prato Alb...	0	Adelphi...	2000	294.468	262.216	0.248	0.167	
26	☒	8	Prato Alb...	11	College...	2000	149.546	222.124	0.187	0.077	
27	☒	8	Prato Alb...	12	Prato Alb...	2000	228.658	341.931	0.31	0.105	
28	☒	8	Prato Alb...	5	Urbanas...	2000	178.51	425.427	0.236	0.08	
29	☒	9	Adelphi...	0	Prato Alb...	2000	169.226	464.88	0.117	0.085	
30	☒	10	Ann Ar...	7	Prato Alb...	2000	222.428	183.898	0.264	0.112	
31	☒	10	Ann Ar...	11	College...	2000	71.45	187.964	0.225	0.036	
32	☒	10	Ann Ar...	12	Prato Alb...	2000	229.566	263.198	0.257	0.111	
33	☒	11	College...	0	Prato Alb...	2000	153.682	338.436	0.245	0.077	
34	☒	11	College...	10	Ann Ar...	2000	32.878	256.618	0.054	0.026	
35	☒	11	College...	11	College...	2000</					

9.4 Routes View



9.5 Traffic Matrix Design



9.6 Traffic Normalization

TRAFFIC MATRIX DESIGN

Traffic generation: general traffic models—
Select a traffic pattern: ▼ Apply this Apply batch

Traffic generation: population-distance traffic model—
Number of nodes: 14

Model parameters—
☒ Euclidean distance (X, Y)
☐ Haversine distance (lon, lat)
 Random factor: 0
 Population offset: 0
 Distance power: 1
 Distance offset: 0
 Normalize by max. population? ☒
 Normalize by max. distance? ☒
Apply Apply batch

Topology information—

ID	Name	X	Y	Population	Level
0	Seattle (WA)	-122.292	47.616	620778	1
1	Yaleville (CA)	-122.181	37.442	62522	1
2	San Diego (CA)	-117.239	32.805	1326960	1
3	San Jose city (CA)	-121.883	37.339	189869	1
4	Boulder (CO)	-105.579	40.377	98869	1
5	Houston (TX)	-95.36	29.767	2145960	1
6	Lincoln (NE)	-96.697	40.823	262341	1
7	Urbana-Champaign (IL)	-88.227	40.113	81291	1
8	Pittsburgh (PA)	-80	40.436	307484	1
9	Atlanta (GA)	-84.383	33.75	432427	1
10	Ann Arbor (MI)	-83.754	42.289	1149425	1
11	Illinois (IN)	-76.477	42.444	33954	1
12	Princeton (NJ)	-74.664	40.34	28572	1
13	College Park (MD)	-76.942	36.96	25867	1

Level matrix—

Node	Level
0	1
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	1
13	1

TM 0

	Node 0	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7	Node 8	Node 9
Node 0	0	0.042	0.552	0.099	0.034	0.431	0.064	0.015	0.047	0
Node 1	0.042	0	0.135	0.012	0.004	0.053	0.007	0.002	0.005	0
Node 2	0.552	0.135	0	0.276	0.099	1.848	0.185	0.038	0.112	0
Node 3	0.099	0.012	0.276	0	0.03	0.215	0.034	0.007	0.019	0
Node 4	0.034	0.004	0.099	0.03	0	0.156	0.032	0.005	0.013	0
Node 5	0.431	0.053	1.848	0.215	0.156	0	0.508	0.145	0.369	0
Node 6	0.064	0.007	0.185	0.034	0.032	0.529	0	0.026	0.051	0
Node 7	0.015	0.002	0.038	0.007	0.005	0.145	0.026	0	0.032	0
Node 8	0.047	0.005	0.112	0.019	0.013	0.369	0.051	0.032	0	0
Node 9	0.07	0.008	0.183	0.03	0.021	0.832	0.094	0.05	0.174	0
Node 10	0.019	0.002	0.046	0.008	0.005	0.151	0.024	0.032	0.069	0
Node 11	0.004	0	0.03	0.002	0.001	0.03	0.004	0.002	0.004	0
Node 12	0.004	0	0.009	0.002	0.001	0.028	0.004	0.002	0.017	0
Node 13	0.004	0	0.01	0.002	0.001	0.033	0.004	0.002	0.029	0
Total	1.885	0.275	2.983	0.736	0.401	4.32	1.028	0.345	0.981	1

Select a normalization pattern
 1. Total normalization
 2. Rowwise normalization
 3. Columnwise normalization
 4. Normalise to maximum traffic that can be carried
 Select a normalization pattern: ▼ Apply Apply all
 Create a set of traffic matrices from a normal one
 Select a model: ▼ Apply Apply all

10. Software Upgrade Management

From the UTL MP-0072 EMS, you can upgrade software to any card in any node in the topology. The software file can be uploading to SCC via SFTP by using SCC settings -> software upload option.

Select below option under Manage Administration->Software Tools->Software Upgrade to upgrade software to all cards. Browse to the desired software upgrade file and click s/w upgrade. Software will upgrade for all selected cards.

Software Upgrade ▶

- SCC
- BMU
- SCU
- Others

Make sure the file has been uploaded to the SCC. The progress of the upgrade on each card is indicated in the progress columns.

S/W Upgrade For SCC

Select Zone: Select Circle: Select Node/ID:

Select SoftwareUpload File:

Now entries

S.No	Card Name	Rack	Subrack	Slot	Select ☐	Upgrading Status
1	SCC	1	2	1	☒	Selected

Showing 1 to 1 of 1 entries

S/W Upgrade For BMU

Select Zone: Select Circle: Select Node/ID:

Select SoftwareUpload File:

Now entries

S.No	Card Name	Rack	Subrack	Slot	Select ☐	Upgrading Status
1	BMU	1	1	1	☒	Selected

Showing 1 to 1 of 1 entries

S/W Upgrade For SCU

Select Zone: Select Circle: Select Node/ID:

Select SoftwareUpload File:

Now entries

S.No	Card Name	Rack	Subrack	Slot	Select ☐	Upgrading Status
1	SCU	1	3	17	☐	Not Selected
2	SCU	1	6	17	☐	Not Selected

Showing 1 to 2 of 2 entries

S/W Upgrade For Others

Select Zone: India

Select Circle: South

Select NodeIP: 192.168.1.1

Select SoftwareUpload File: Select

Get S/W Files

S/W Upgrade

Showing 1 to 10 of 10 entries

S.No	Card Name	Rack	Subrack	Slot	Select ☐	Upgrading Status
1	10GOMTP11	1	3	1	☐	Not Selected
2	10GOMTP11	1	3	2	☐	Not Selected
3	10GOMTP11	1	3	3	☐	Not Selected
4	10GOMMP81	1	3	5	☐	Not Selected
5	10GOMMP81	1	3	7	☐	Not Selected
6	MUX1	1	4	1	☐	Not Selected
7	DEMUX1	1	5	1	☐	Not Selected
8	OBA1	1	6	9	☐	Not Selected
9	OBA1	1	6	12	☐	Not Selected
10	OPA1	1	6	14	☐	Not Selected

Showing 1 to 10 of 10 entries

First
Previous
1
Next
Last

Acronyms and Abbreviations

A

ALS Automatic laser shutdown.

B

Bandwidth In the data communication area, bandwidth specifies the maximum value of the rate when the data passes through some data channel.

BMU Bay Monitoring Unit.

D

DCN Data Communication Network.

DEMUX Demultiplexer

E

EMS Element Management System

F

FEC Forward Error Correction

FTP File Transfer Protocol

G

GNE Gateway Network Element

H

HA High Availability.

HTTP Hyper-Text Transmission Protocol

I

IP Internet Protocol.

L

LAN Local Area Network.

LCT Local Craft Terminal

M

MUX Multiplexer.

MIB Management Information Base.

N

NE Network Element

NMS Network Management System.

NTP Network Time Protocol

O

OAM Operations, Administration, and Maintenance

OBA Optical Booster Amplifier

OLA Optical Line Amplifier

OMTP11 (Optical Multi Transponder 1-Client 1-Line)

OMTP11T (Optical Multi Transponder 2-Lines)

OMTP12 (Optical Multi Transponder 1-Client 2-Lines)

OMTP22 (Optical Multi Transponder 2-Clients 2-Lines)

OMMP81 (Optical Multi Service Muxponder 8-Clients 1-Line)

OMMP82 (Optical Multi Service Muxponder 8-Clients 2-Lines)

OPA Optical Power Amplifier

OSC Optical Supervisor Channel.

OTN Optical Transmission Network

OTU Optical Transponder Unit

P

PSU Power supply Unit

R

ROADM Reconfigurable optical add-drop multiplexer.

S

SCC System Control Card.

SCU Subrack Control Unit.

SDH Synchronous Digital Hierarchy.

SNMP Simple Network Management Protocol

SQL Structured Query Language.

STM Synchronous Transfer Mode.

T

TCP/IP Transmission Control Protocol/Internet Protocol

TMF TeleManagement Forum.

W

WAN Wide Area Network.

WDM Wavelength Division Multiplexing